



**HKU
Med**

School of Clinical Medicine
Department of Medicine
香港大學內科學系

Chest pain on exertion: *Ischaemic heart disease* *Angina pectoris*

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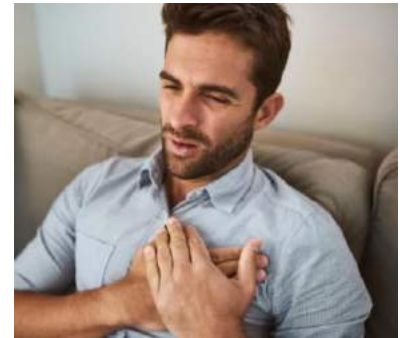
CTM, M/49 (1)

C/O: Presents for evaluation of chest pain on exertion for 3 months

- **No past history cardiovascular diseases.**
- **On and off retrosternal discomfort while walking back to his office after lunch**
- **Constricting in character and relieve after resting**

Family Hx: Father died of CAD at age of 65

Social Hx: Chronic smoker and non-drinker



Ischaemic Heart Disease (IHD)

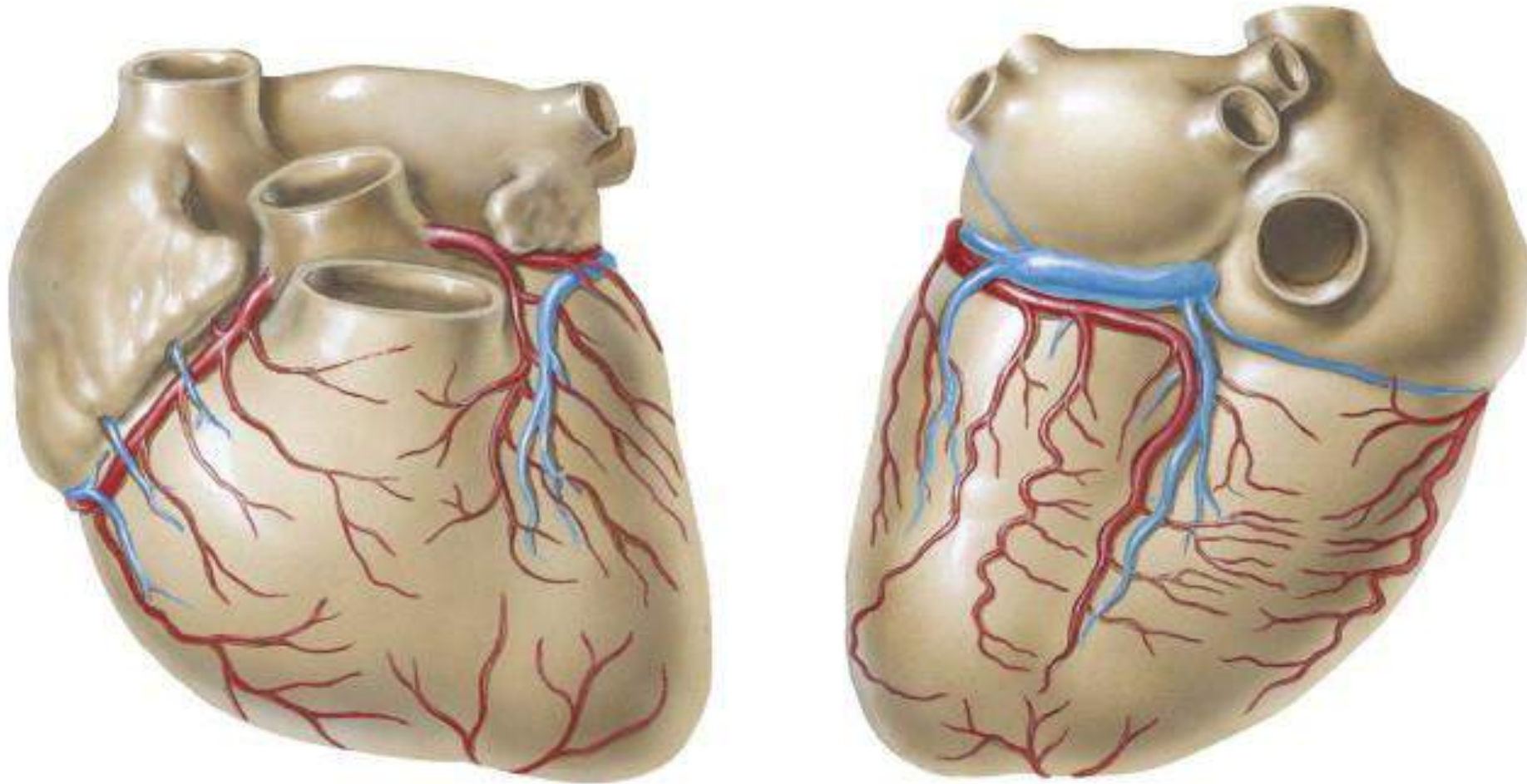
◆ Pathophysiology and Etiologies

◆ Diagnosis and Assessment

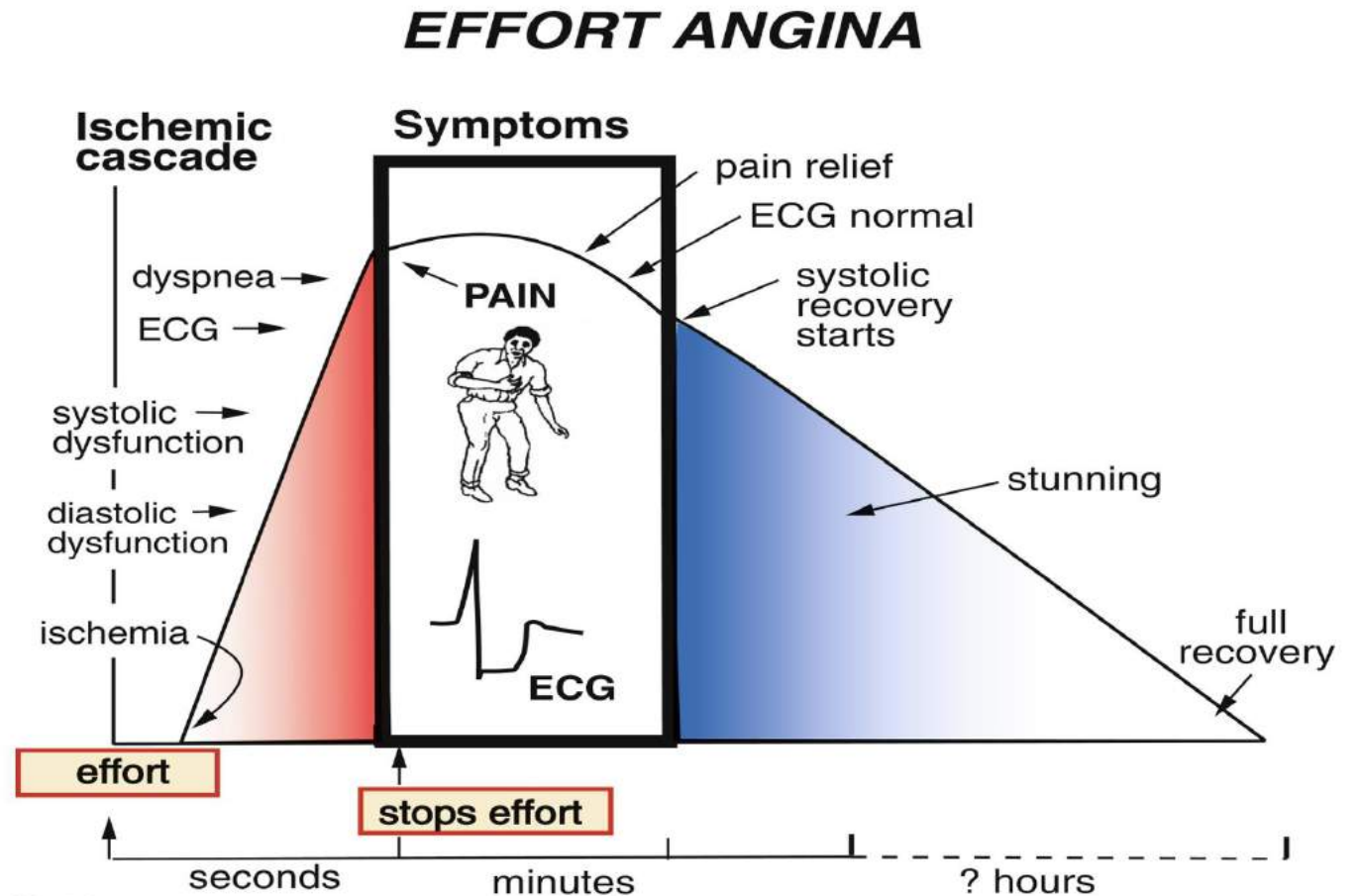
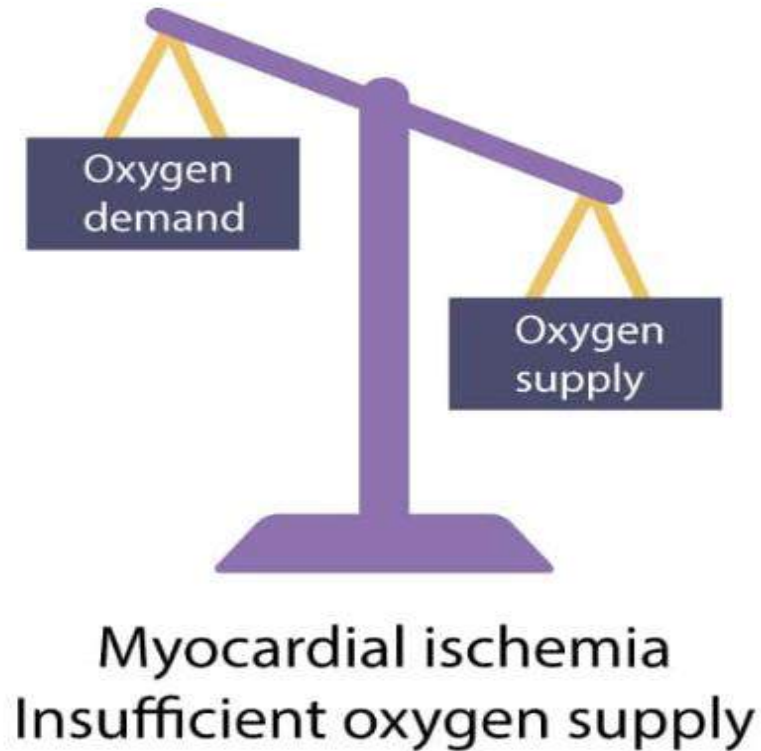
◆ Management

Blood Supply to the Heart

ARTERIOVENOUS VARIATIONS

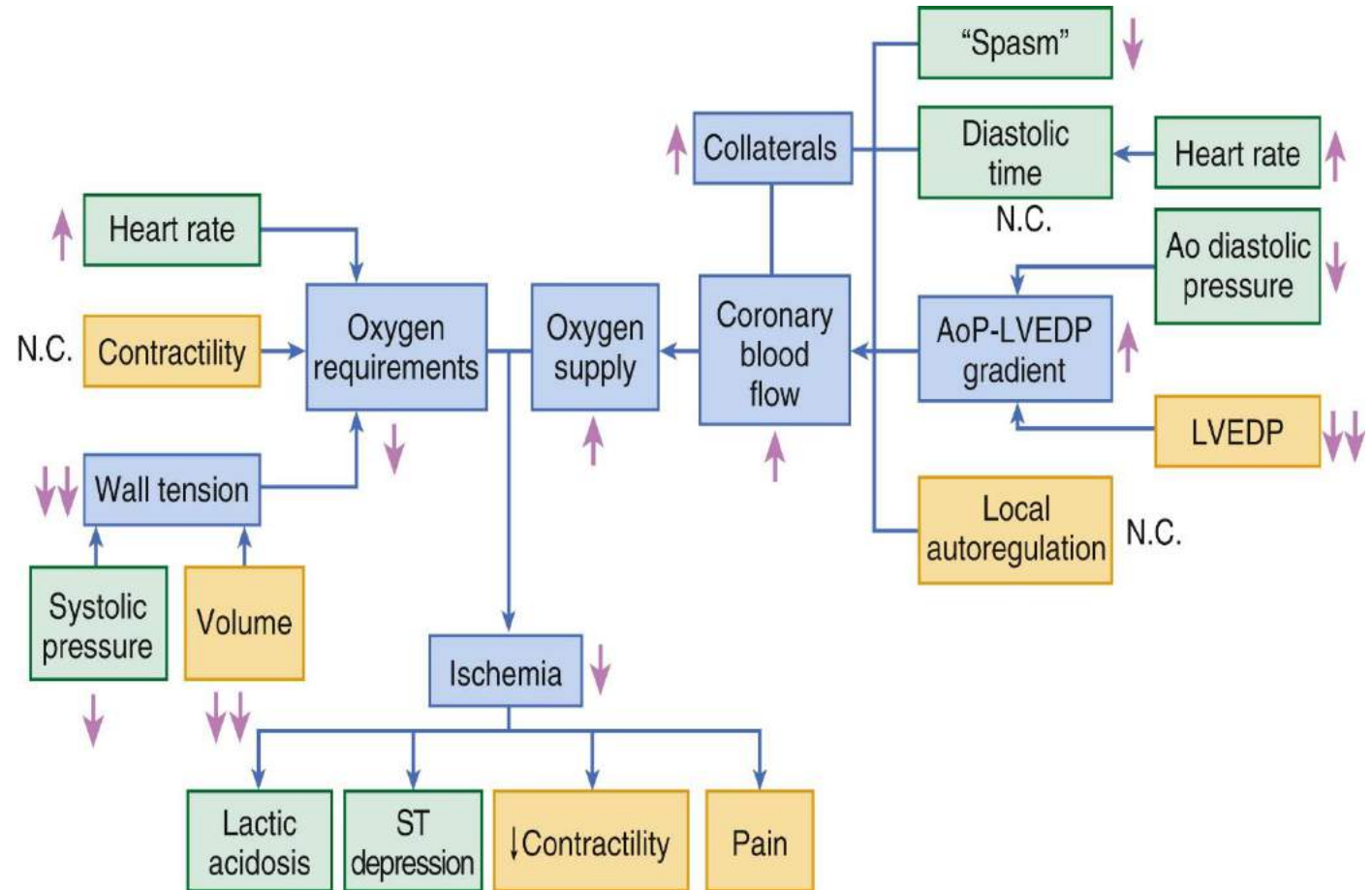
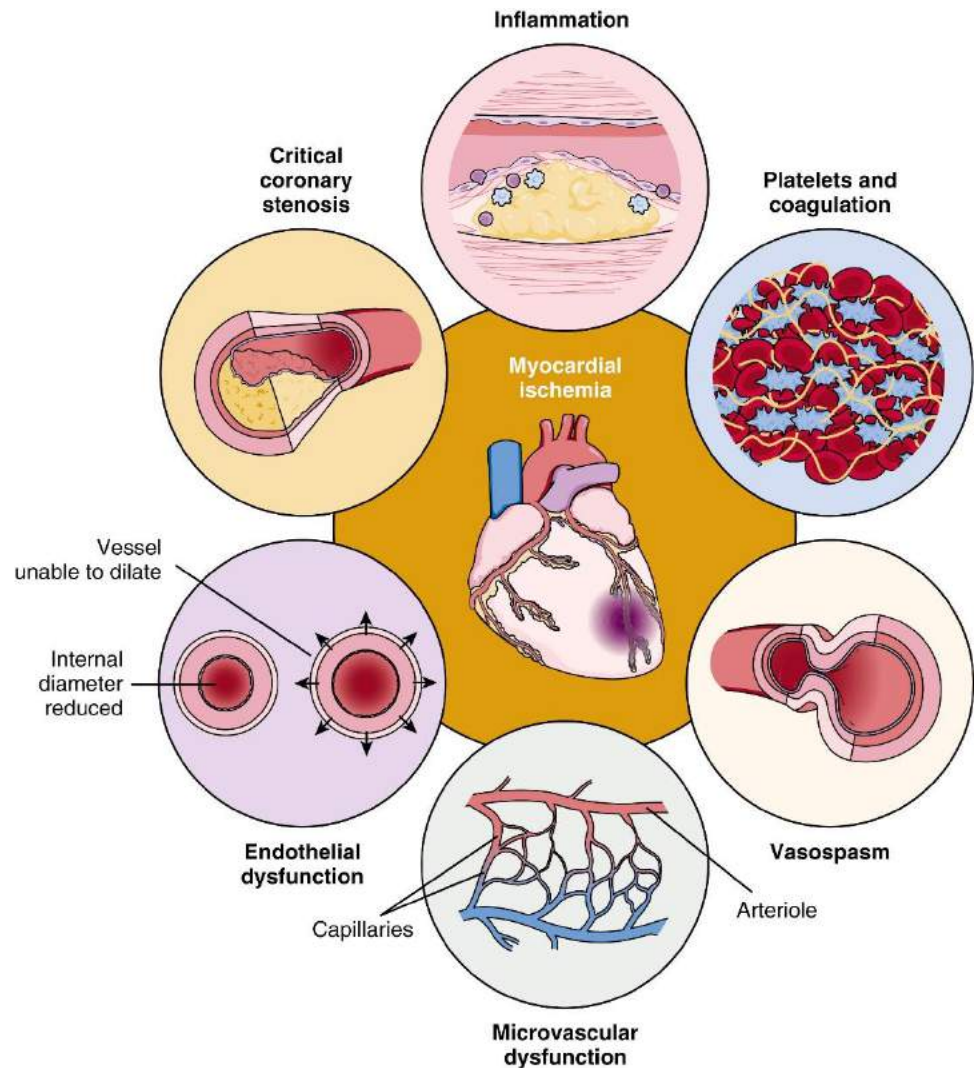


Pathophysiology of IHD

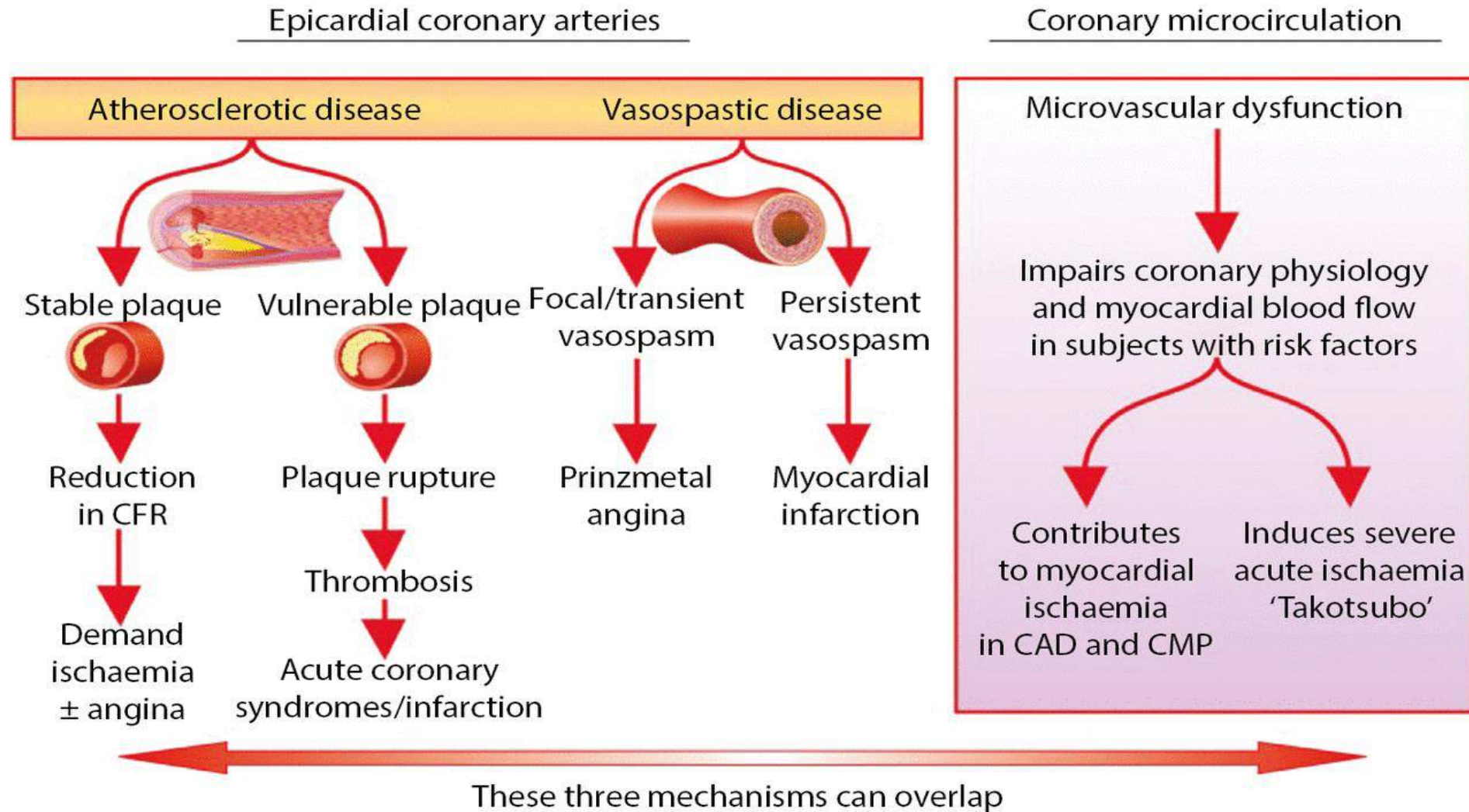


Angina pectoris is generally defined as a discomfort in the chest or adjacent areas that occurs predictably and reproducibly at a certain level of exertion and is relieved with rest or nitroglycerin.

Pathophysiology of IHD



Mechanisms of Myocardial Ischemia



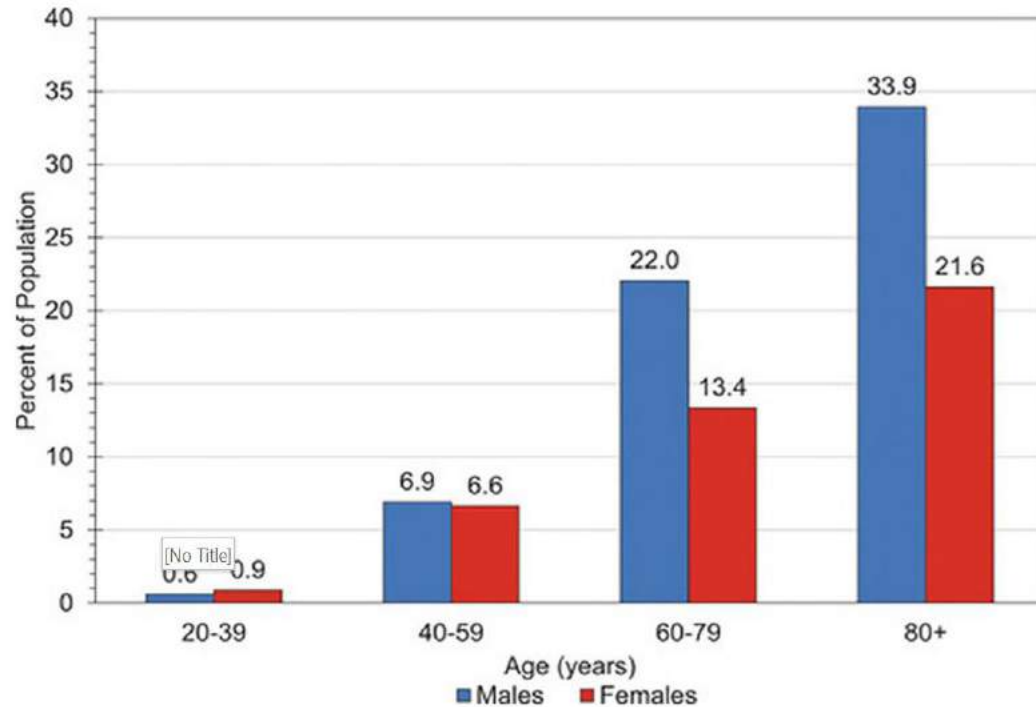
Mechanisms of Ischemic Chest Pain

- The mechanisms of cardiac pain and the neural pathways involved are poorly understood.
- It is presumed that angina pectoris results from ischemic episodes that excite chemosensitive and mechanosensitive receptors in the heart.
- The afferent fibers traverse the nerves that connect to the **upper five thoracic sympathetic ganglia and the upper five distal thoracic roots** of the spinal cord
- Within the spinal cord, cardiac sympathetic afferent impulses may converge with impulses from somatic thoracic structures, which may be the basis for **referred cardiac pain**, such as to the chest.
- **Silent ischemia** in diabetic patients, for example, has been proposed to be related to failed development of the cardiac sensory system because of reduced nerve growth factor.

IHD- Epidemiology

- Angina is most common in middle-aged and elderly men. Among persons 60 to 79 yrs of age, approximately 25% of men and 16% of women have coronary heart disease, and these figures rise to 37% and 23% among men and women >80 yrs of age, respectively.
- The incidence of coronary heart disease and angina in women after menopause is similar to that of men.
- Although the survival rate has steadily improved over time, IHD remains the number one cause of death in men and women (27% of deaths).
- The initial manifestation of IHD is angina pectoris in **50%**, and about **50%** of patients presenting to the hospital with acute coronary syndrome have preceding angina.
- In US, **the annual rate of MI in patients with symptomatic angina of 3 to 3.5%/yr.**
- **Within 12 months of initial diagnosis, 10% to 20% of patients with diagnosis of stable angina progress to MI or unstable angina.**

IHD- Epidemiology

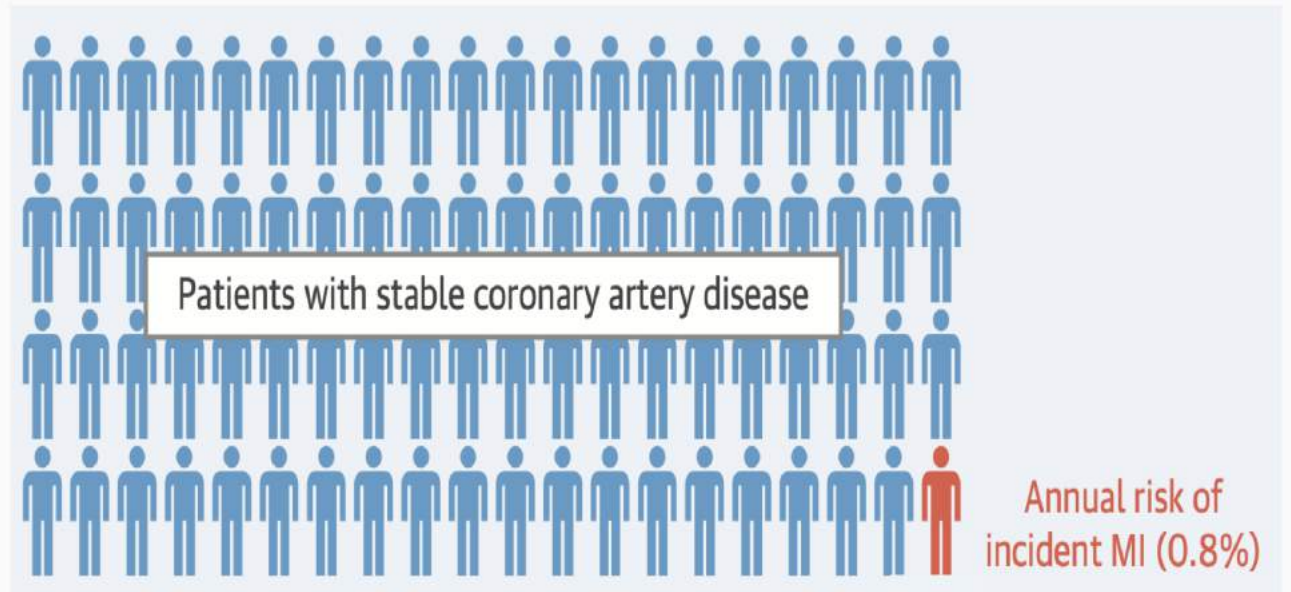


Prevalence of CHD in US 2015-2018

Virani SS, et al. JACC 2023

Incidence of myocardial infarction in stable CAD

CORONOR Registry n= 4184



Lemesle G et al. 2017;69:2149-56

Etiologies of IHD

ATHEROSCLEROSIS OF CORONARY ARTERY



PRIMARY CARDIAC CAUSE

Coronary artery abnormalities

- Coronary spasm
- Coronary arteritis
- Coronary dissection
- Coronary artery anomalies
- Radiation-induced coronary disease

Myocardial bridging

Aortic stenosis

Hypertrophic cardiomyopathy

Dilated cardiomyopathy

Tachycardia

PRIMARY NONCARDIAC CAUSE

Anemia

Sickle cell disease

Hypoxemia

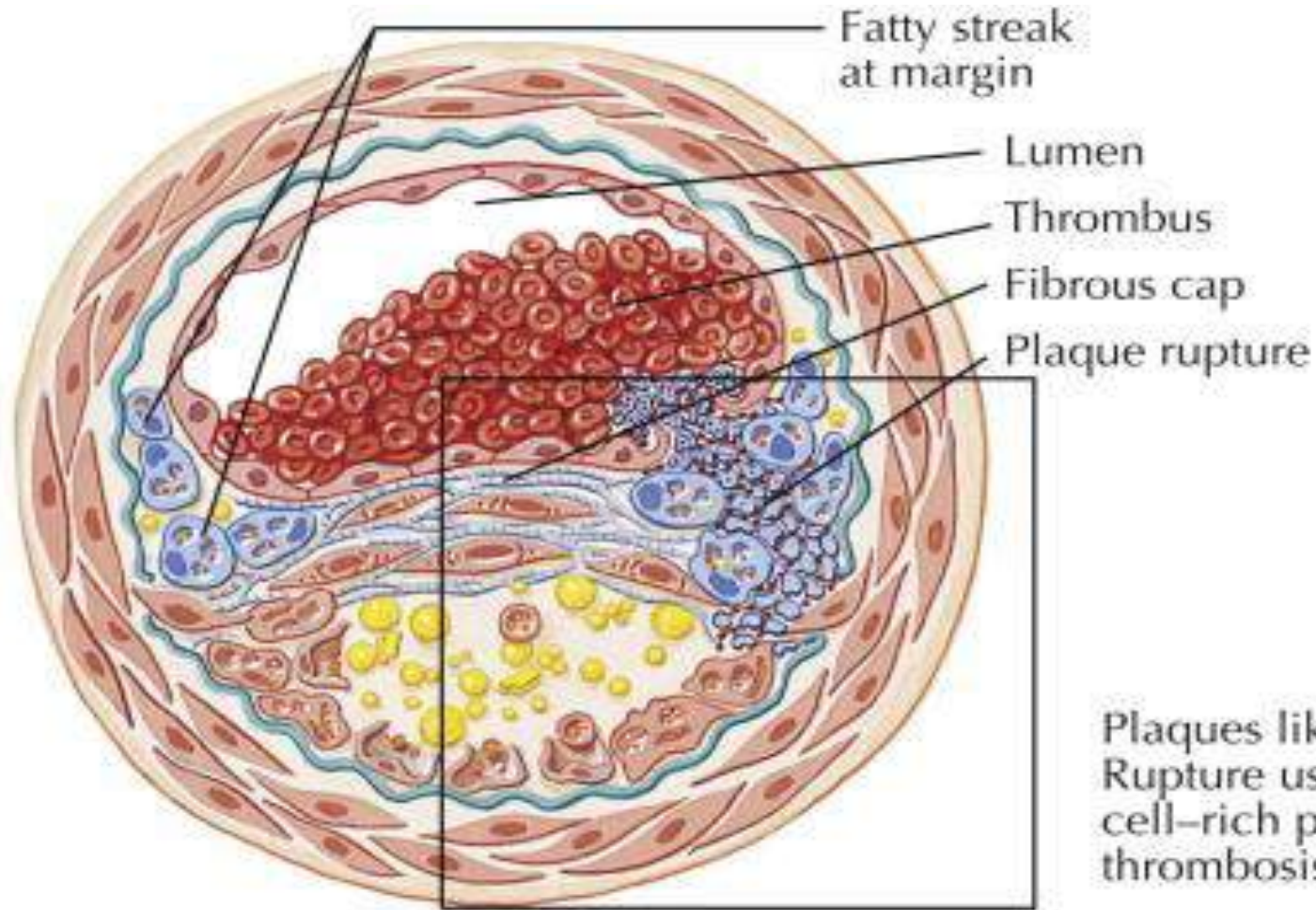
Carbon monoxide poisoning

Hyperviscosity (e.g., polycythemia)

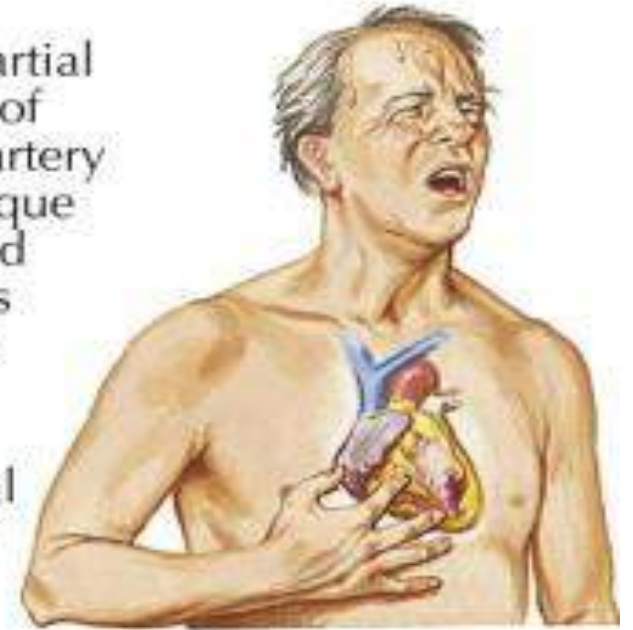
Hyperthyroidism

Pheochromocytoma

Pathogenesis of Atherosclerosis

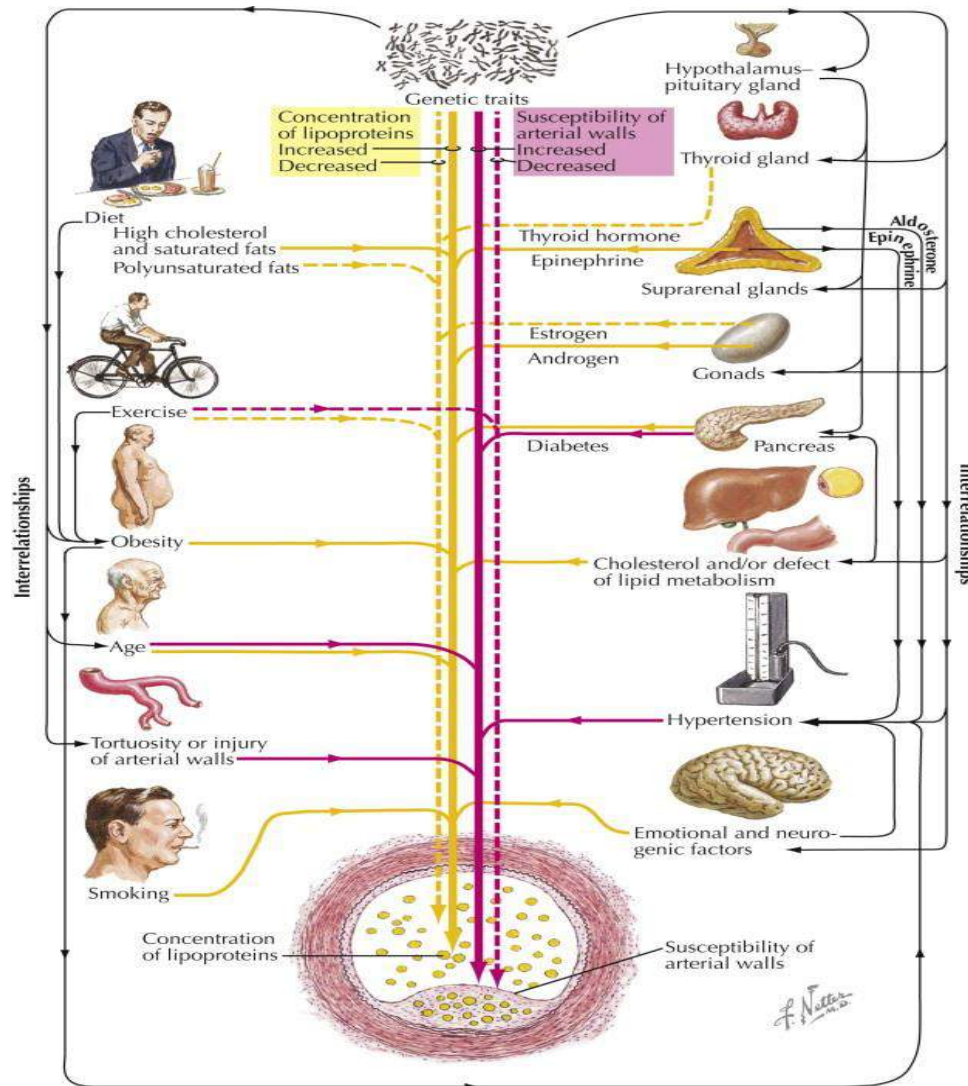


Total or partial occlusion of coronary artery due to plaque rupture and thrombosis can cause angina or frank myocardial infarction.

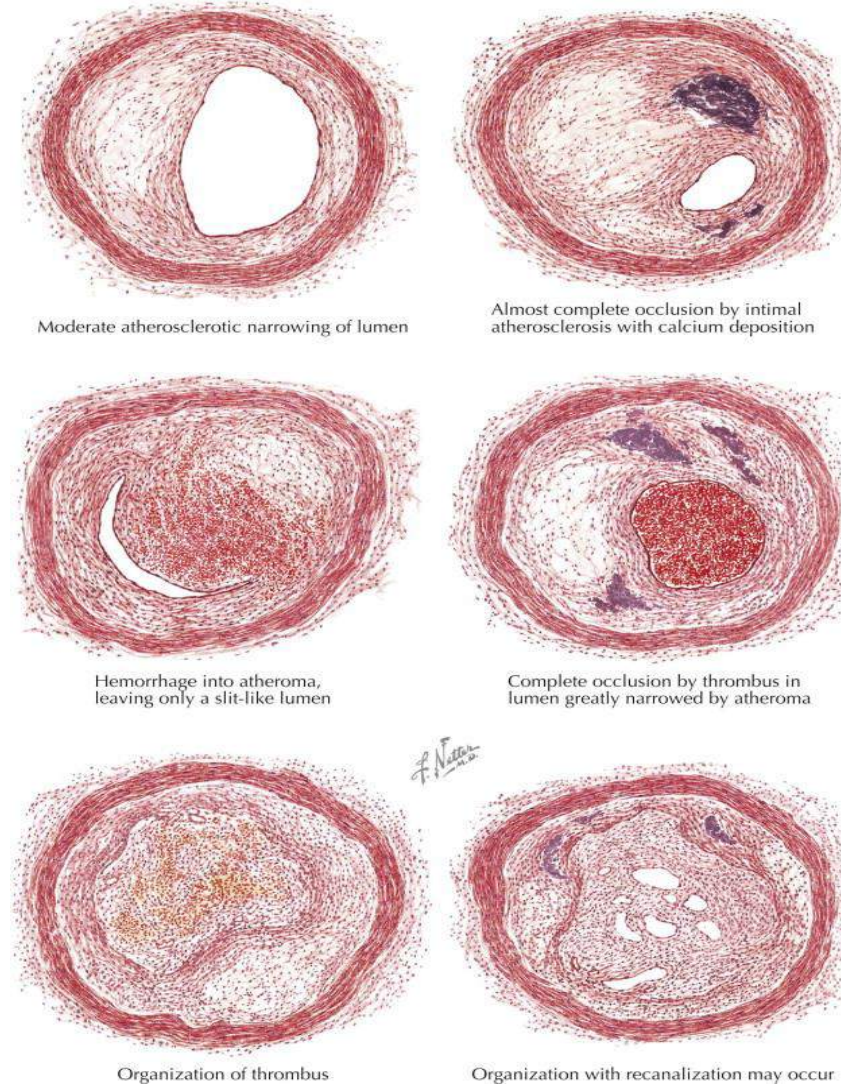


Plaques likely to rupture are termed unstable. Rupture usually occurs in lipid-rich and foam cell-rich peripheral margins and may result in thrombosis and arterial occlusion.

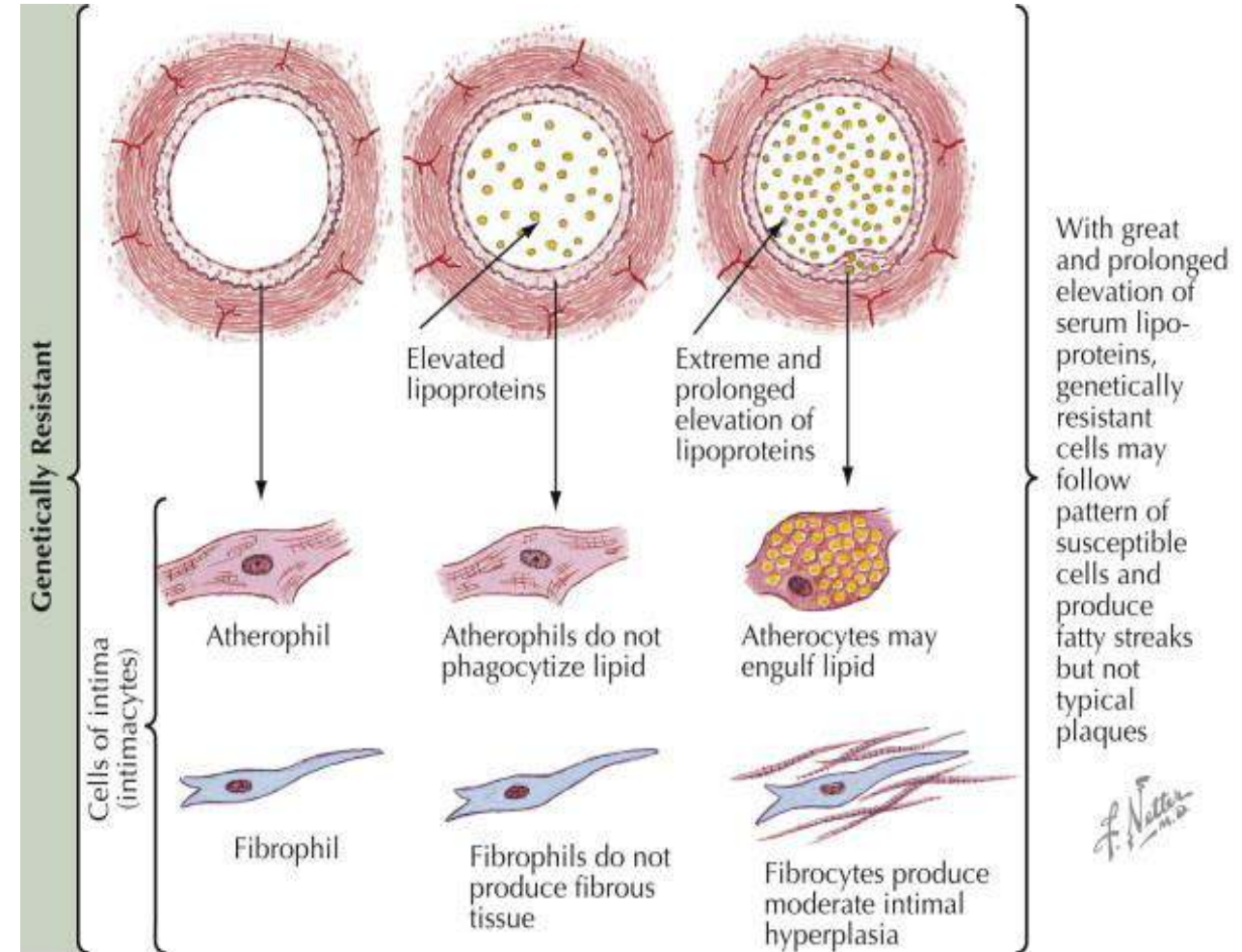
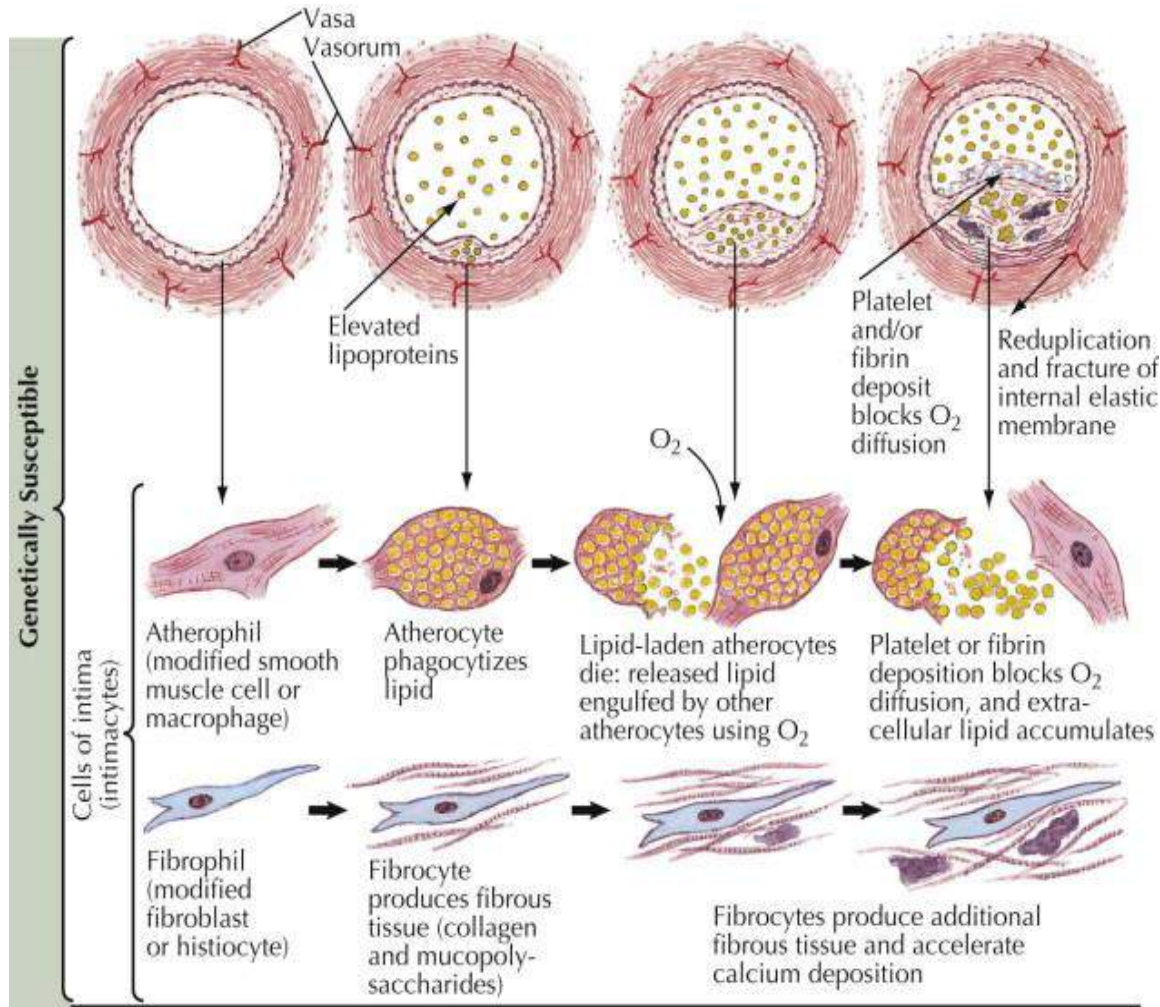
Pathogenesis of Atherosclerosis



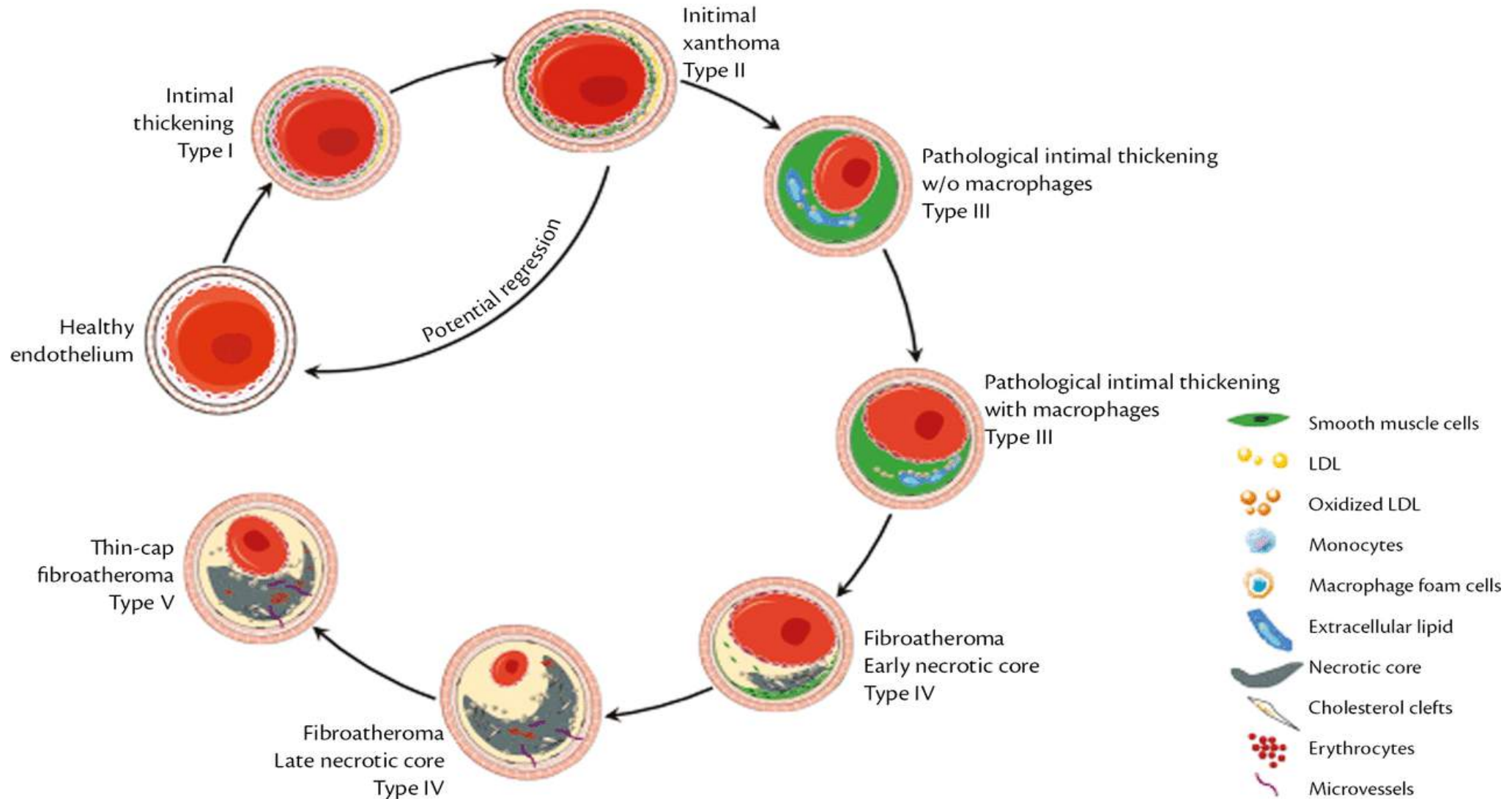
TYPES AND DEGREES OF CORONARY ATHEROSCLEROTIC NARROWING OR OCCLUSION



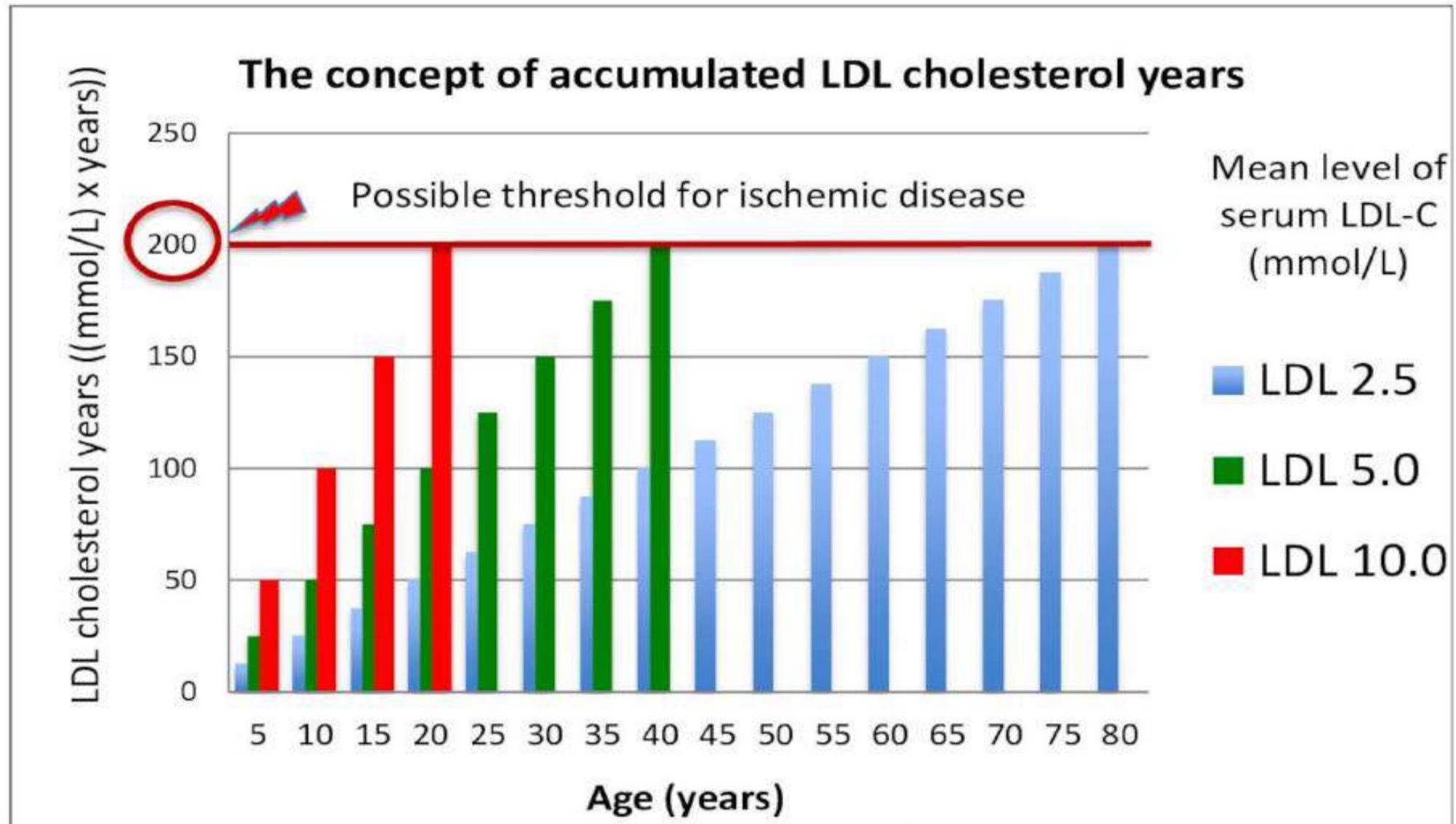
Pathogenesis of Atherosclerosis



Pathogenesis of Atherosclerosis



Accumulated LDL-C Years and Risk of CAD



The Emerging Threat of CAD

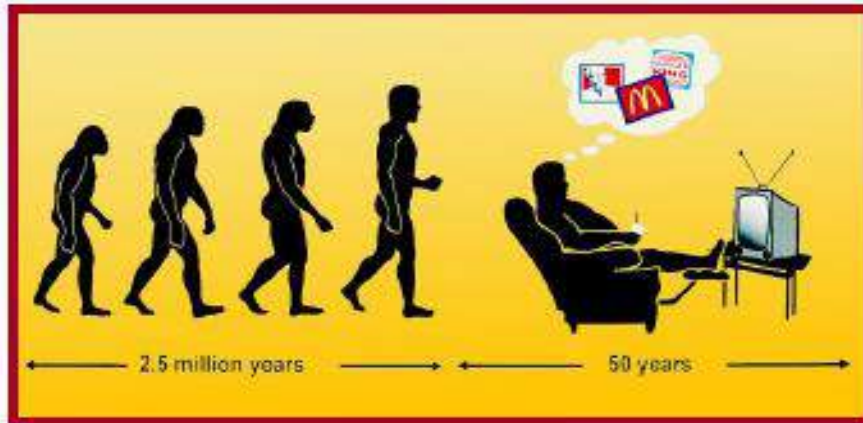
CVD is the number one killer in the 21st century

Smoking
High cholesterol
High Blood Pressure

Obesity
Metabolic Syndrome
Type 2 diabetes

1950s-1970s

1990s-2000s



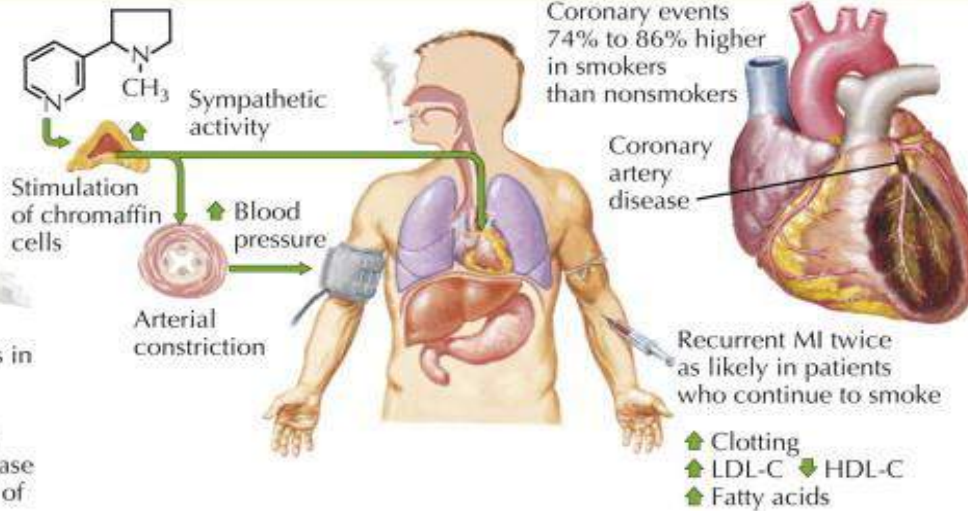
Etiologies: Diet and Substances Intakes

Tobacco



One half of all coronary deaths in women attributed to smoking

From a cardiologic standpoint, impact on coronary artery disease makes tobacco the most lethal of abused substances.



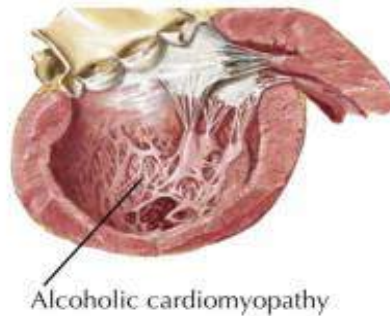
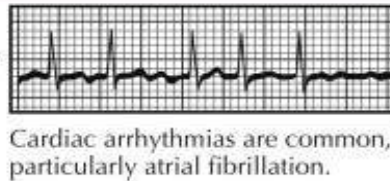
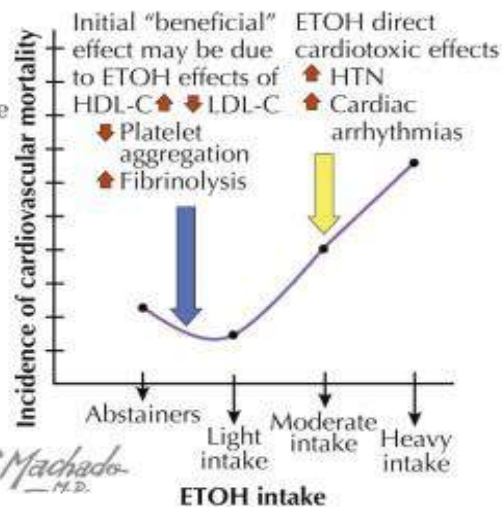
Alcohol



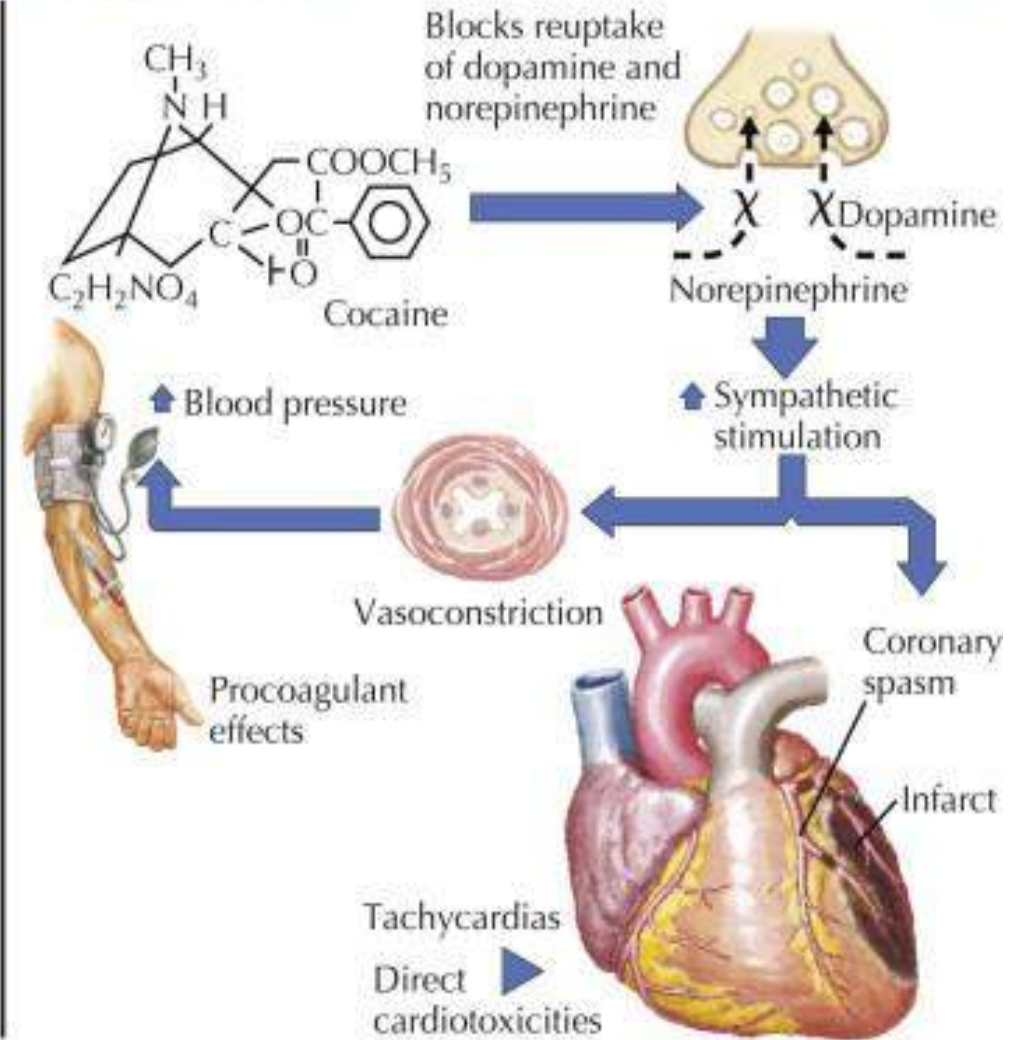
Comorbidities of alcohol and tobacco both

- HTN
- Blood pressure

The effect of alcohol on the heart is complex, showing a mix of adverse and beneficial effects.



Cocaine abuse



Ischaemic Heart Disease (IHD)

◆ Pathophysiology and Etiologies

◆ **Diagnosis and Assessment**

◆ Management



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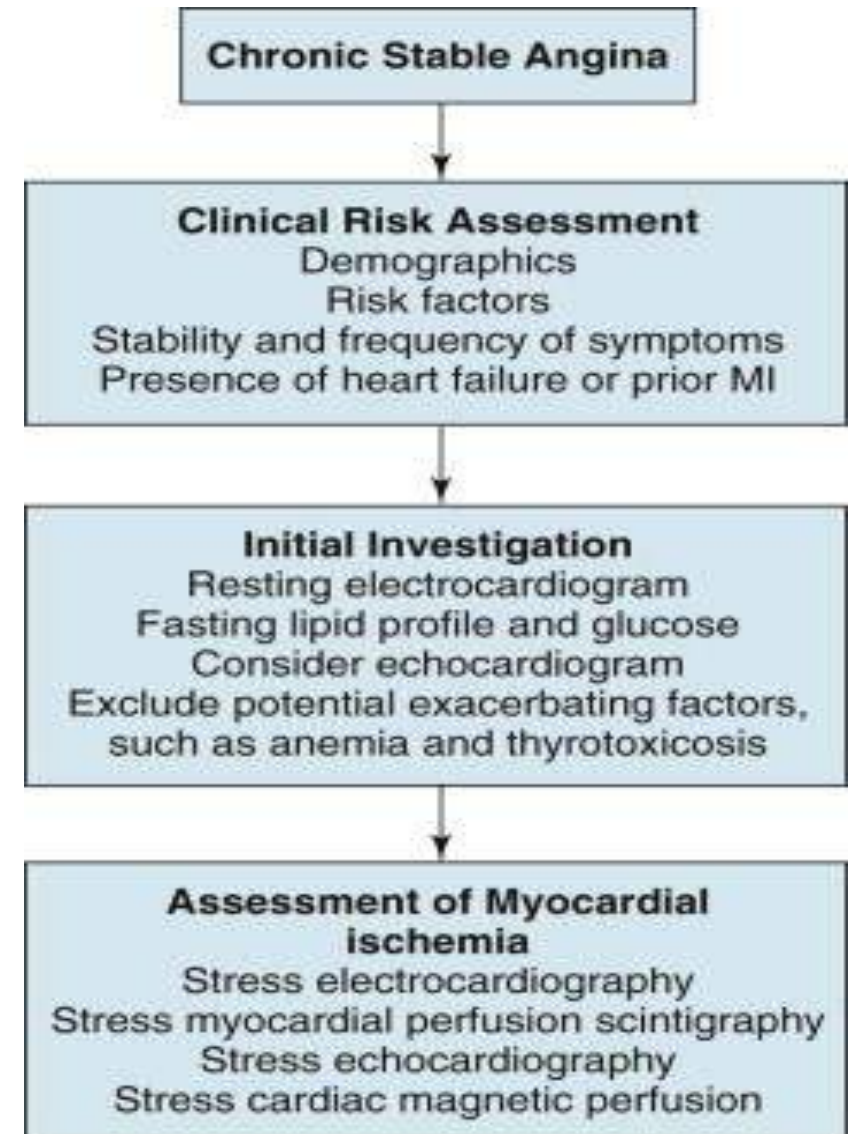
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Diagnosis and Assessment of IHD

◆ History

◆ Physical Examination

◆ Investigations



History:

Clinical Characteristics of *Angina Pectoris/Anginal equivalents*

Clarify the details of the pain using *SOCRATES*

Site – Retrosternal or slightly to left, occ. at extrathroacic sites

Onset – Gradual increase and fading away, usually 2-10 mins

Character – Tightness, pressure, squeezing, heaviness, burning, aching, fullness, “heavy weight” or “band across chest

Radiation – Neck throat, lower jaw, (L) shoulder or arm (ulnar distribution), occ to (R) arm, interscapular, epigastrium or back

Associations – Shortness of breath, dizziness, syncope

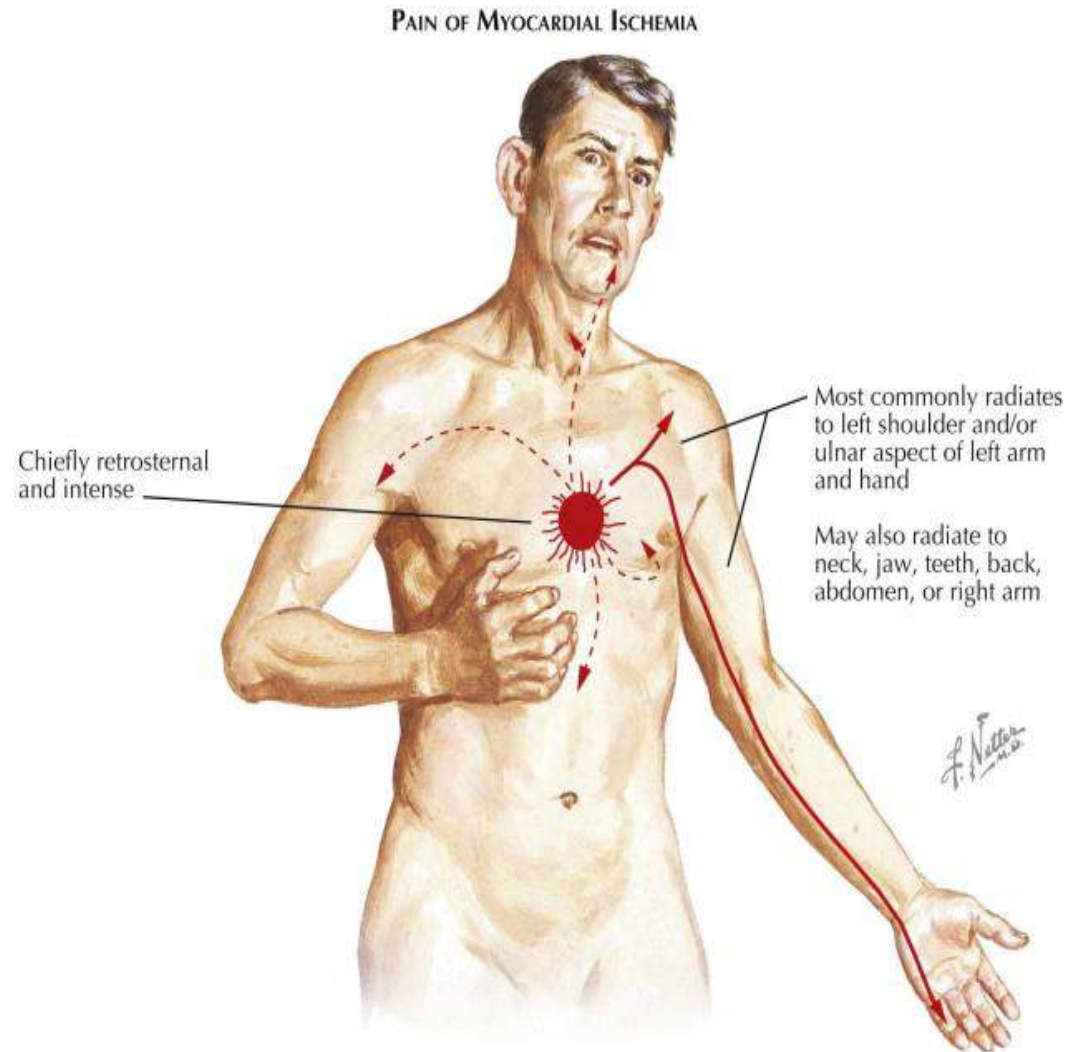
Time course – *worsening / improving / fluctuating*

Exacerbating / Relieving factors – 4 “E”- Eating, Exertion, Emotion, Environment (hot, cold) vs. Rest, cessation of activity/stress, NTG

Severity – *on a scale of 0-10, how severe is the pain?*

History:

Clinical Characteristics of *Angina Pectoris*/*Anginal equivalents*



Common descriptions of pain



Viselike



Constricting



Crushing weight and/or pressure

Other manifestations of myocardial ischemia

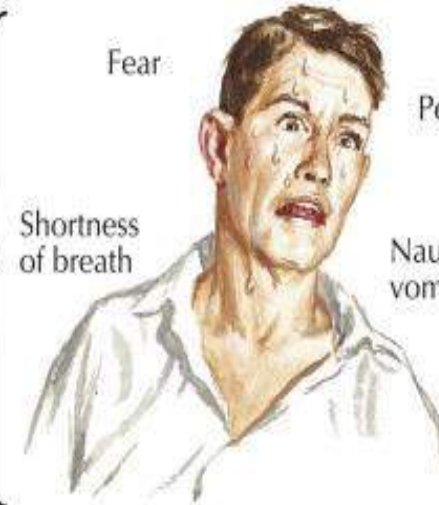
Fear

Shortness of breath

Perspiration

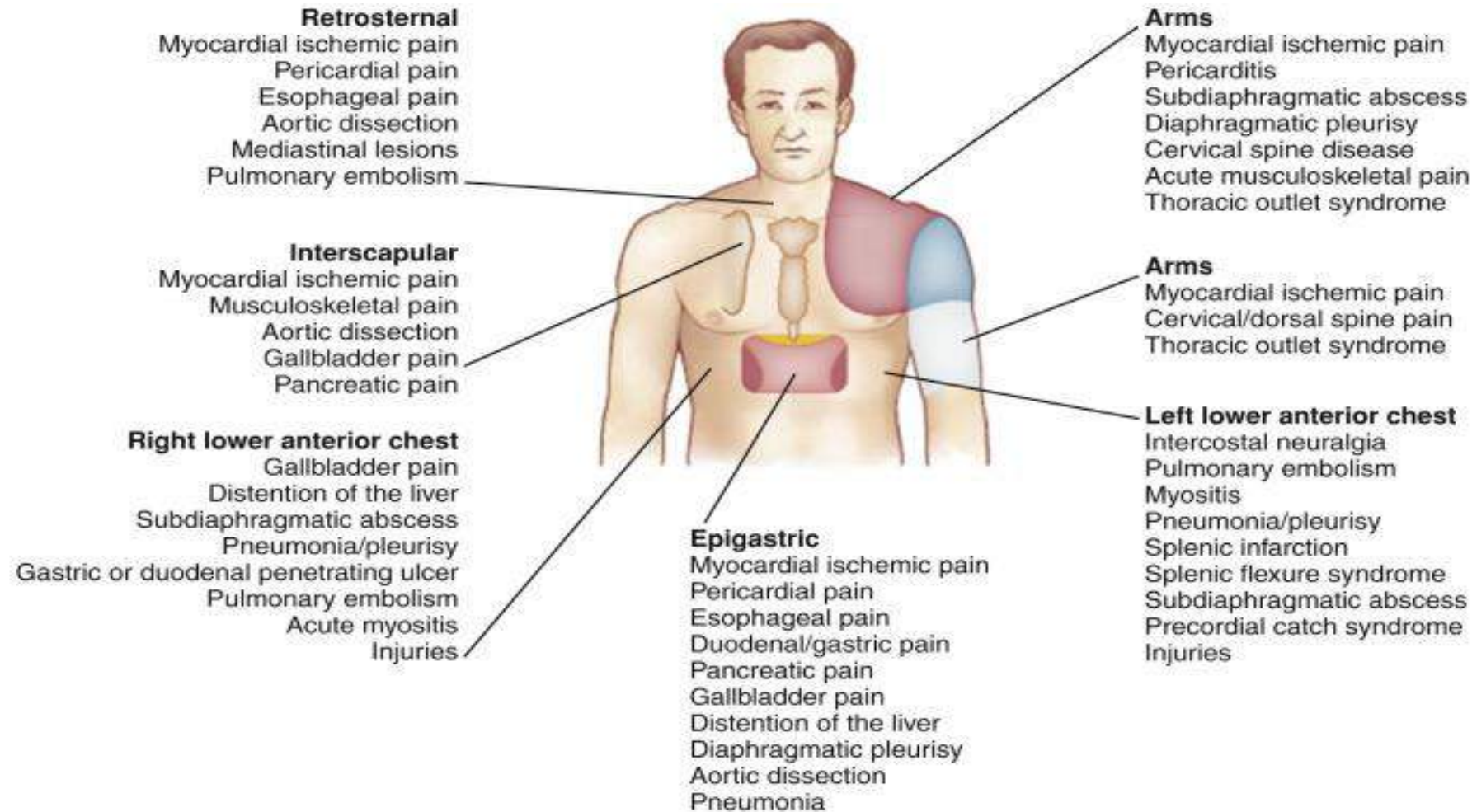
Nausea, vomiting

Weakness, collapse, coma



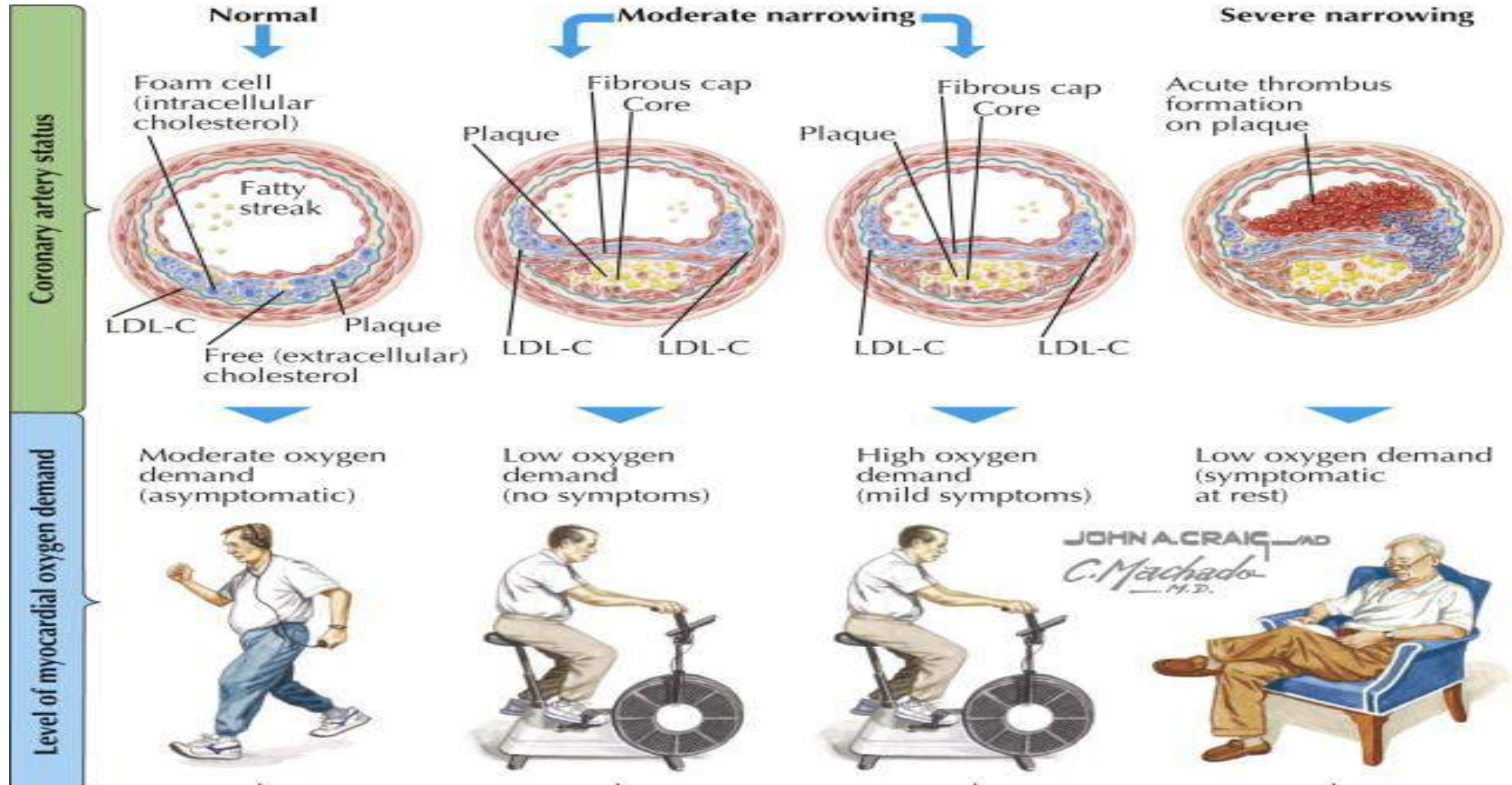
History:

Location of Discomfort and Cause of Chest Symptoms



History:

Severity of Chest Symptoms



History:

Non-Chest Pain Symptoms of IHD

DYSPNEA

Dyspnea on exertion

Dyspnea at rest

Paroxysmal nocturnal dyspnea

Temporal change of increasing exertional dyspnea with declining effort tolerance

NON-CHEST LOCATIONS OF DISCOMFORT (EITHER EXERTIONAL OR AT REST)

Neck or mandibular discomfort or pain

Throat tightness

Shoulder discomfort

Upper arm or forearm discomfort (more often left-sided)

Interscapular or infrascapular discomfort

MID-EPIGASTRIC OR ABDOMINAL

Mid-epigastric burning, often postprandially

Sharp abdominal pain (atypical, but more common in women)

Right upper quadrant discomfort (may mimic gallbladder disease or pancreatitis)

Nausea or vomiting (often associated with increased vagal tone secondary to inferior myocardial ischemia or infarction)

DIAPHORESIS

EXCESSIVE FATIGUE AND WEAKNESS

Often a discernible prodrome of increasing fatigue with declining effort tolerance

DIZZINESS AND SYNCOPE

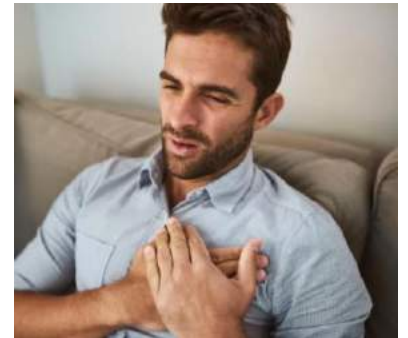
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Clinical Likelihood of CAD

Patients with angina and/or dyspnoea and suspected coronary artery disease

Pre-test probability of coronary artery disease



	Typical		Atypical		Non-anginal		Dyspnoea ^a	
Age	M	W	M	W	M	W	M	W
30-39	3%	5%	4%	3%	1%	1%	0%	3%
40-49	22%	10%	10%	6%	3%	2%	12%	3%
50-59	32%	13%	17%	6%	11%	3%	20%	9%
60-69	44%	16%	26%	11%	22%	6%	27%	14%
70+	52%	27%	34%	19%	24%	10%	32%	12%

^a In addition to the classic Diamond and Forrester classes, patients with dyspnoea only or dyspnoea as the primary symptom are included. The dark green shaded regions denote the groups in which non-invasive testing is most beneficial (pre-test probability >15%). The light green shaded regions denote the groups with pre-test probability of CAD between 5-15% in which the testing for diagnosis may be considered after assessing the overall clinical likelihood based on modifiers of pre-test probability.

www.escardio.org/guidelines

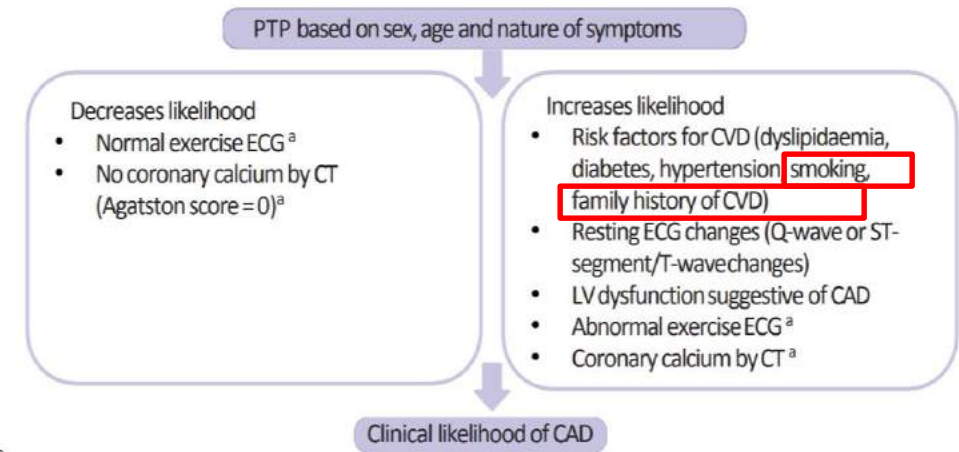
ESC Guidelines on the diagnosis and management of chronic coronary syndromes (European Heart Journal 2019; 10.1093/eurheartj/ehz425)

©ESC

5

Patients with angina and/or dyspnoea and suspected coronary artery disease

Determinants of clinical likelihood of CAD



^a if available.

www.escardio.org/guidelines

ESC Guidelines on the diagnosis and management of chronic coronary syndromes (European Heart Journal 2019; 10.1093/eurheartj/ehz425)

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Demographic data and symptoms

Cardiovascular risk factors



Clinical Likelihood of CAD



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CTM, M/49 (2)

P/E:

- **Blood pressure 130/85**
- **Xanthelasma on bilateral upper eyelids, no tendon xanthoma**
- **Normal cardiovascular examination**

Ix:

- **ECG- NAD**
- **Blood tests: L/RFT/TFT- NAD, FBS 5.6mmol/L,
TC 6.7 mmol/L, LDL-C 4.8 mmol/L, HDL-C 1.2 mmol/L
Lp(a) 150mg/dL**



Physical Examination:

Differential Diagnosis of Chest Pain

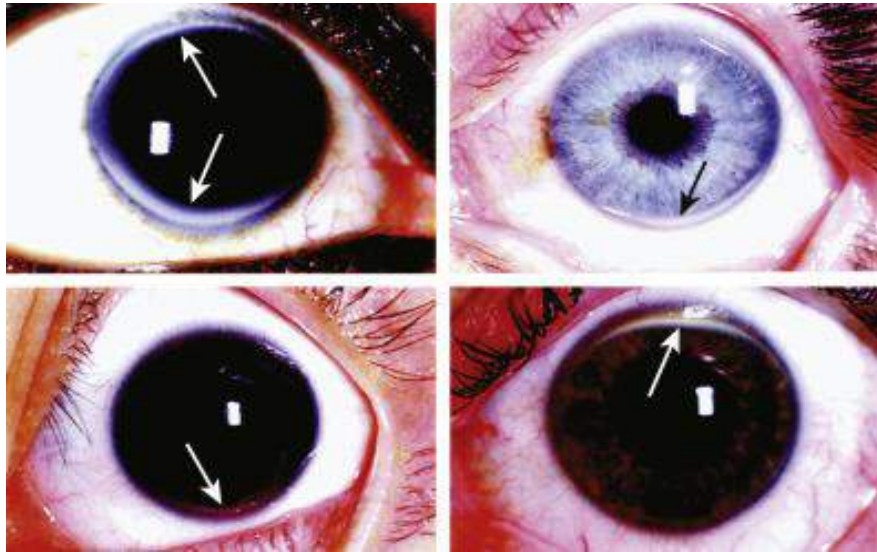
- Majority of cases : with normal physical findings
- **Looks for risk factors for CAD:** corneal arcus, xanthomas, retinal arteriolar changes in hypertension and DM.
- **Looks for complications of IHD:** cardiomegaly, heart failure and mitral regurgitation.
- **Looks for implicating or contributing conditions** for IHD, such as thyroid enlargement or severe anemia.



Physical Examination:

Goal	Category	Findings/Comments
Assessing risk factors	Blood pressure	>15 mm Hg arm blood pressure disparity = increased risk for peripheral arterial disease (PAD) and cardiovascular death
	Weight	Body mass index $>30 \text{ kg/m}^2$ = obesity
	Lipid abnormalities	Cutaneous xanthomas Xanthelasma on the eyelids Corneal arcus
	Diabetes mellitus	Acanthosis nigricans Skin tags
	Tobacco abuse and chronic obstructive pulmonary disease (COPD)	Odor of tobacco Staining of teeth, fingers, or nails Premature skin wrinkling Prolonged expiration, wheezing, distant breath sounds
	Miscellaneous	Ear lobe creases (Frank sign)
Assessing for complications	Congestive heart failure	Jugular venous distention, S ₃ , S ₄ , displaced point of maximal impulse, hepatomegaly, pulmonary/peripheral edema Ischemic mitral regurgitation Low-output cardiac failure
	Arrhythmias	Ectopy, atrial fibrillation
	PAD	Carotid bruits Peripheral pulses Peripheral skin discoloration or hair loss
Assessing for other causes of angina and dyspnea	Aortic stenosis	Late peaking systolic murmur Pulsus parvus et tardus
	Hypertrophic cardiomyopathy	Harsh crescendo-decrescendo systolic murmur dynamic with provocation
	Pulmonary hypertension	Loud P2, right S4, TR murmur, RV lift/heave

Physical Examination:



Xanthelasma on bilateral upper and lower eyelids, and Corneal arcus indicative of underlying lipid abnormalities.



Ear lobe crease, also called *Frank sign*, a potential marker of increased risk for coronary artery disease



Acanthosis is associated with insulin resistance and diabetes mellitus

Non-Invasive Investigations

LIPID LEVELS

Low-density lipoprotein (LDL) and high-density lipoprotein (HDL) cholesterol

Triglyceride level, Lipoprotein (a)

METABOLIC EVALUATION

Fasting plasma glucose concentration

Serum creatinine level

Thyroxine level

*Hemoglobin A_{1c} in patients with known or suspected diabetes

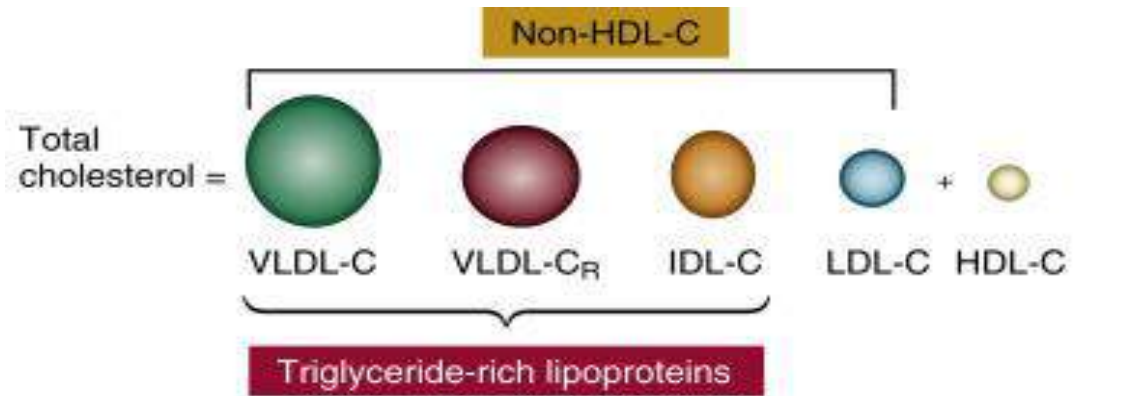
MARKERS OF CARDIAC FUNCTION

*High-sensitivity C-reactive protein

*Brain natriuretic peptide

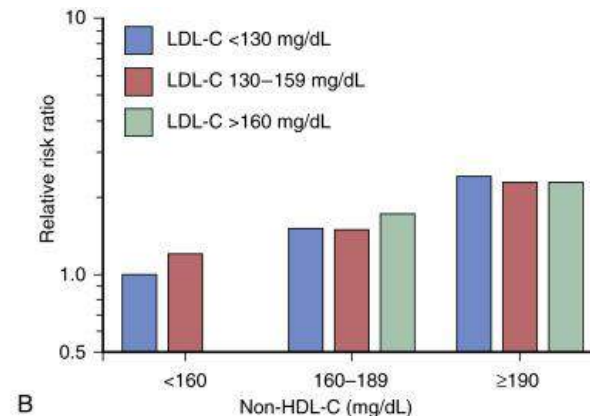
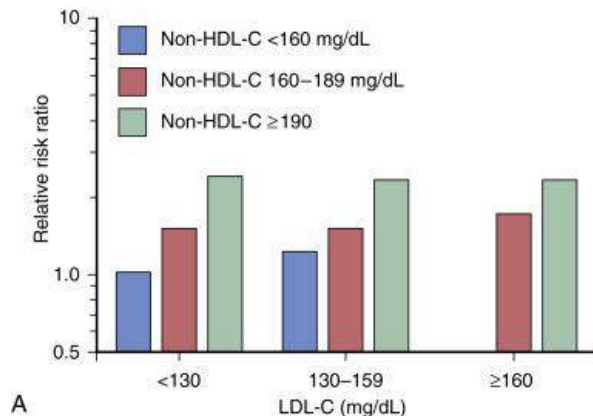
Non-Invasive Investigations

Blood Testing: Lipid profile



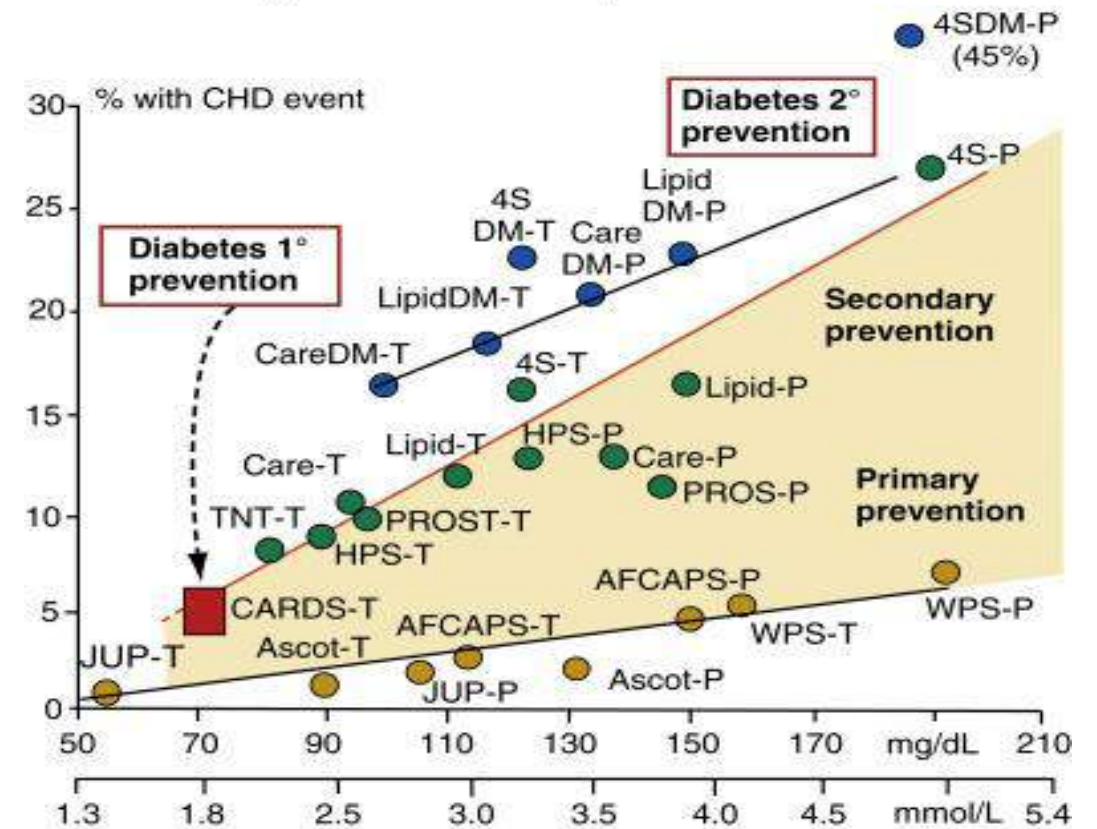
LDL-C = Total cholesterol – HDL-C – Triglyceride/5 (Friedewald)

Non-HDL-C = Total cholesterol – HDL-C



EFFECTS OF LDL-LOWERING

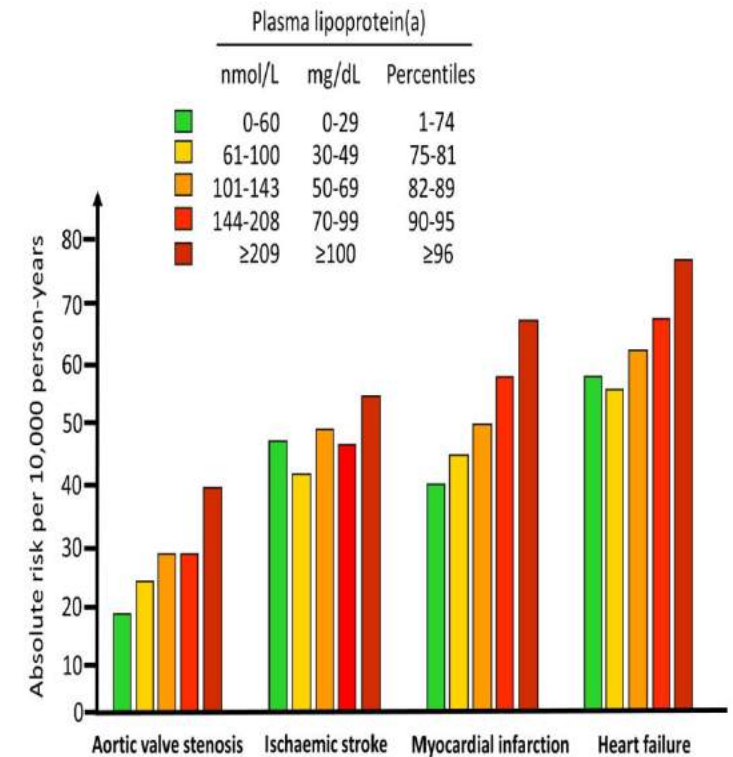
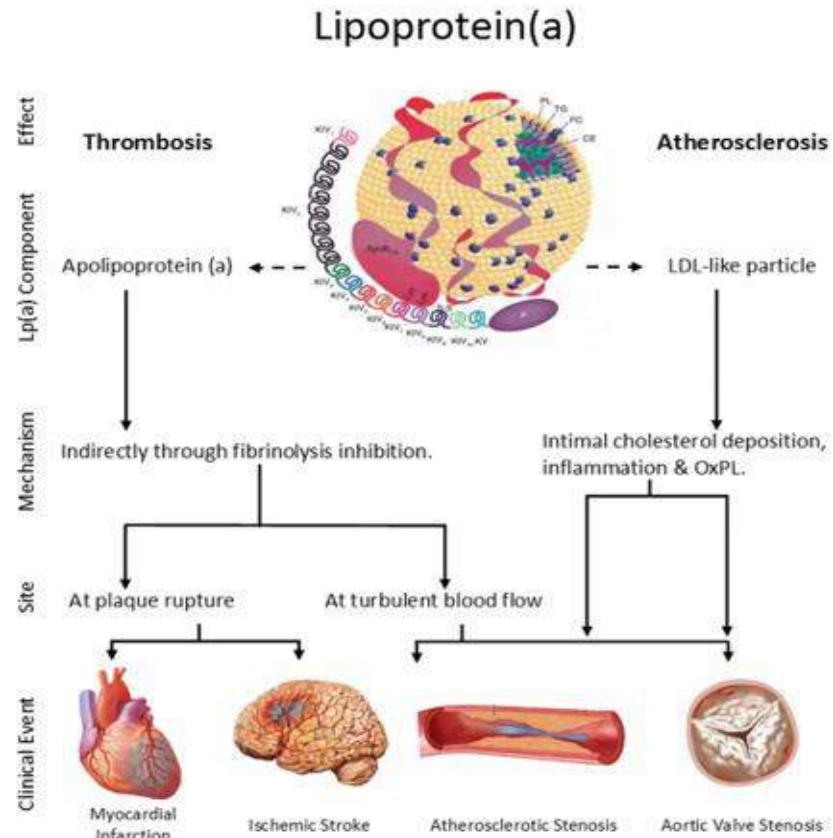
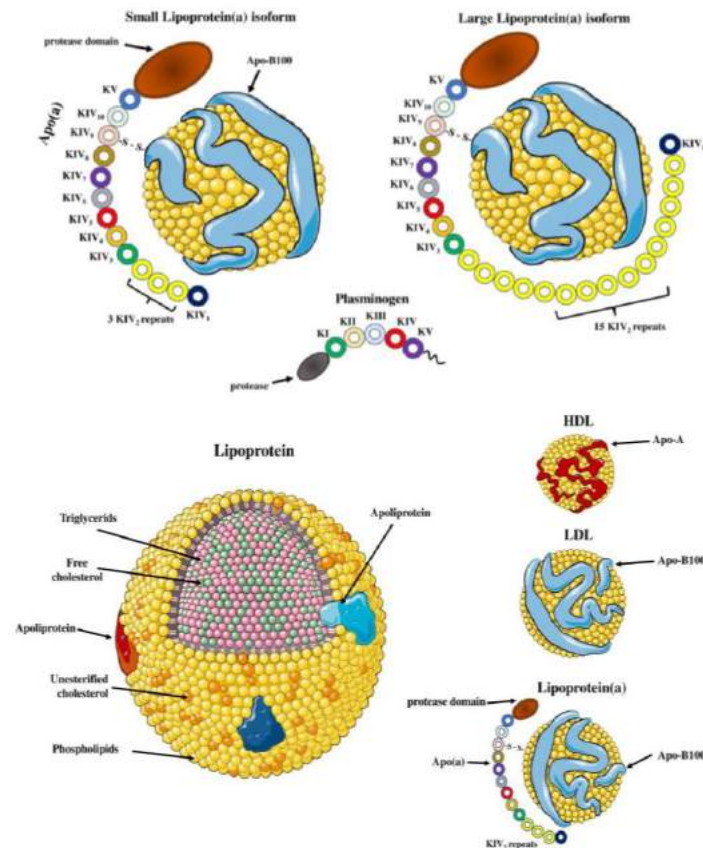
The LDL - hypothesis: the lower, the better



Chronic Coronary Artery Disease: A Companion to Braunwald's Heart Disease

Non-Invasive Investigations

Blood Testing: Lipid profile



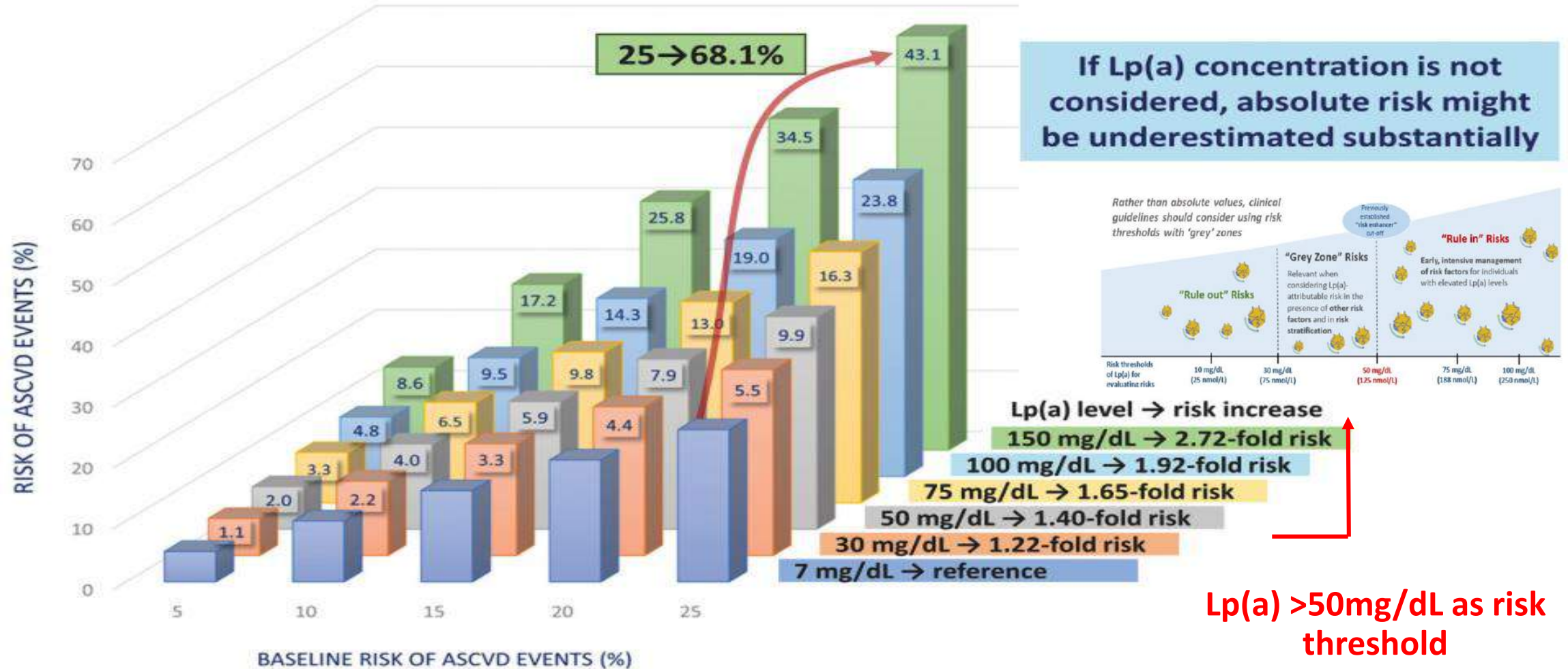
Lp(a) is a subtype of LDL-C with apo(a); and major determinant of level is genetic variation

Catherine J, et al. Endotext 2023

Lampsas S, et al. Molecules 2023

Non-Invasive Investigations

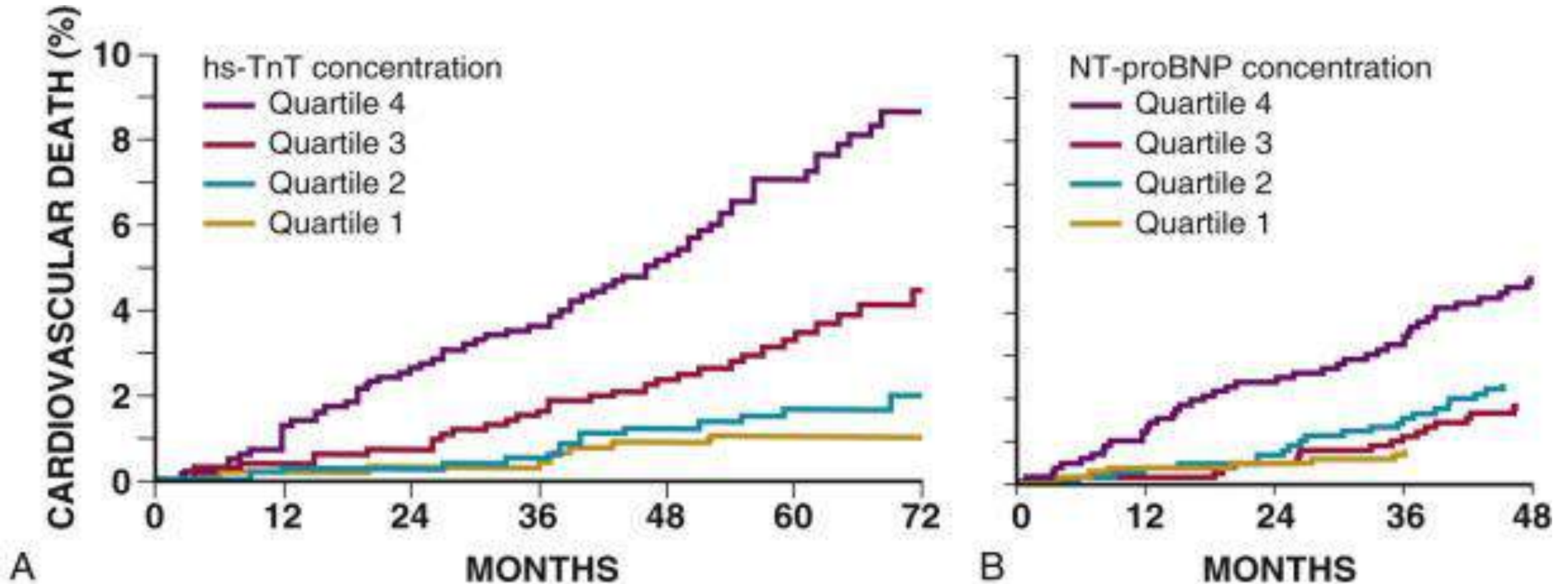
Blood Testing: Lipid profile



Non-Invasive Investigations

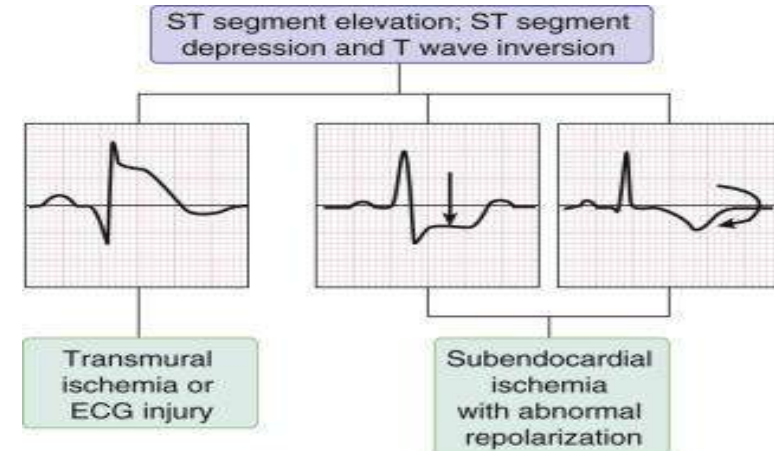
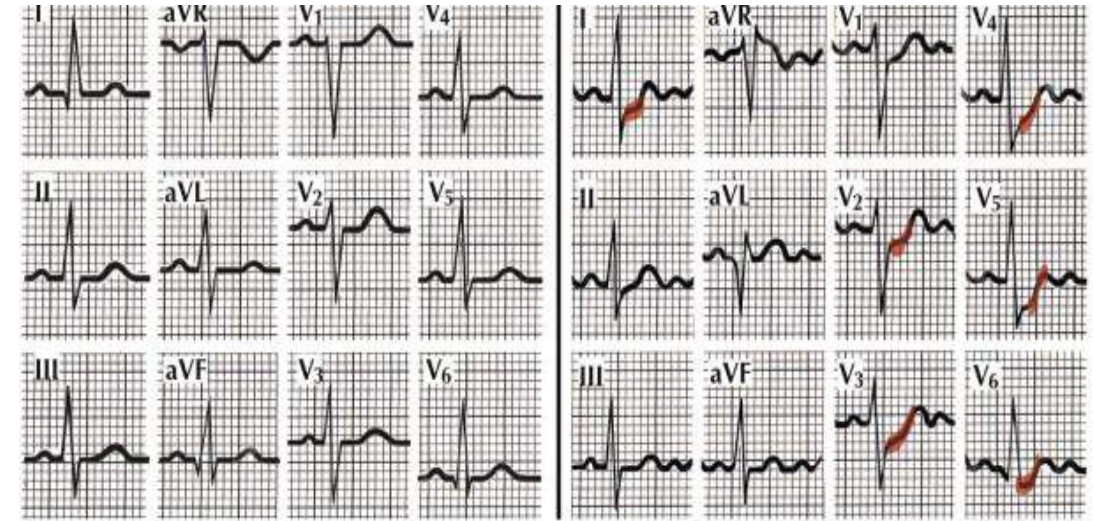
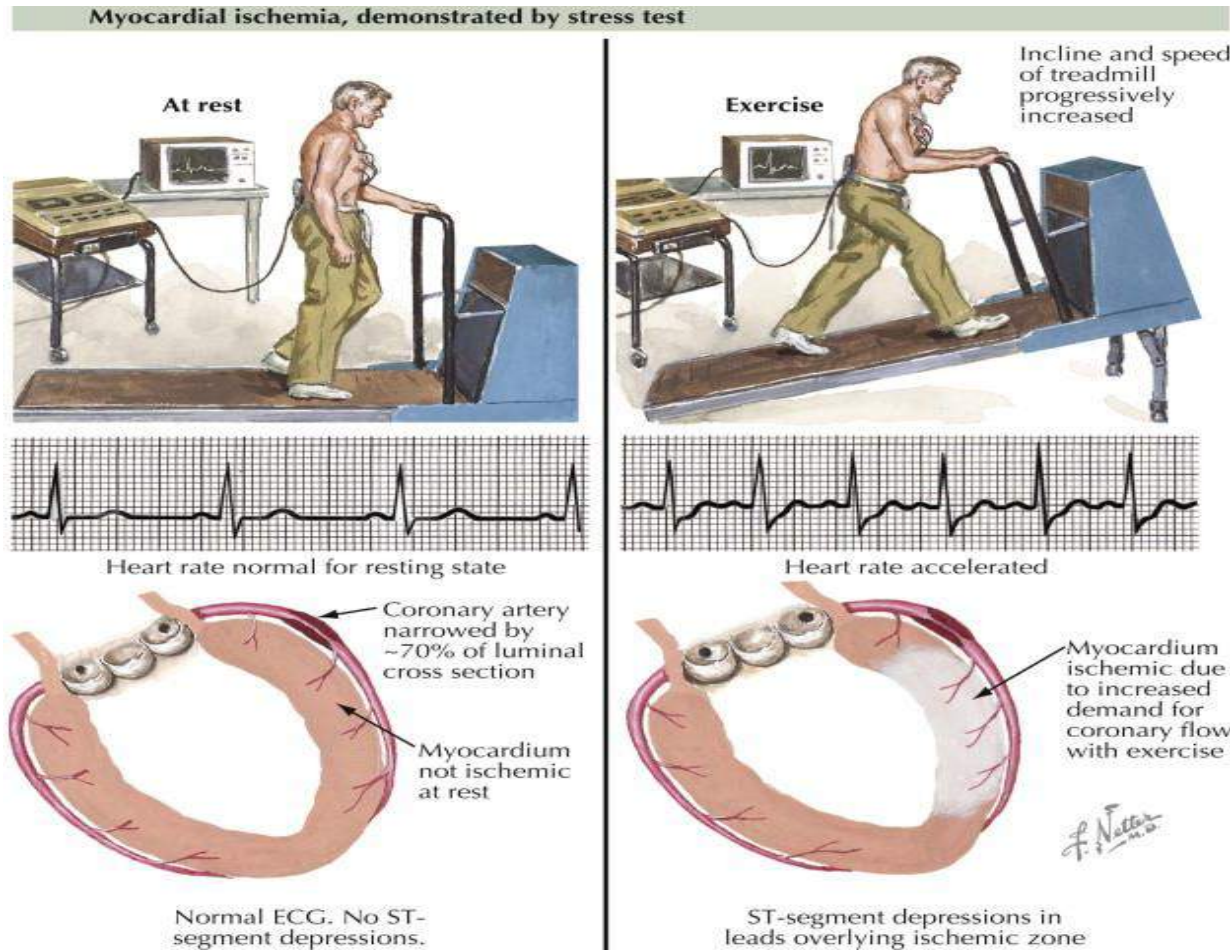
Blood Testing

Biomarkers of Myocyte Injury, Ischemia, and Hemodynamic Stress in IHD



Non-Invasive Investigations

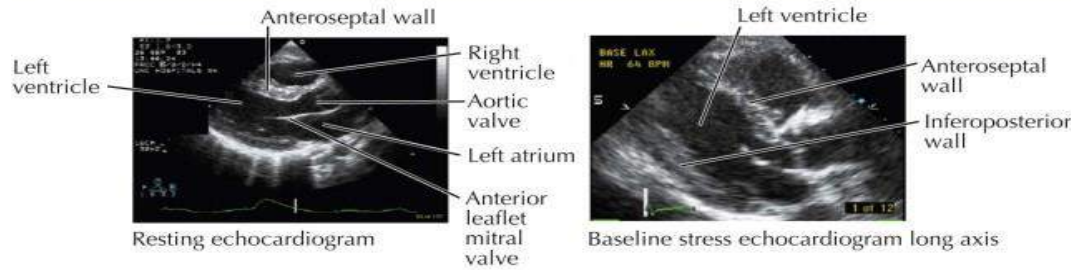
Exercise Stress Testing



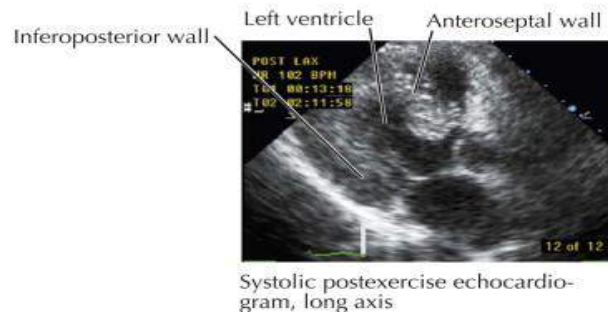
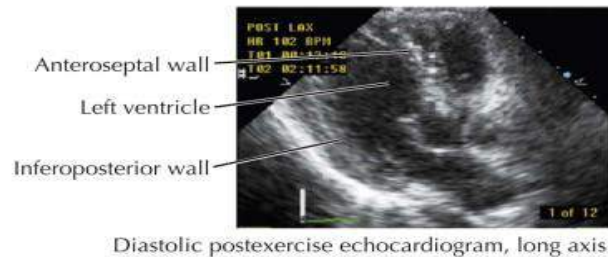
Non-Invasive Investigations

Exercise Stress Echo

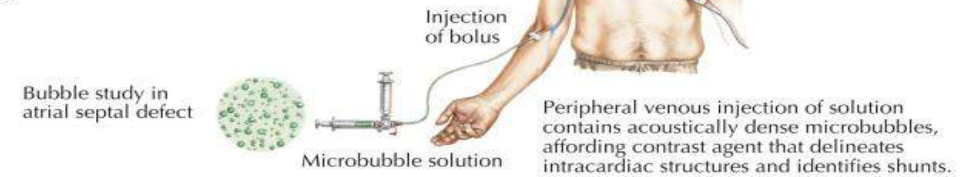
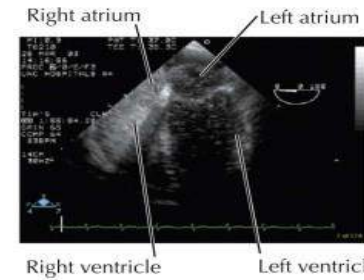
Exercise echocardiography



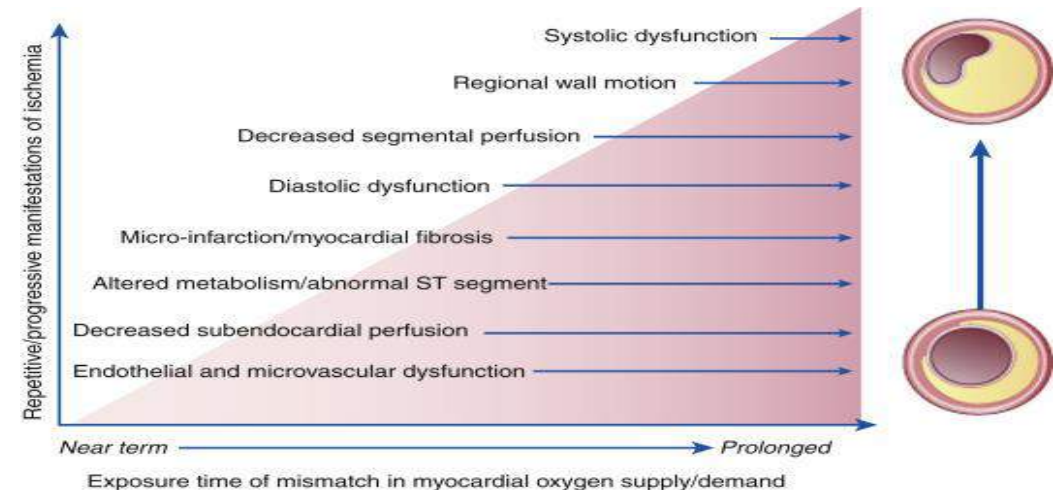
Exercise performed to elicit ischemic signs and postexercise echocardiogram used to evaluate ventricular function, wall motion, and thickness. Often correlated with stress echocardiography



Contrast echocardiography



Peripheral venous injection of solution contains acoustically dense microbubbles, affording contrast agent that delineates intracardiac structures and identifies shunts.

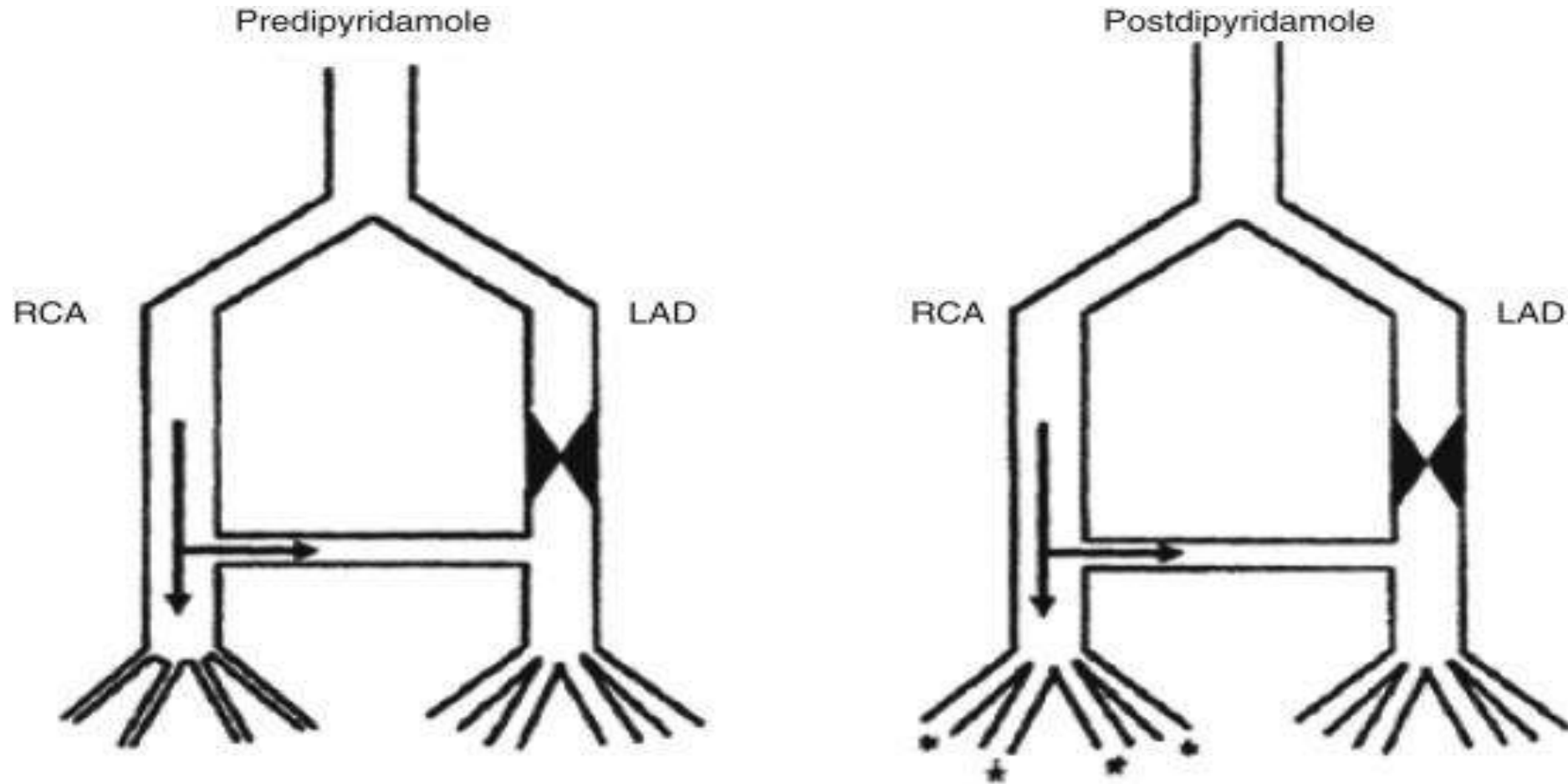


Chronic Coronary Artery Disease: A Companion to Braunwald's Heart Disease

Netter Collection of Medical Illustration: CVS 2014

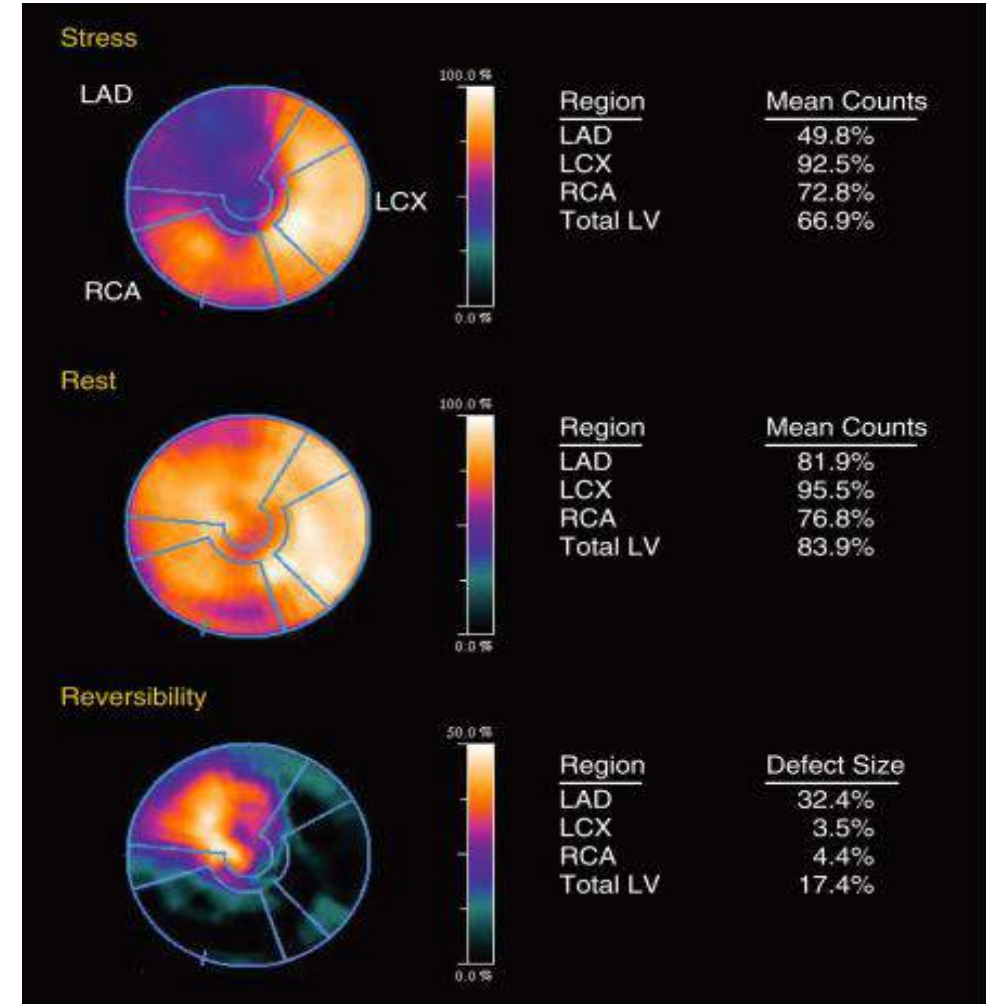
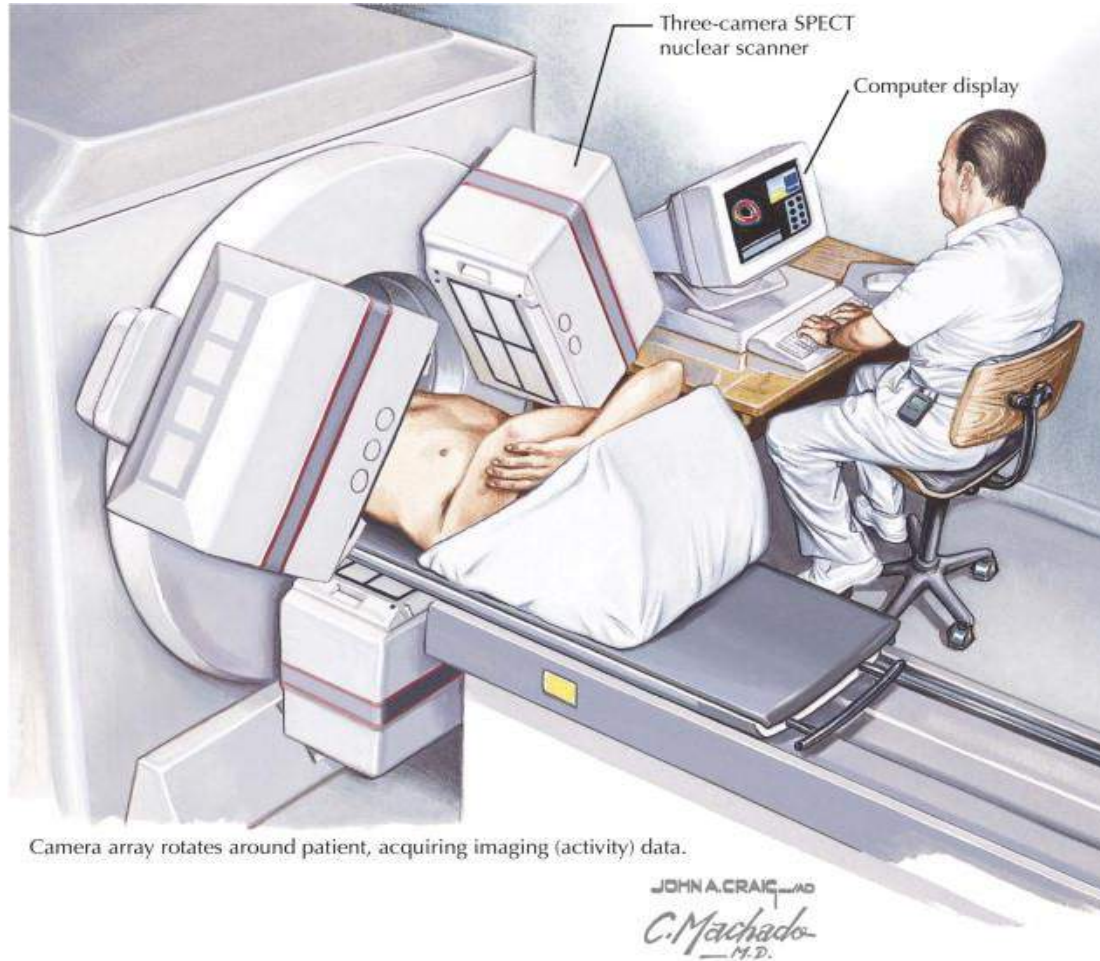
Non-Invasive Investigations

Principle of Pharmacological Stress



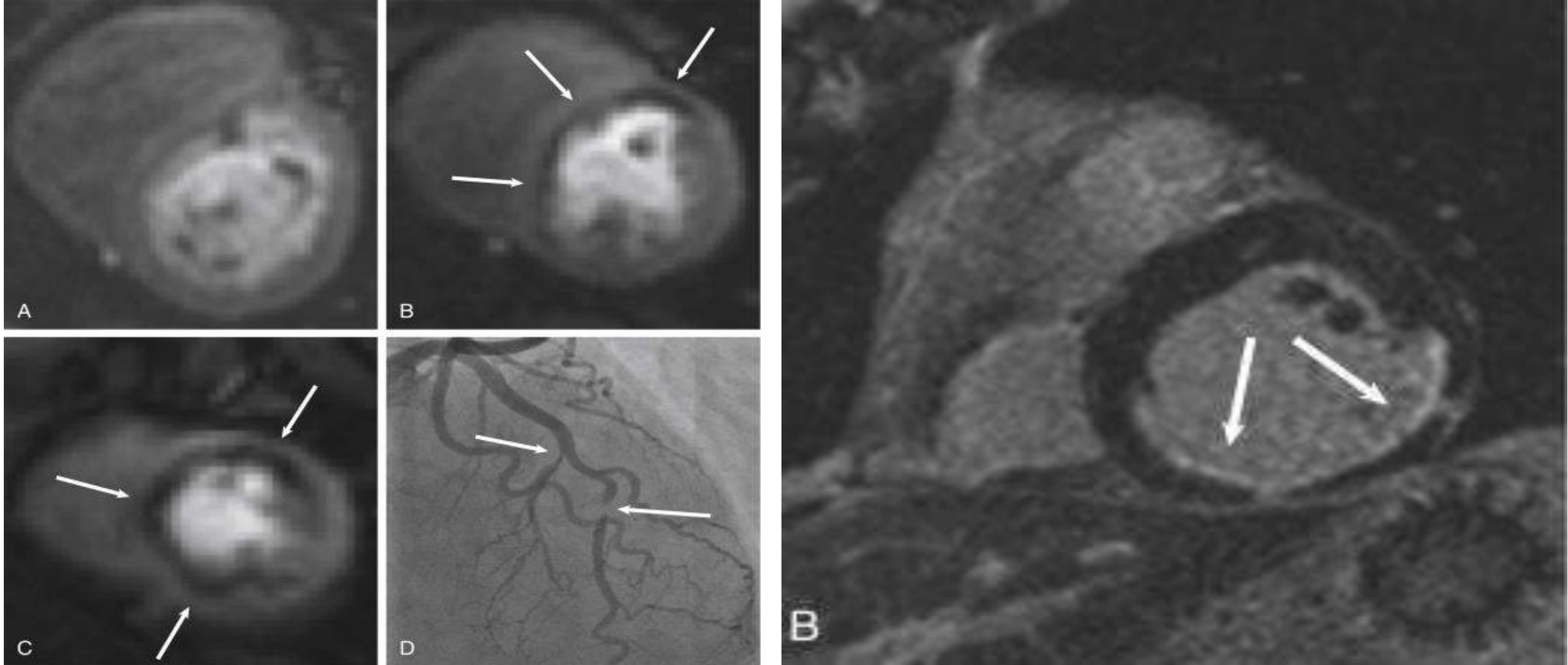
Non-Invasive Investigations

SPECT



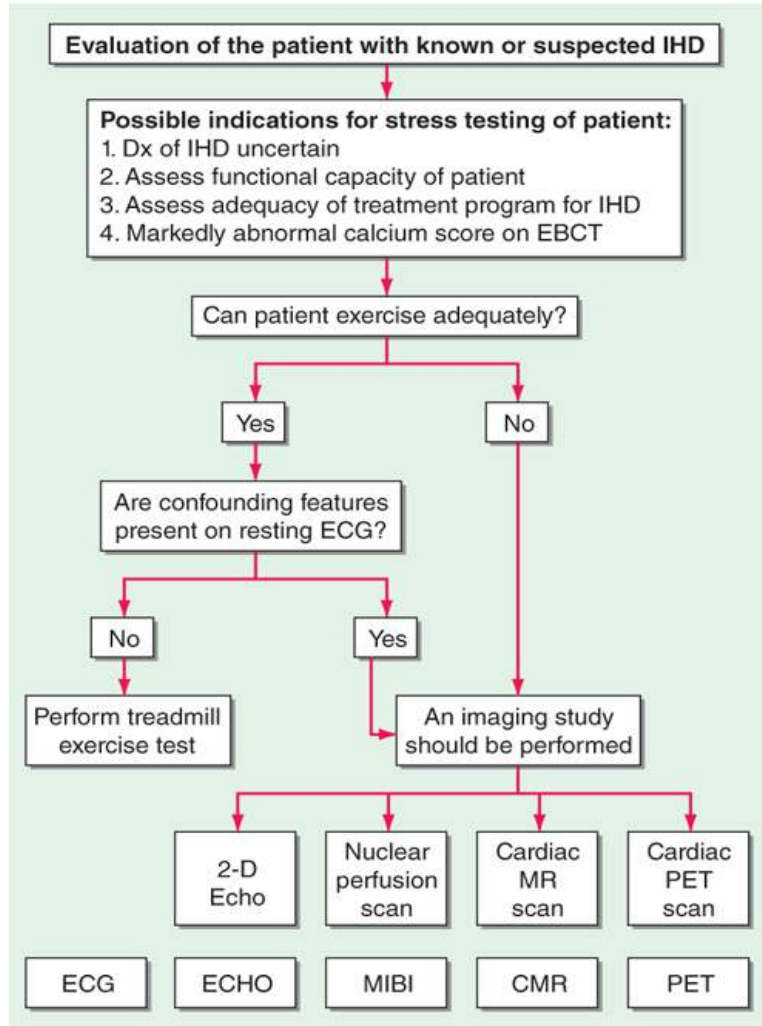
Non-Invasive Investigations

Cardiac MRI Perfusion Scan



Non-Invasive Investigations

Choices of Stress Testing



	STRESS ECG	MYOCARDIAL PERFUSION SCINTIGRAPHY	STRESS ECHO	CARDIAC MRI PERFUSION
Technical difficulty	+	++	+++	+++
Ease of interpretation	+++	++	+++	++
Diagnostic sensitivity (%)	50-80	65-90	65-90	70-90
Diagnostic specificity (%)	80-95	90-95	90-95	90-95
Risk stratification	++	++	++	+
Identification of hibernating myocardium	-	++	+++	+++
Identification of ischemic territory	+	++	++	+++
Limitations	Conduction or repolarization anomalies	Radiation exposure	Diagnostic images not possible in all patients	Claustrophobia and incompatibility with devices
Cost	+	+++	++	+++

Harrison's Principles of Internal Medicine



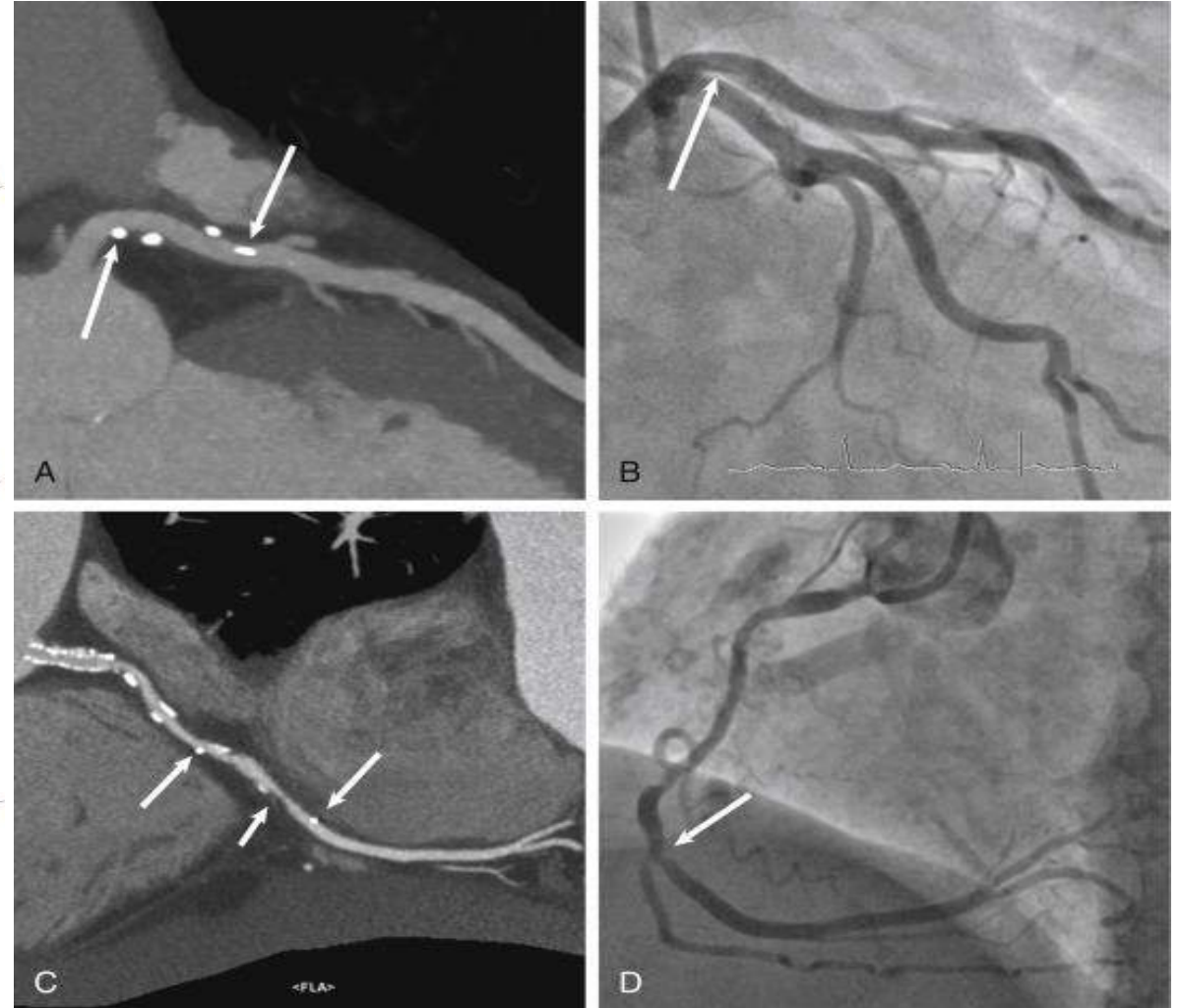
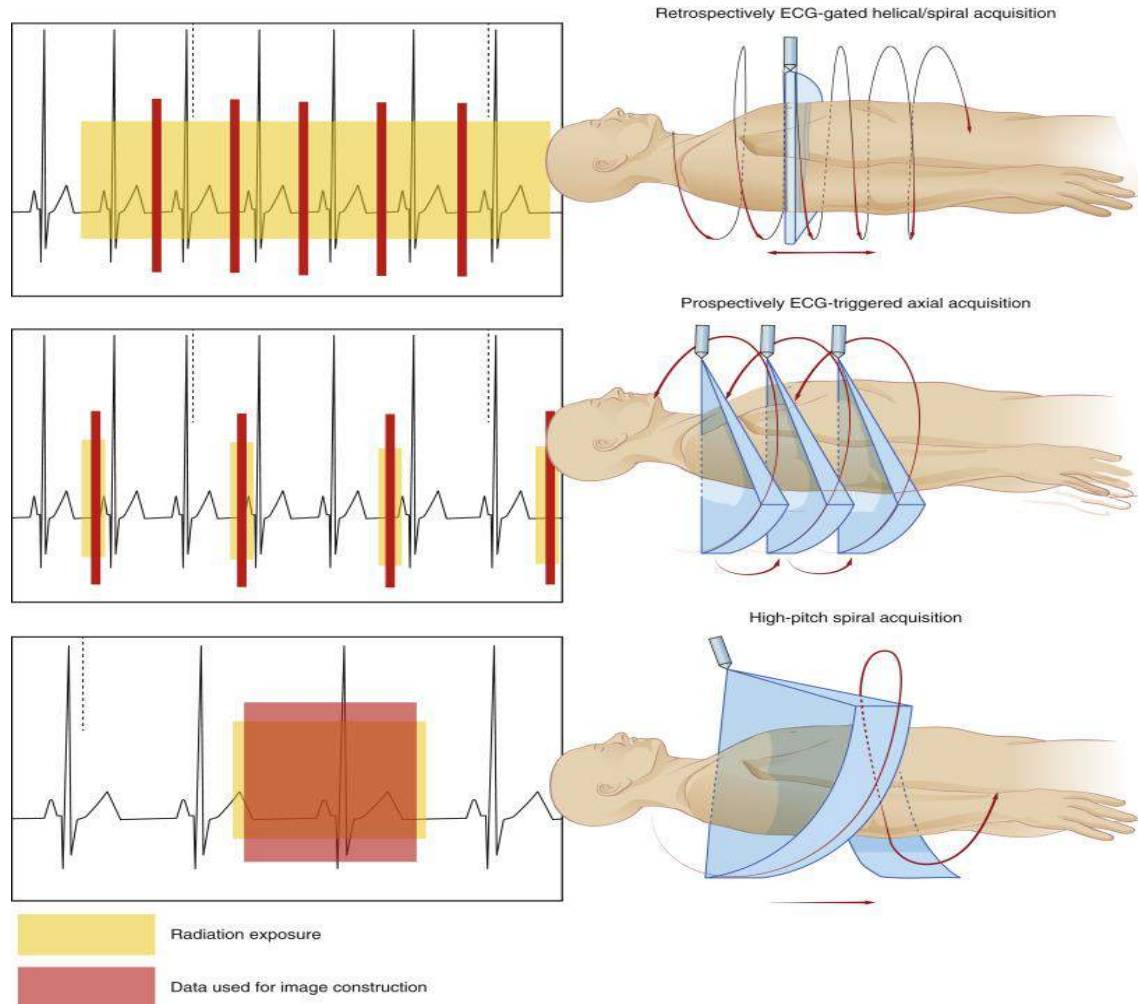
HKU Med

School of Clinical Medicine
Department of Medicine
香港大學內科學系

Cardiovascular Therapeutics: A Companion to Braunwald's Heart Disease

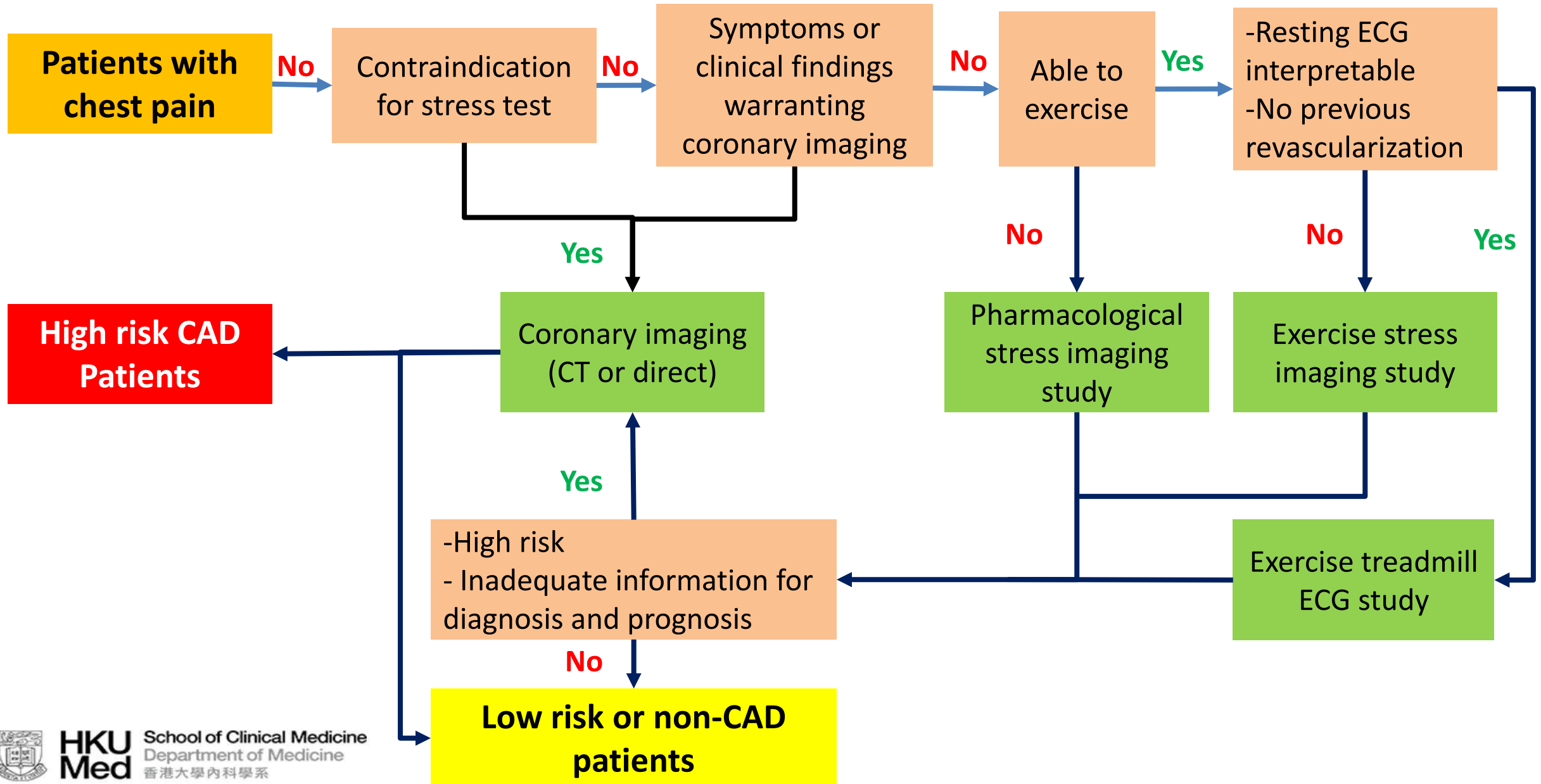
Non-Invasive Investigations

CT Coronary Angiogram



Chronic Coronary Artery Disease: A Companion to Braunwald's Heart Disease

Non-Invasive Investigations

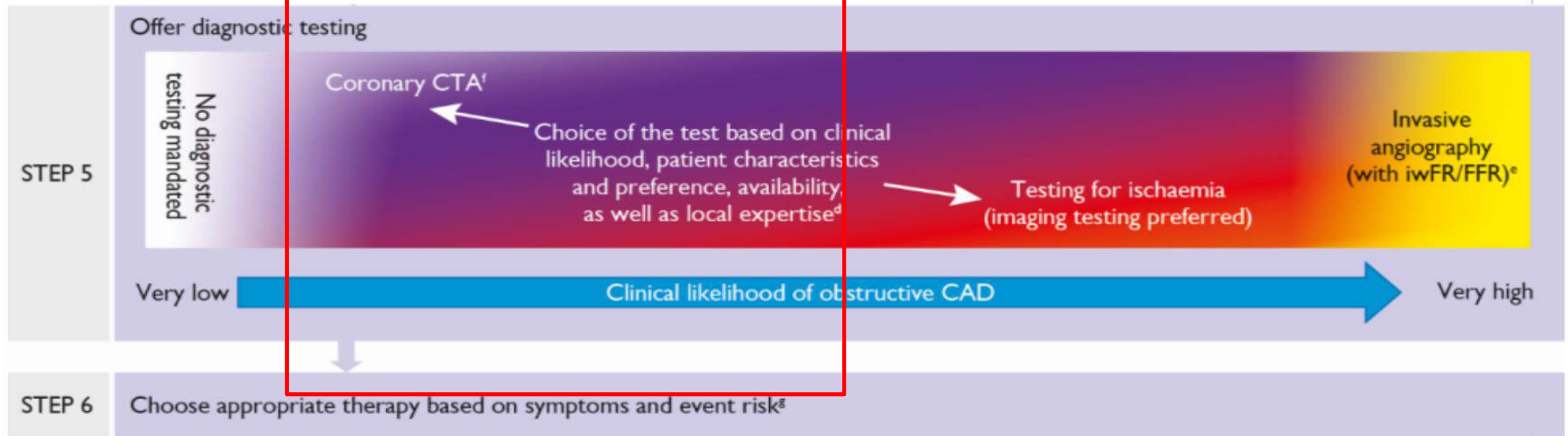


Clinical Likelihood of CAD

Patients with angina and/or dyspnoea and suspected coronary artery disease



Diagnostic approach (2)



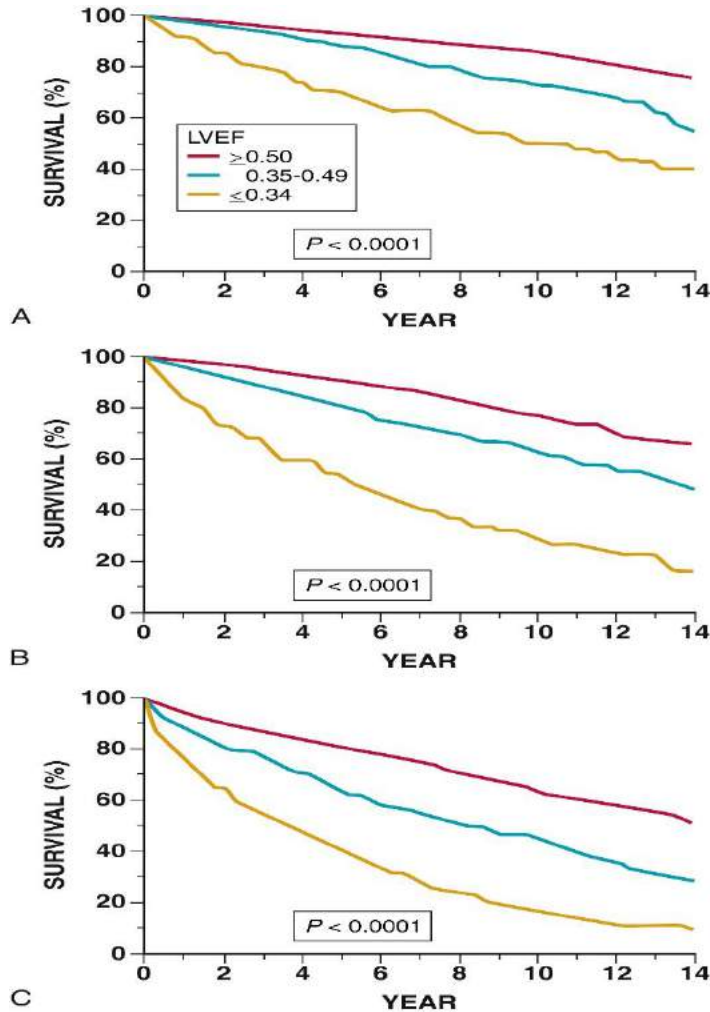
^d Ability to exercise, individual test-related risks, and likelihood of obtaining diagnostic test result. ^e High clinical likelihood and symptoms inadequately responding to medical treatment, high event risk based on clinical evaluation (such as ST-segment depression, combined with symptoms at a low workload or systolic dysfunction indicating CAD), or uncertain diagnosis on non-invasive testing. ^f Functional imaging for myocardial ischaemia if coronary CTA has shown CAD of uncertain grade or is non-diagnostic. ^g Consider also angina without obstructive disease in the epicardial coronary arteries (see section 6 of full text).

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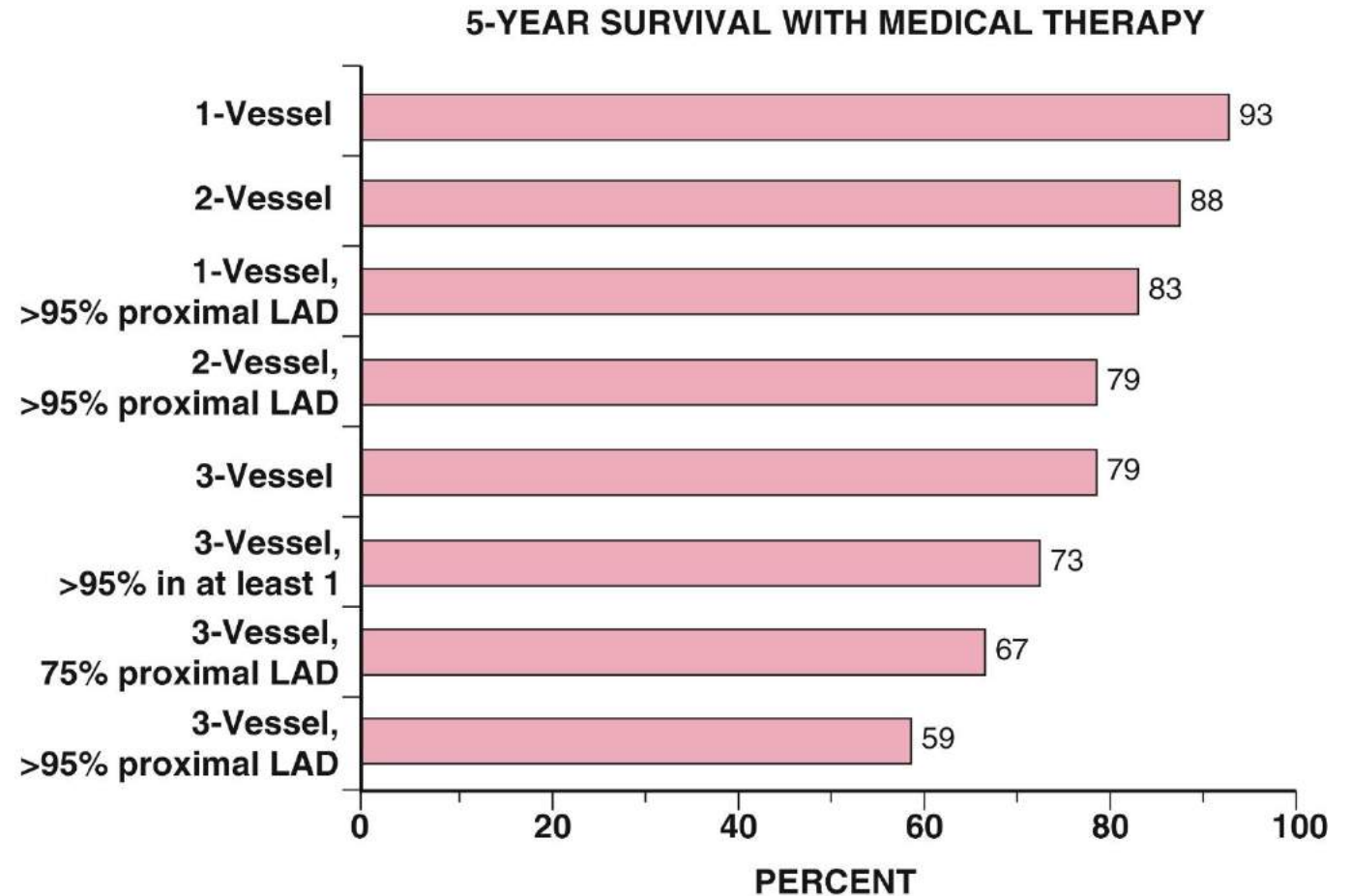
www.escardio.org/guidelines

ESC Guidelines on the diagnosis and management of chronic coronary syndromes
(European Heart Journal 2019; 10.1093/eurheartj/ehz425)

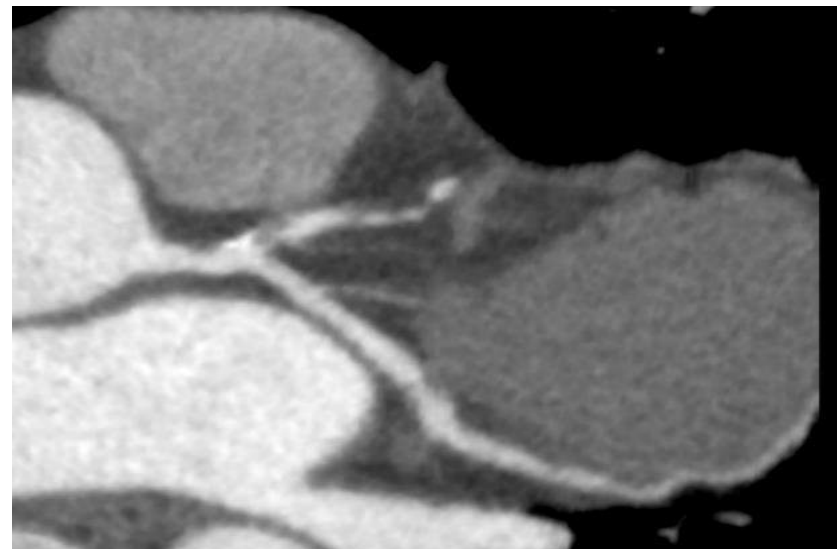
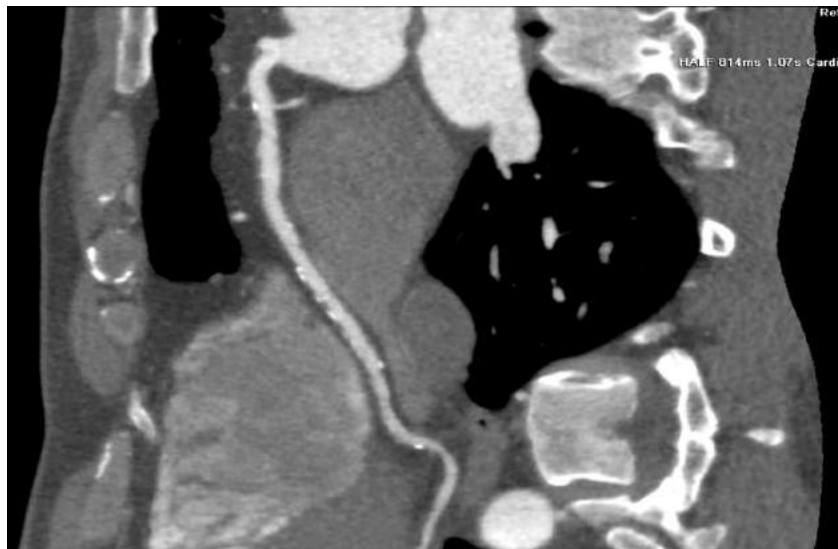
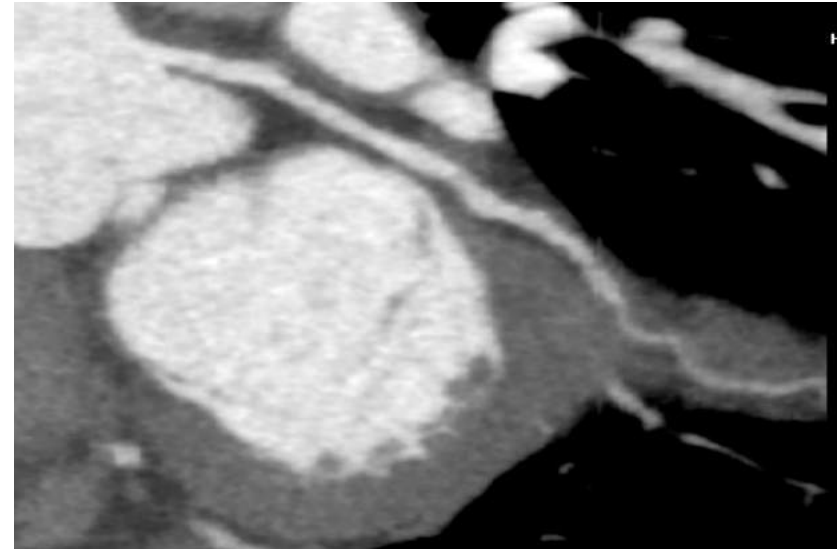
High Risk CAD



EXTENT OF CAD

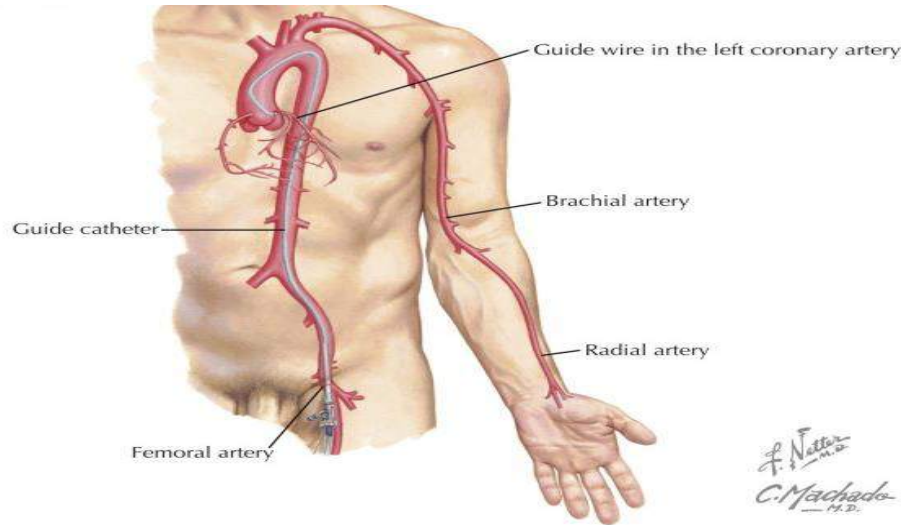


CTM, M/49 (3)



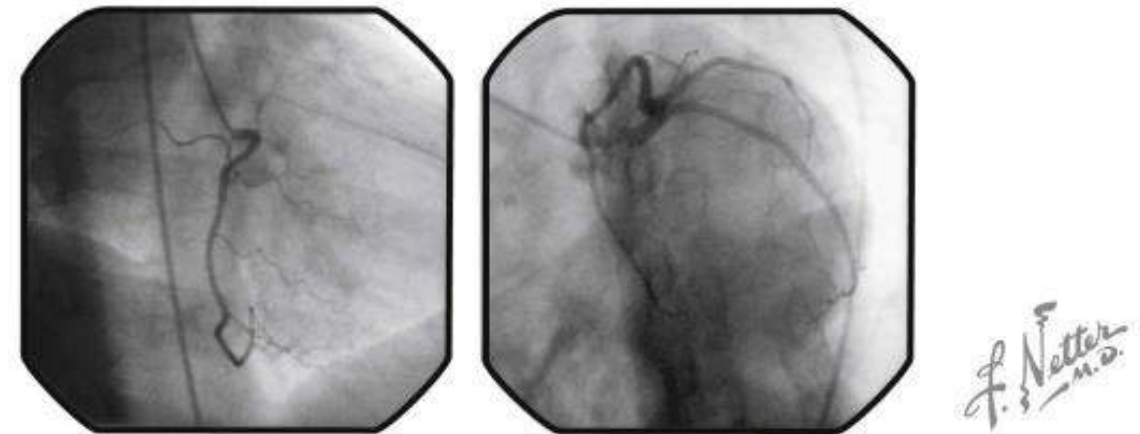
Invasive Investigations

Coronary Angiogram



- Unacceptable angina despite medical therapy (for consideration of revascularization)
- Non-invasive testing results with high-risk features
- Angina or risk factors for CAD in the setting of depressed LVEF

For diagnostic purposes, in the individual in whom the results of non-invasive testing are unclear



CTM, M/49 (4)



Ischaemic Heart Disease (IHD)

- ◆ Pathophysiology and Etiologies
- ◆ Diagnosis and Assessment
- ◆ **Management**

Management:

Principles of Management

Aims of Pharmacological Management

- Reduce angina symptoms and exercise-induced ischaemia.
- Prevent cardiovascular events.

Optimal Treatment

- Treatment that satisfactorily controls symptoms and prevents cardiac events.
- With maximal patient adherence and minimal adverse events.

Management:

Principles of Management

“Optimal” Anti-Ischaemic Treatment in CCS Patients

- No universal definition of which drug/combination is “optimal” .
- Anti-ischaemic drug therapies must be adapted to each patient’s characteristics and preferences.
- Initial drug therapy usually consists of one or two antianginal drugs, as necessary, plus drugs for secondary prevention of CVD.
 - depends on expected tolerance related to the individual patient’s profile and comorbidities,
 - potential drug interactions with co-administered therapies,
 - patient’s preferences after being informed of potential adverse effects,
 - drug availability.

Management:

General Measures

1. Rule out and control aggravating conditions

- Associated non-cardiac diseases
- Associated cardiac disease
- Use of drugs aggravating angina

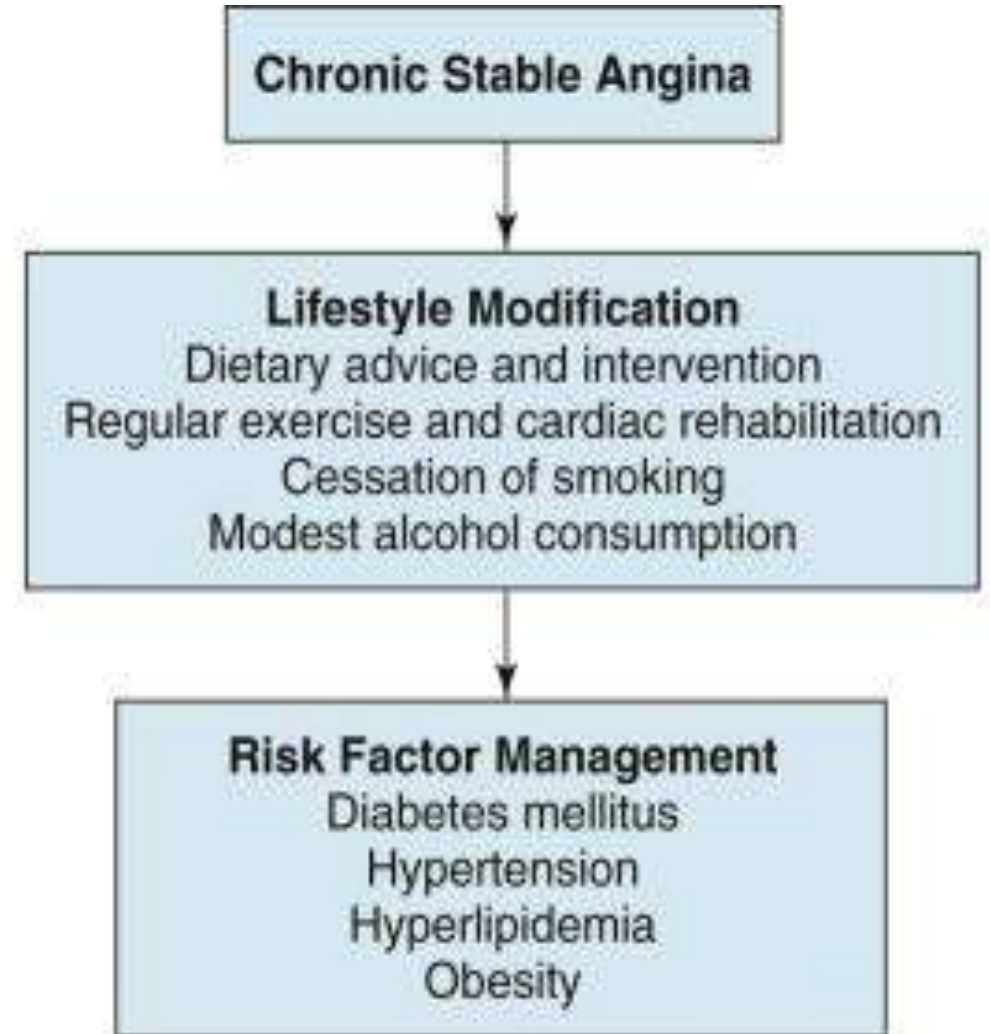
2. Smoking cessation

3. Dietary counseling for body weight and lipid control

4. Exercise prescription

5. Treat to targets

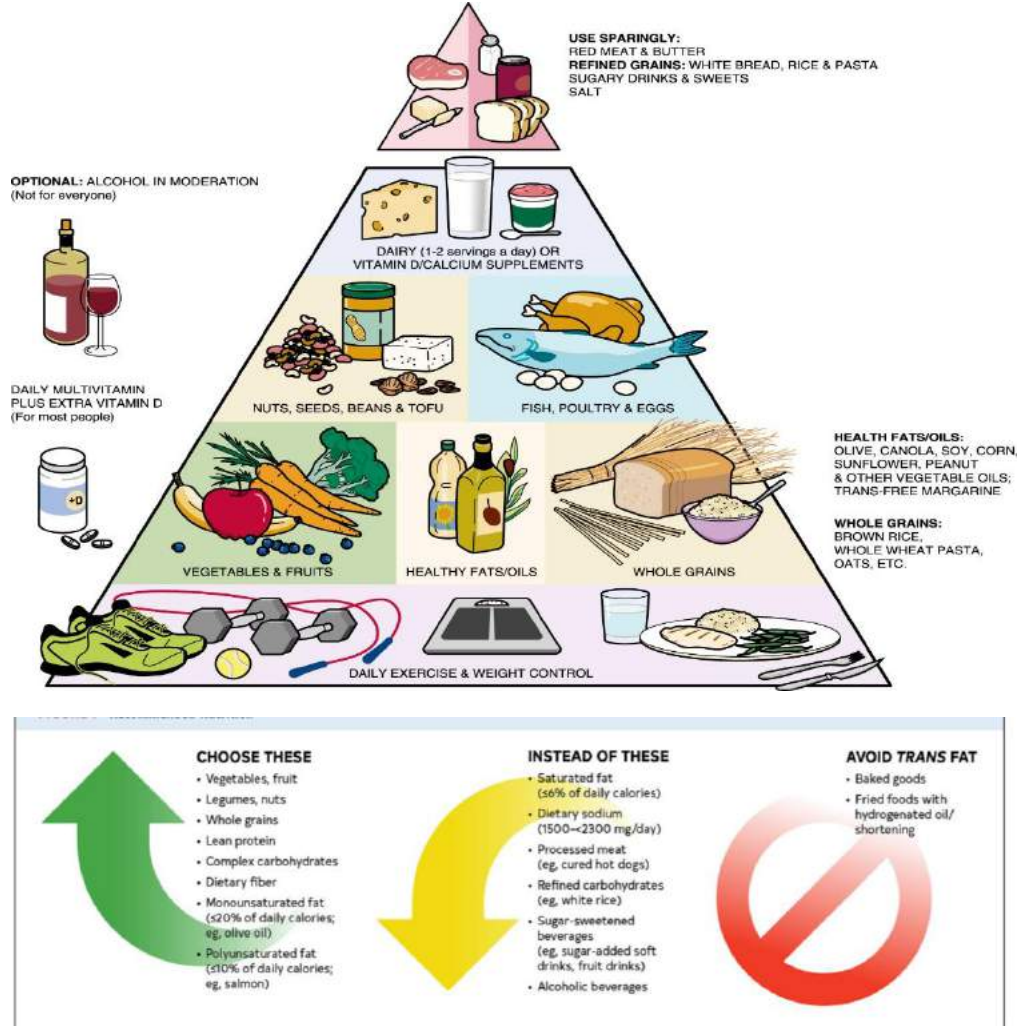
- Hypertension
- Blood lipids
- Diabetes



Cardiovascular Therapeutics: A
Companion to Braunwald's Heart Disease

Management:

General Measures: Lifestyle Modifications

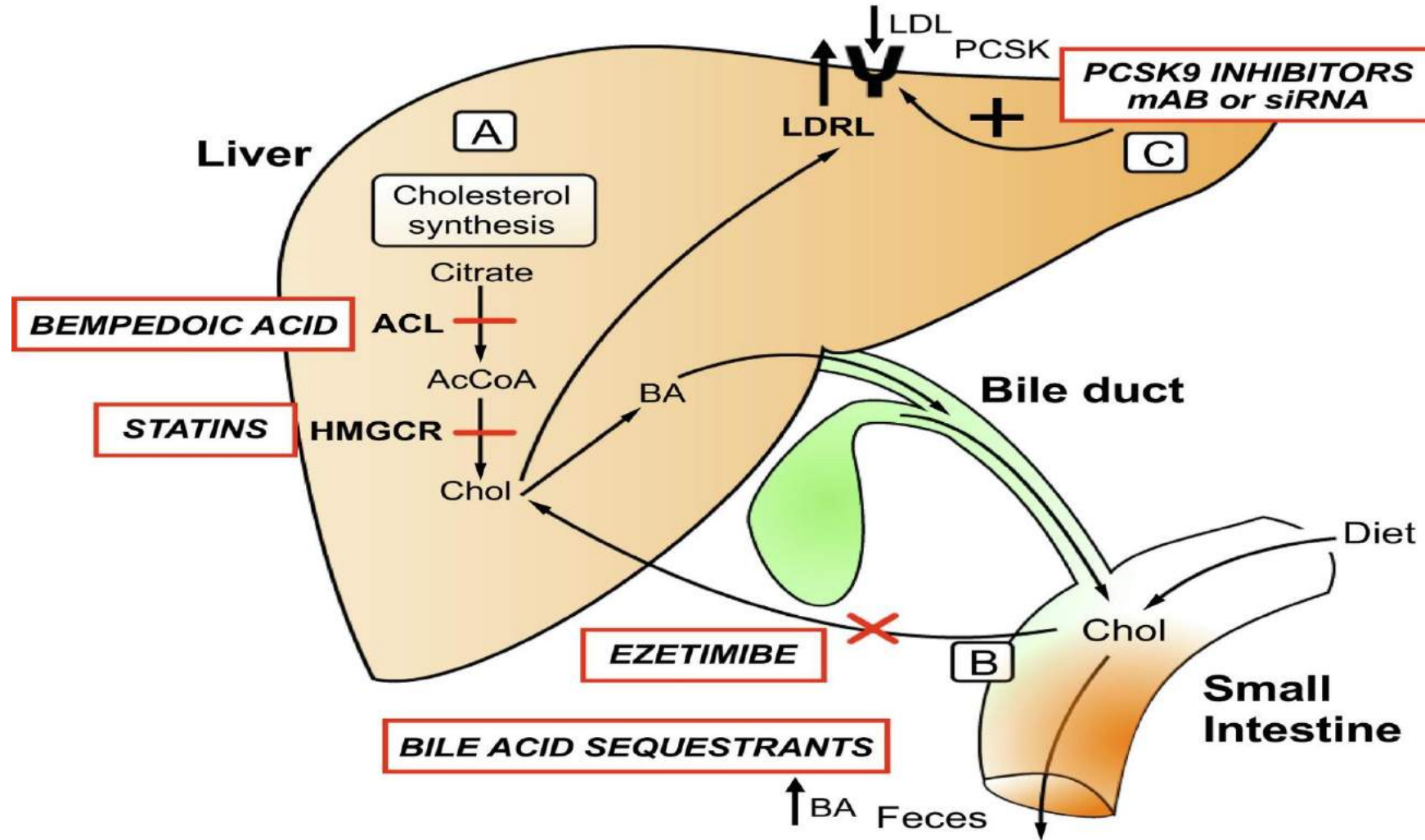


- Diet in upper 40% of population**
 - Good fat: Low saturated and trans fats, high polyunsaturated fat, high fish oil
 - Good carbohydrates: Low glycemic load, high fiber (whole grains)
 - High folate (vegetables, fruit)
- Not currently smoking**
- Moderate alcoholic beverage drinking (1 drink every other day to daily)**
- Regular exercise (0.5 h daily; e.g., 2 mph walking)**
- BMI <25 kg/m² (optimal, <21 kg/m²)**

JACC 2023

Management:

Lipid Lowering Therapies



Management:

Lipid Lowering Therapies

Lipid-Modifying Therapies	LDL-C	NON-HDL-C	HDL-C	Triglycerides	Lp(a)
Statins*	↓ 18–55%	↓ 15–51%	↑ 5–15%	↓ 7–30%	↔
Bile-acid sequestrants*	↓ 15–30%	↓ 4–16%	↑ 3–5%	↑ 0–10%	
Cholesterol absorption inhibitors*	↓ 13–20%	↓ 14–19%	↑ 3–5%	↓ 5–11%	
PCSK9 inhibitors*	↓ 61–62%	↓ 52%	↑ 5–7%	↓ 12–17%	↓ 25%
ApoB antisense	↓ 25–37%		↑ 2–15%	↓ 9–26%	↓ 21–33%
MTP inhibitor	↓ 44–50%	↓ 44–50%	↓ 12–↑1%	↓ 29–45%	↓ 15–19%
Nicotinic acid	↓ 5–25%	↓ 8–23%	↑ 15–35%	↓ 20–50%	↓ 20–40%
Fibric acids	↓ 5–↑20%	↓ 5–19%	↑ 10–20%	↓ 20–50%	
Long-chain omega-3 fatty acids*	↓6–↑25%	↓5–14%	↓ 5–↑7%	↓ 19–44%	

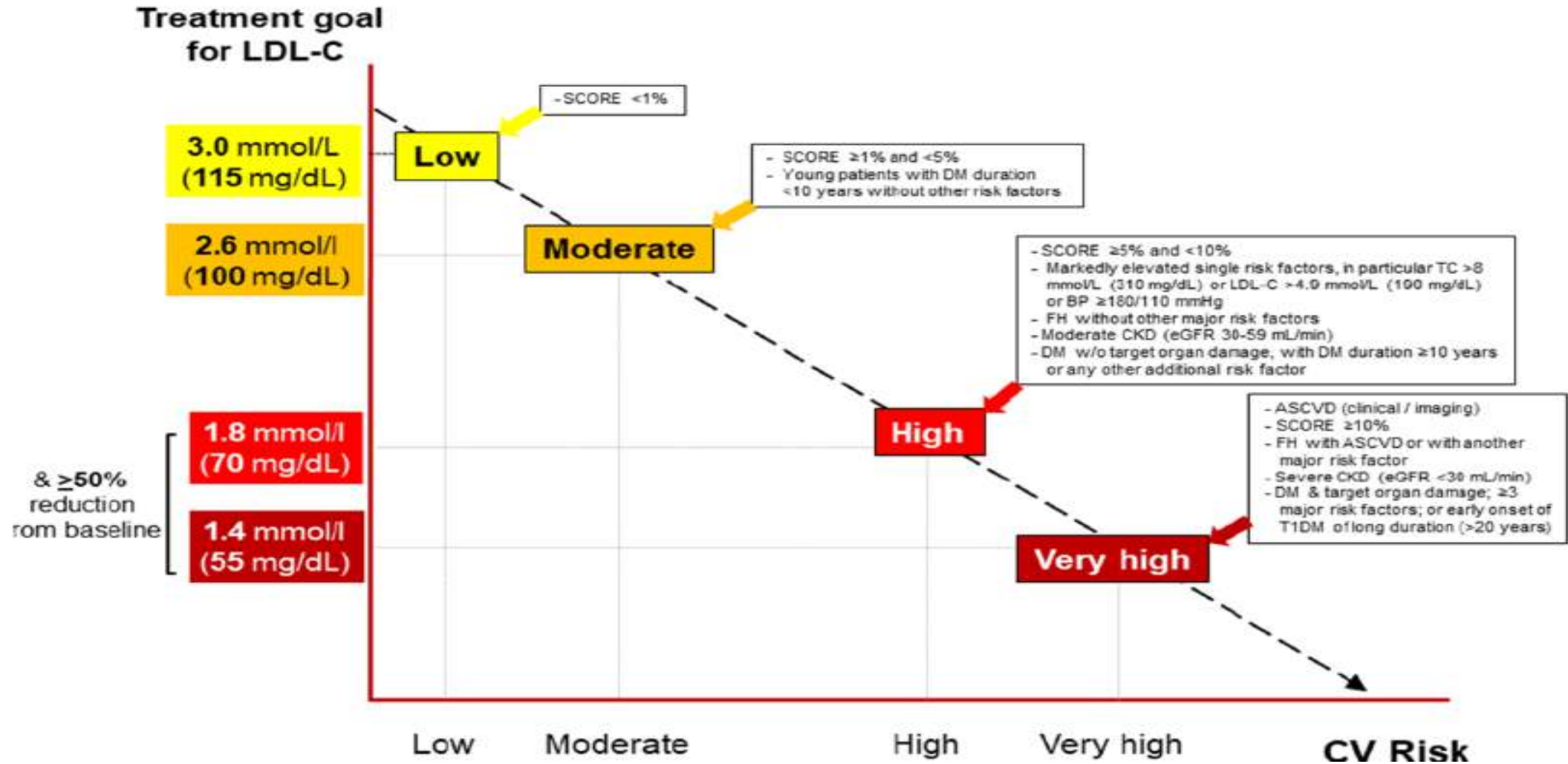
***Proven to reduce CVD events**



**HKU
Med**

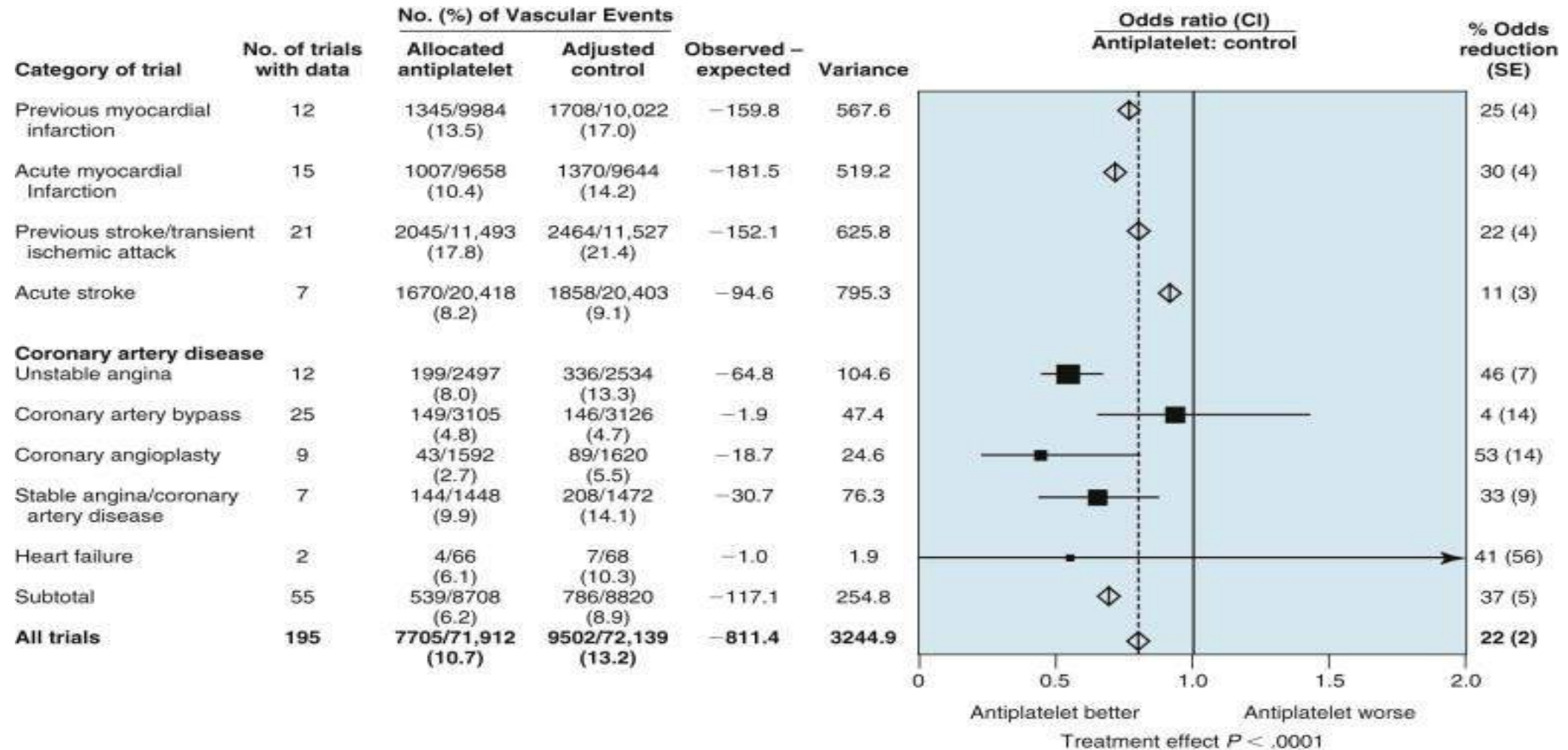
School of Clinical Medicine
Department of Medicine
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Management: Lipid Lowering Therapies



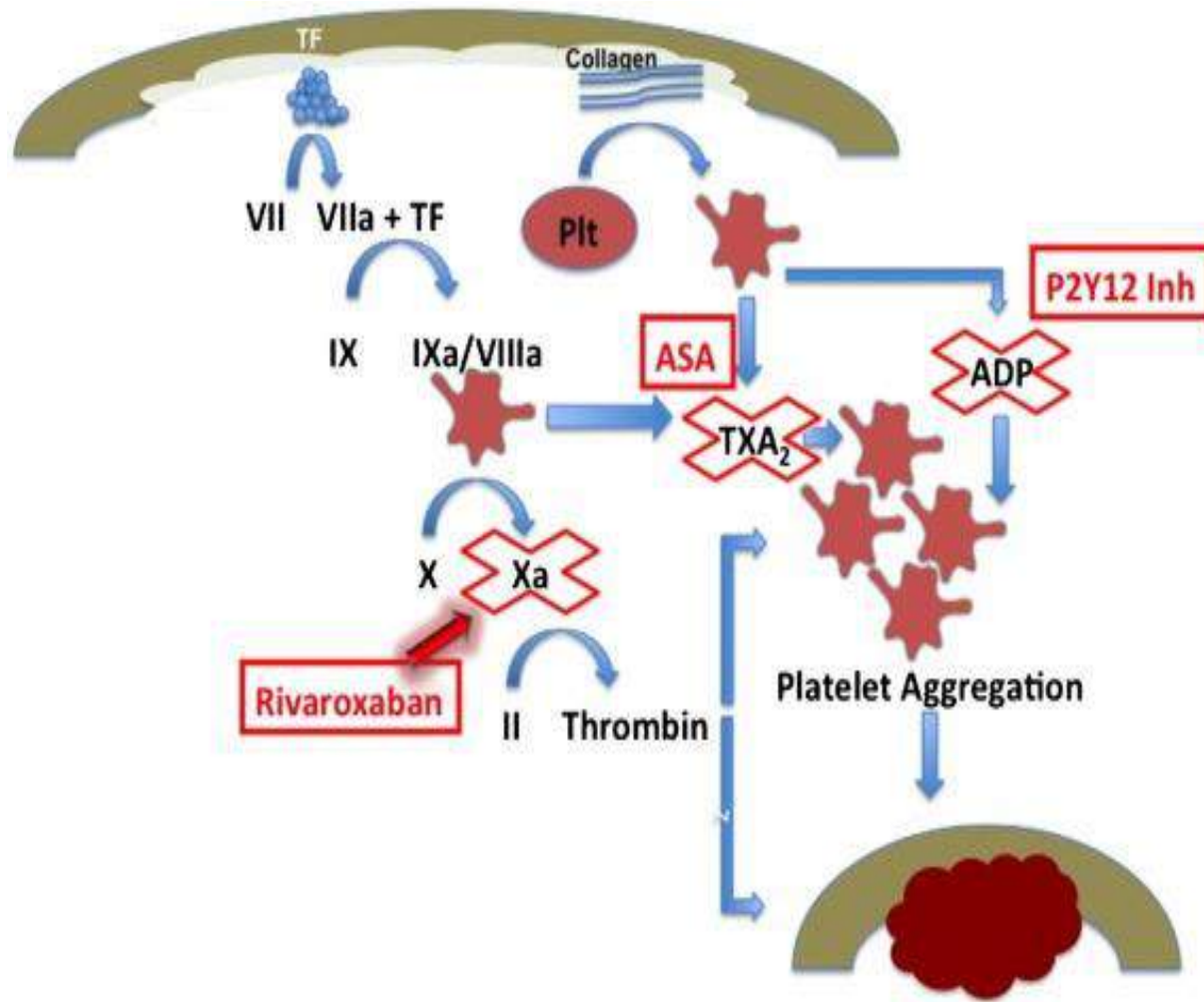
Management:

Anti-platelet Agents



Management:

Anti-platelet Agents/Anti-coagulant



Indications:

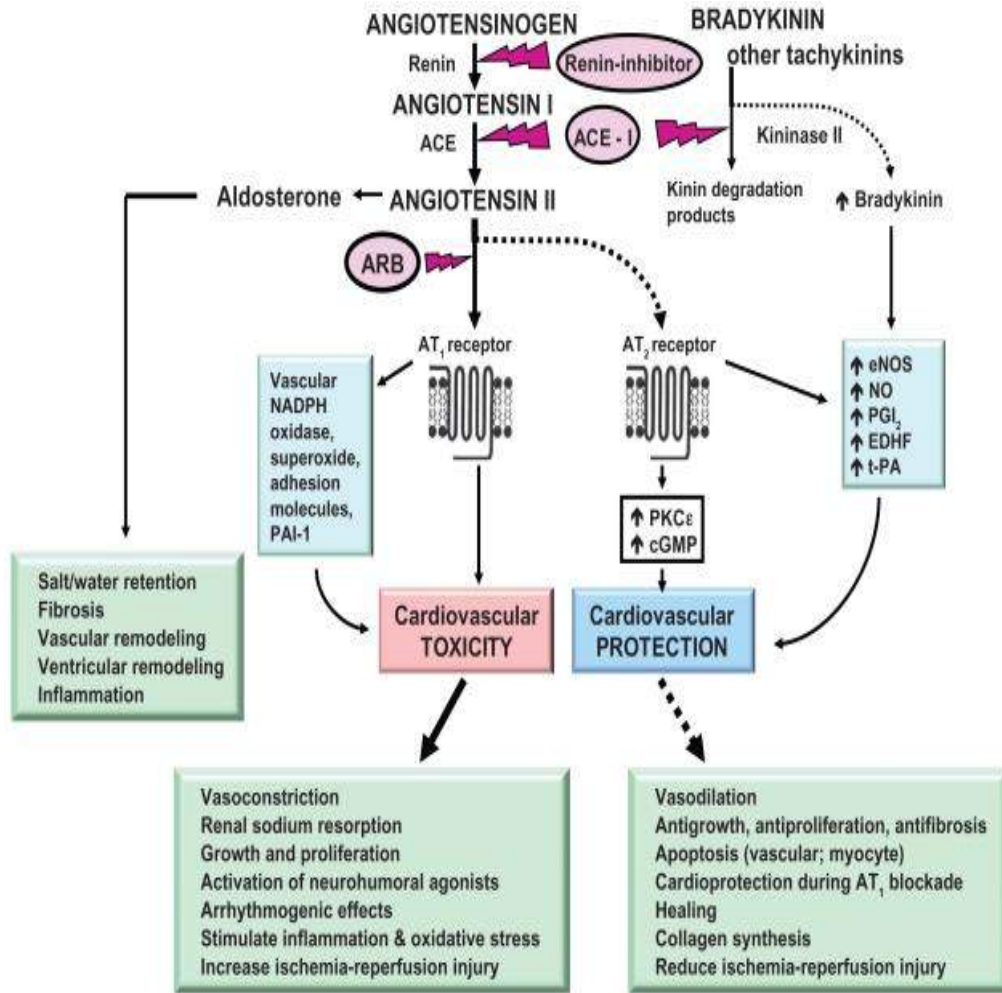
- **Aspirin** in the absence of contraindications
- **Clopidogrel** when aspirin is contraindicated
- **Aspirin + P2Y₁₂ blocker** after PCI
- **Rivaroxaban 2.5 mg** twice daily added to low-dose (81-100 mg daily) aspirin for high risk CAD/PAD patients

Side Effects:

- GI upset (aspirin), bleeding

Management:

CVD Protection: ACEI/ARB



Indications:

- ACE inhibitors should be prescribed in all patients with stable ischemic heart disease who also have hypertension, diabetes mellitus, left ventricular ejection fraction 40% or less, or CKD, unless contraindicated.
- ARBs are recommended for patients with SIHD who have hypertension, diabetes mellitus, LV systolic dysfunction, or CKD and have indications for, but are intolerant of, ACE inhibitors.

Side Effects:

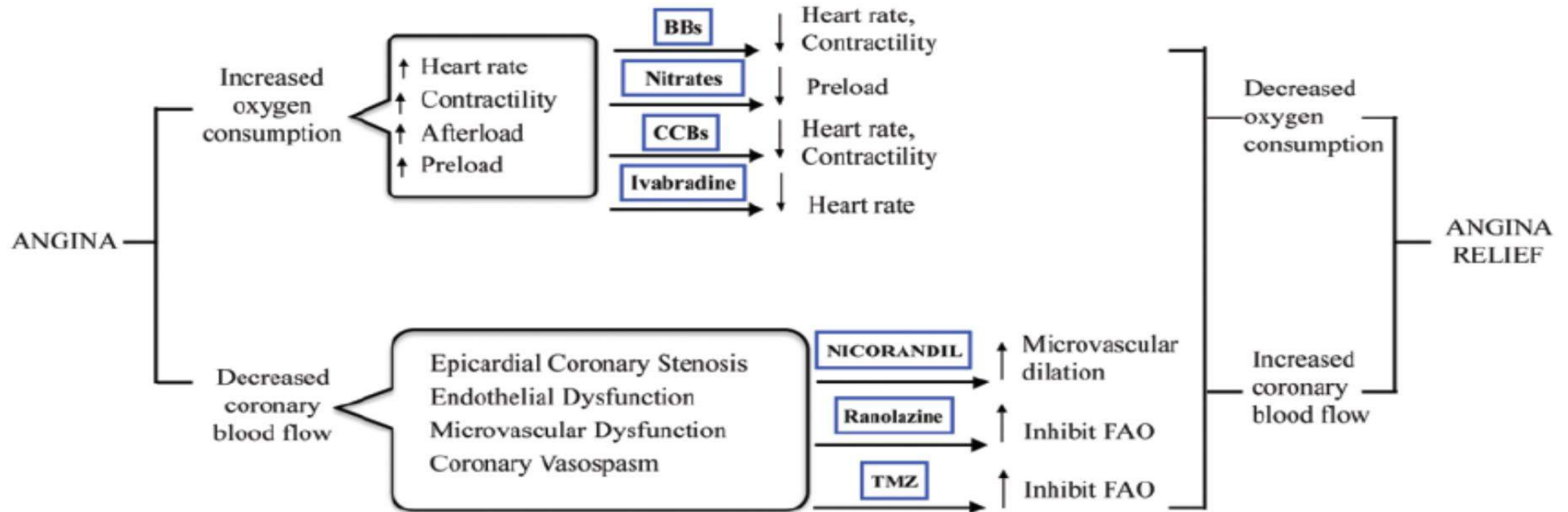
- cough (ACEI), syncope, hypotension, angioedema hyperkalaemia

Contraindications:

- High grade bilateral renal artery stenosis

Management:

Anti-ischemic Agents

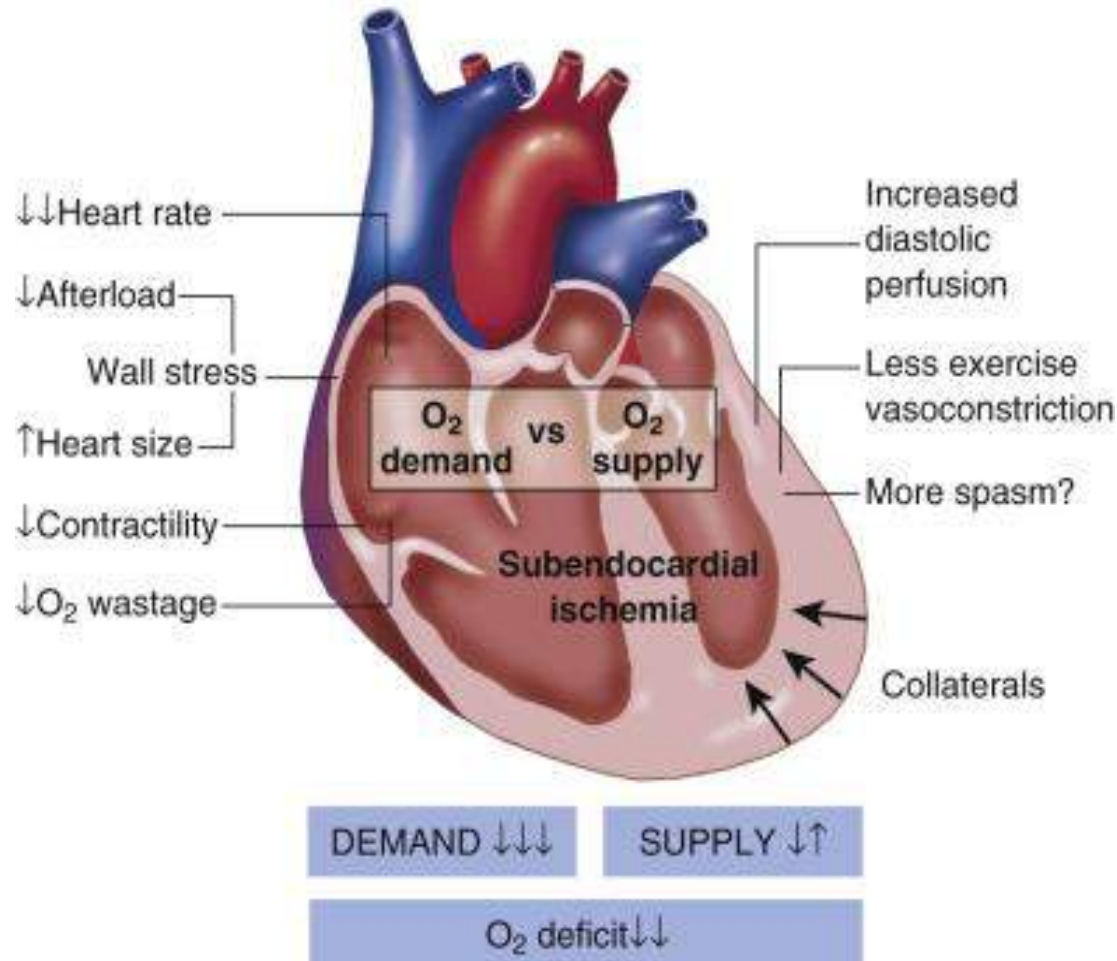


Padala SK, et al. J Cardiovasc Pharmacol Ther 2017;22:499-510

Management:

Anti-ischemic Agents: β -Blockers

BETA BLOCKADE EFFECTS ON ISCHEMIC HEART



Indications:

- first-line monotherapy or in combination with CCB or nitrates, are used to prevent or reduce the frequency of angina in patients with CAD.
- for patients with systolic LV failure (EF<40%), heart failure and prior MI
- ventricular rate control in atrial fibrillation

Side Effects:

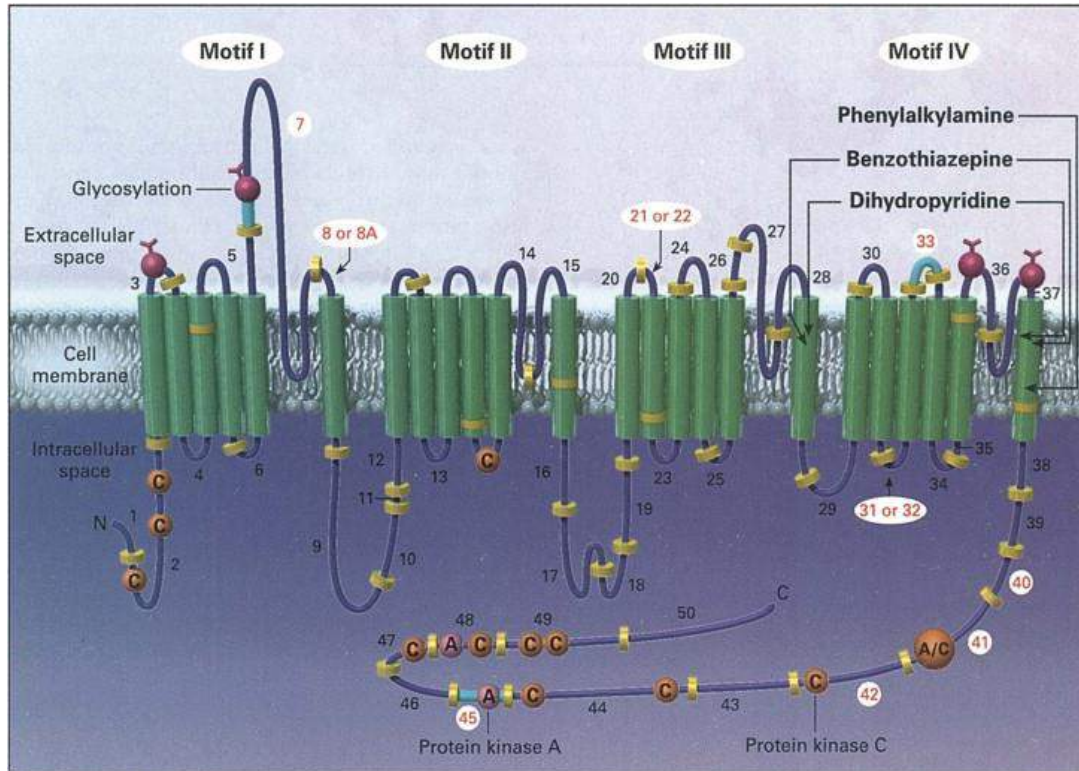
- bradycardia, syncope, hypotension, bronchial spasm, impaired glucose tolerance

Contraindications:

- AV block, sinus node dysfunction
- Bronchial asthma
- Vasospasm angina, PAD, Raynaud phenomenon
- Depression

Management:

Anti-ischemic Agents: CCBs



3 types of calcium antagonists—phenylalkylamine (verapamil-like), benzothiazepine (diltiazem-like), and dihydropyridine (nifedipine-like).

Indications:

- as monotherapy (β -Blockers contraindicated or intolerant) or in combination with β -Blockers or nitrates, are used to prevent or reduce the frequency of angina in patients with CAD
- Vasospasm angina
- ventricular rate control in atrial fibrillation

Side Effects:

- Bradycardia (non-dihydropyridine CCBs), syncope, hypotension, peripheral edema, headache, dizziness, and constipation

Contraindications:

- AV block, sinus node dysfunction and HF (non-dihydropyridine CCBs)

Cardiovascular Therapeutics: A Companion to Braunwald's Heart Disease
Chronic Coronary Artery Disease: A Companion to Braunwald's Heart Disease

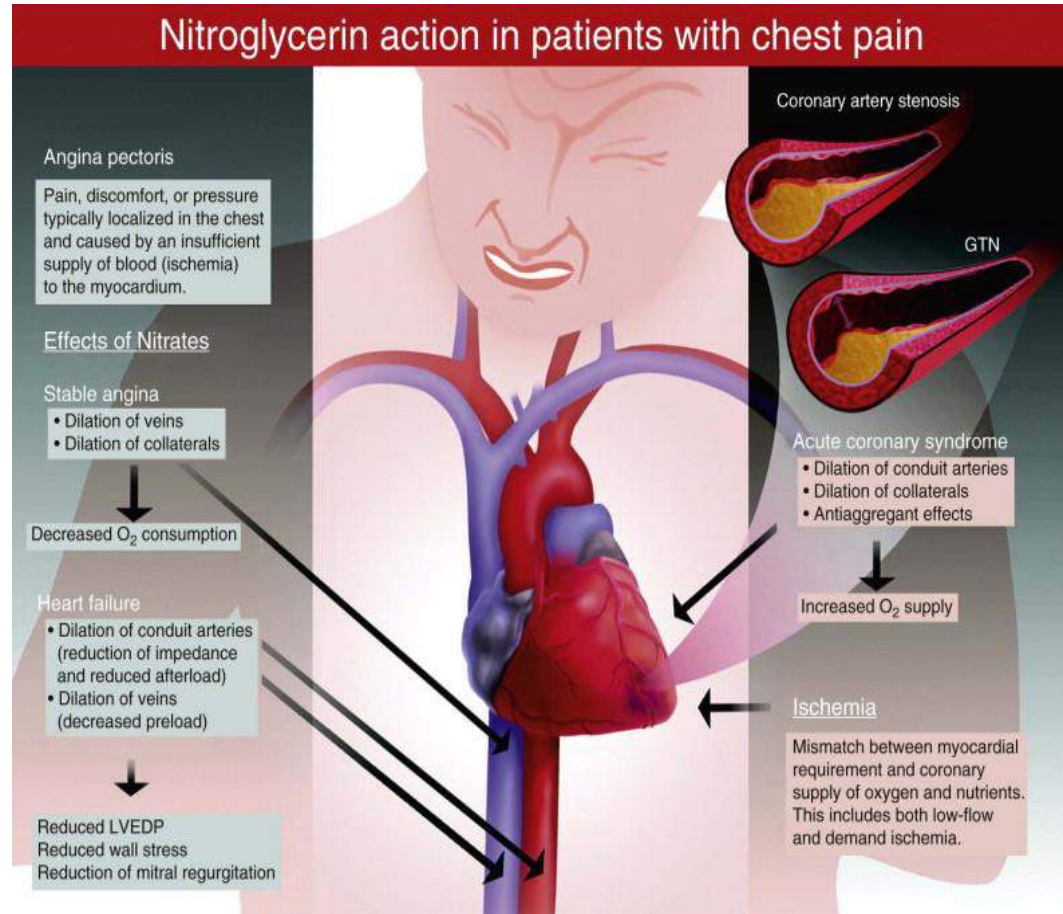
Management:

Anti-ischemic Agents: CCBs

	AMLODIPINE	DILTIAZEM	NIFEDIPINE	VERAPAMIL
Heart rate	↑/–	↓	↑	↓
Sinoatrial node conduction	–	↓	–	↓
Atrioventricular node conduction	–	↓	–	↓
Myocardial contractility	↓/–	↓	↓/–	↓↓
Neurohormonal activation	↑/–	↑	↑	↑
Vascular dilation	↑↑	↑	↑↑	↑
Coronary flow	↑	↑	↑	↑

Management:

Anti-ischemic Agents: Nitrate



Venodilatation

coronary vasodilatation

Indications:

Short-acting nitrates

- to relieve acute anginal pain
- use prophylactically to improve exercise tolerance and prevent exercise-induced ischemia.

Long-acting nitrates

- as monotherapy or in combination with β -blockers or CCBs, are used to prevent or reduce the frequency of angina in patients with CAD.

Side Effects:

- hypotension, syncope, tachycardia, headache, and facial flushing. Less common is methemoglobinemia

Contraindications:

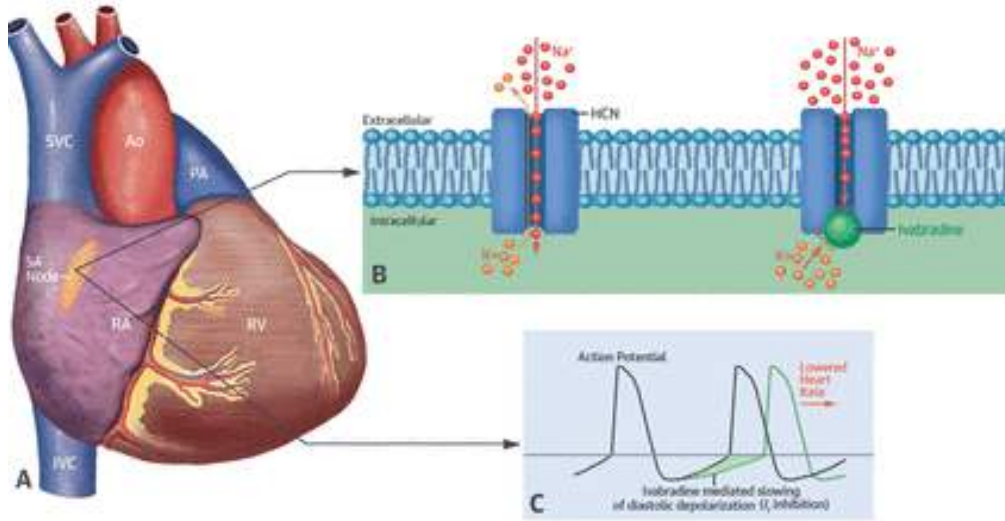
- HOCM
- Concomitant administration on the same day with selective phosphodiesterase inhibitors (PDE-5) such as sildenafil, tadalafil, and vardenafil.
- SBP < 90 mm Hg.

Cardiovascular Therapeutics: A Companion to Braunwald's Heart Disease
Chronic Coronary Artery Disease: A Companion to Braunwald's Heart Disease

Management:

Anti-ischemic Agents: Novel Agents

Ivabradine

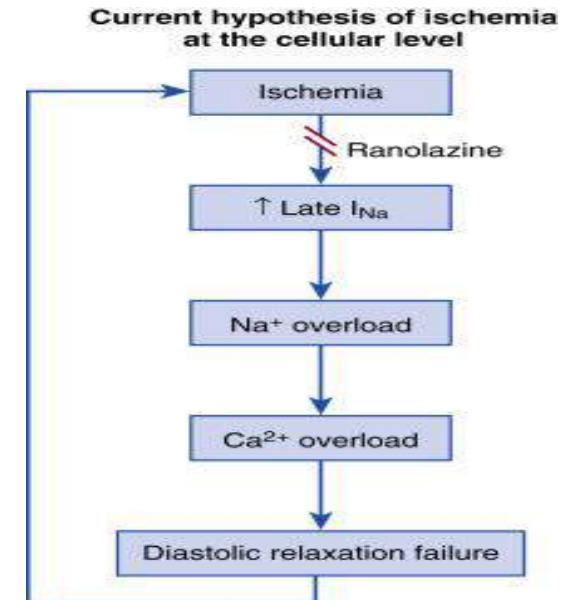


Koruth, J.S. et al. J Am Coll Cardiol. 2017;70(14):1777-84.

Side effects:

- phosphenes, bradycardia, and AF
- headache, uncontrolled blood pressure, dizziness, ventricular extrasystoles, and first-degree AV block

Ranolazine



Side effects:

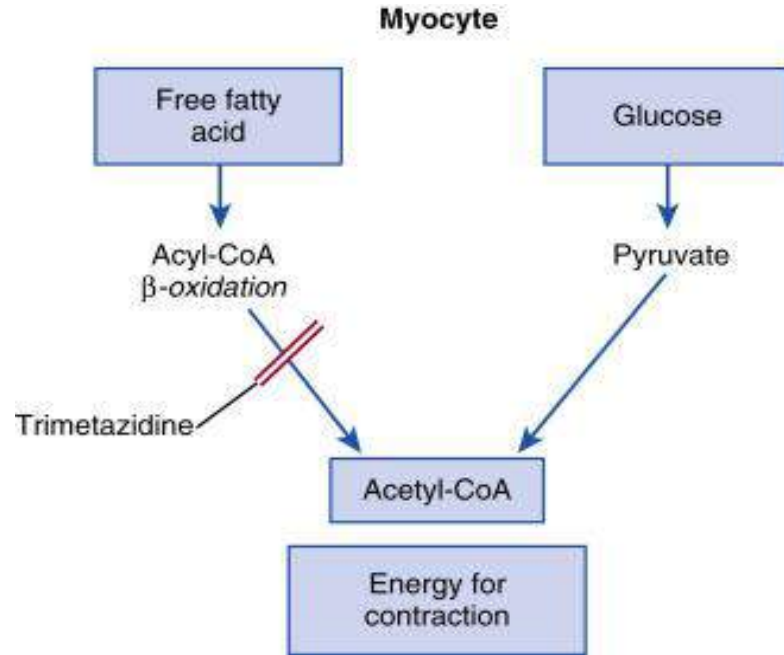
- prolongs QT, nausea, headache, dizziness and constipation, syncope and postural hypotension
- contraindicated in liver and renal failure

Management:

Anti-ischemic Agents: Novel Agents

Trimetazidine

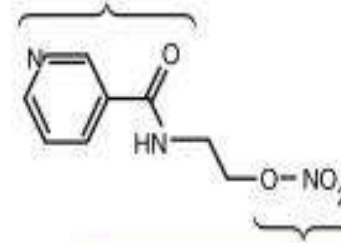
Nicorandil



Side effects:

- nausea, vomiting, fatigue, dizziness, and myalgia.
- induce or increase parkinsonian symptoms

Activation of ATP-sensitive K⁺ channels
• Dilation of coronary resistance arterioles



Nitrate-associated effects
• Vasodilation of coronary epicardial arteries

Side effects:

- caution in patients taking corticosteroids due to a small risk of gastrointestinal perforation
- sulfonylureas may antagonize the effects of nicorandil

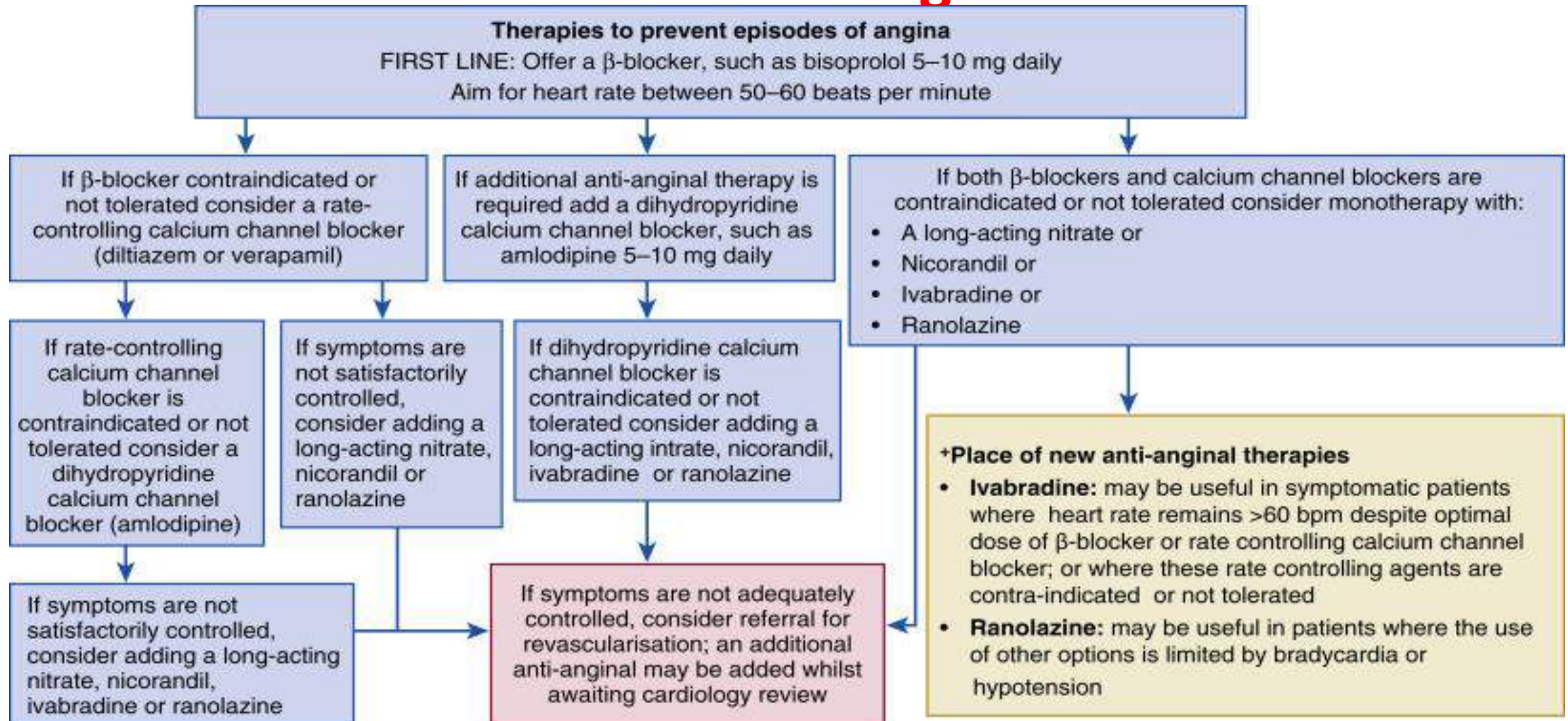
Management:

Anti-ischemic Agents

Class Of Medication	Mechanism of Action	Clinical Effects In Nste-Acs
Traditional Therapies		
Beta blockers	Decrease heart rate, blood pressure, and contractility through antagonism of beta 1 receptors	Decrease mortality ⁵¹
Nitrates	Decrease preload through venodilation; vasodilate coronary arteries	No benefit on mortality
Calcium channel blockers	May vasodilate, reduce heart rate, or decrease contractility depending on specific drug	No clear benefit on mortality or reinfarction Increased reinfarction rate when short-acting nifedipine is used alone
Newer and Experimental Therapies		
Ranolazine	Inhibits late inward sodium current	Decreases recurrent ischemia and arrhythmias
Trimetazidine	Shifts myocardial metabolism from fatty acid to glucose use	Decreases short-term mortality
Nicorandil	Activates ATP-sensitive K ⁺ channels and dilates arterioles; may have ischemic precondition-like effect	Decreases arrhythmias and transient ischemia

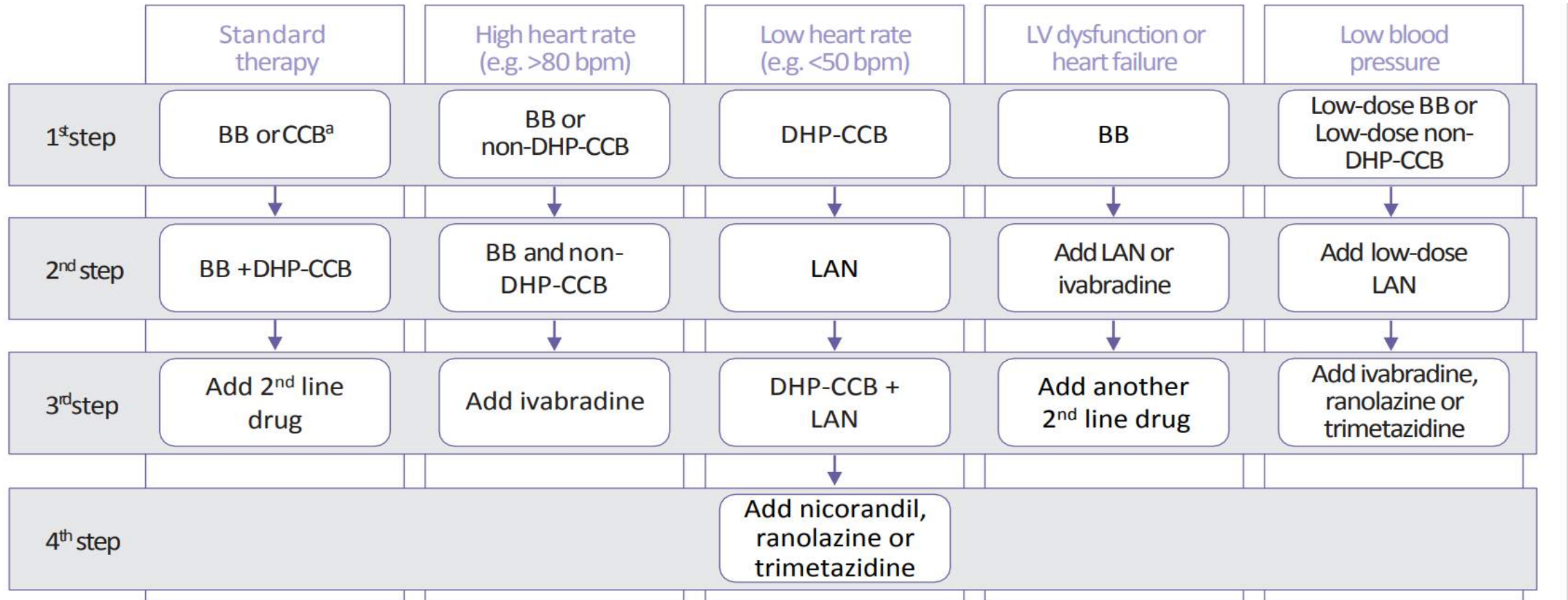
Management:

Anti-ischemic Agents



Management:

Anti-ischemic Agents

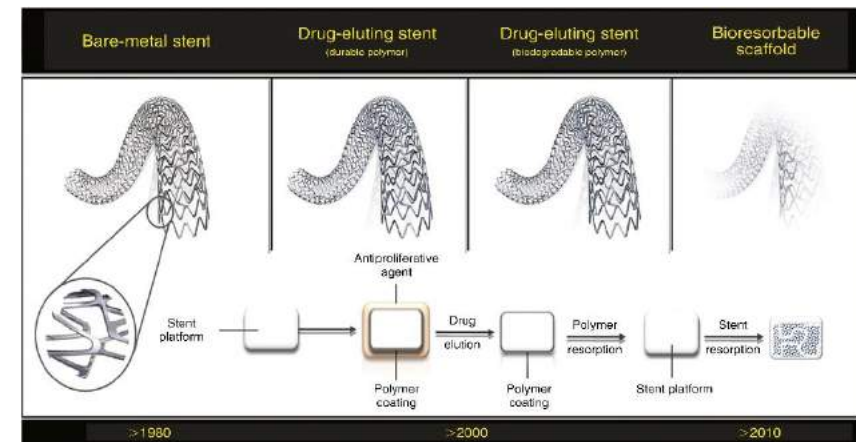
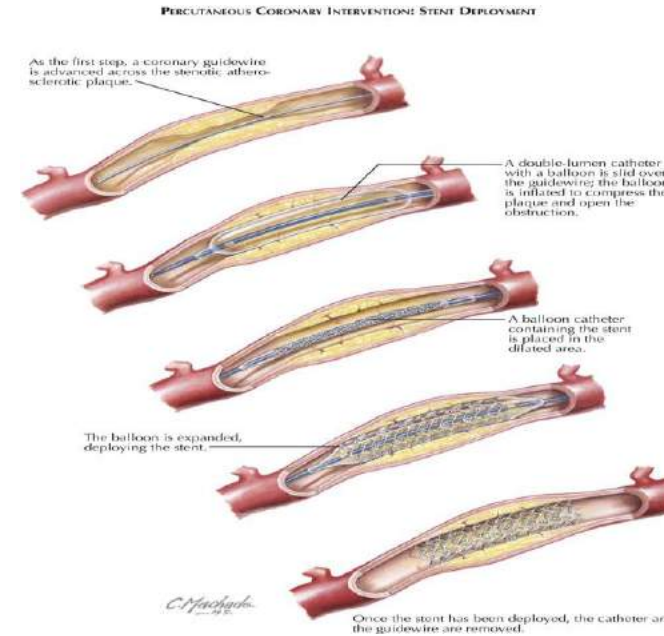
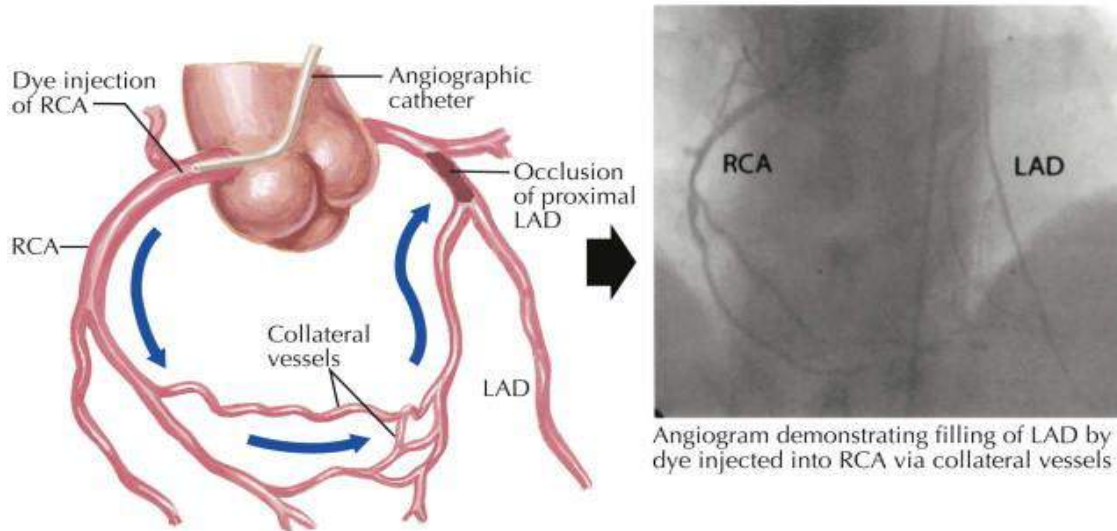
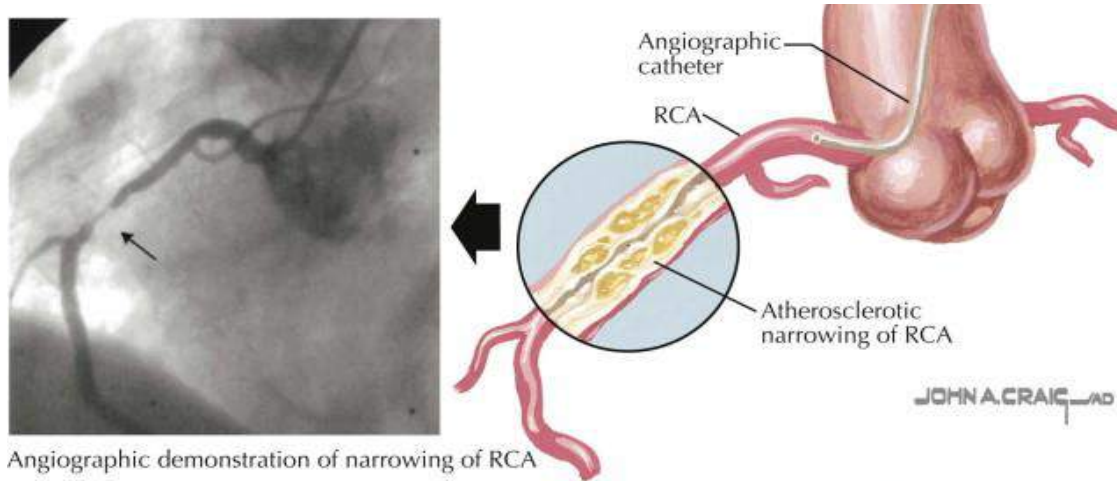


^a Combination of a BB with a DHP-CCB should be considered as first step; combination of a BB or a CCB with a second-line drug may be considered as a first step.

www.escardio.org/guidelines

2019 ESC Guidelines for the diagnosis and management of chronic coronary syndromes
European Heart Journal (2019) – doi:10.1093/eurheartj/ehz425

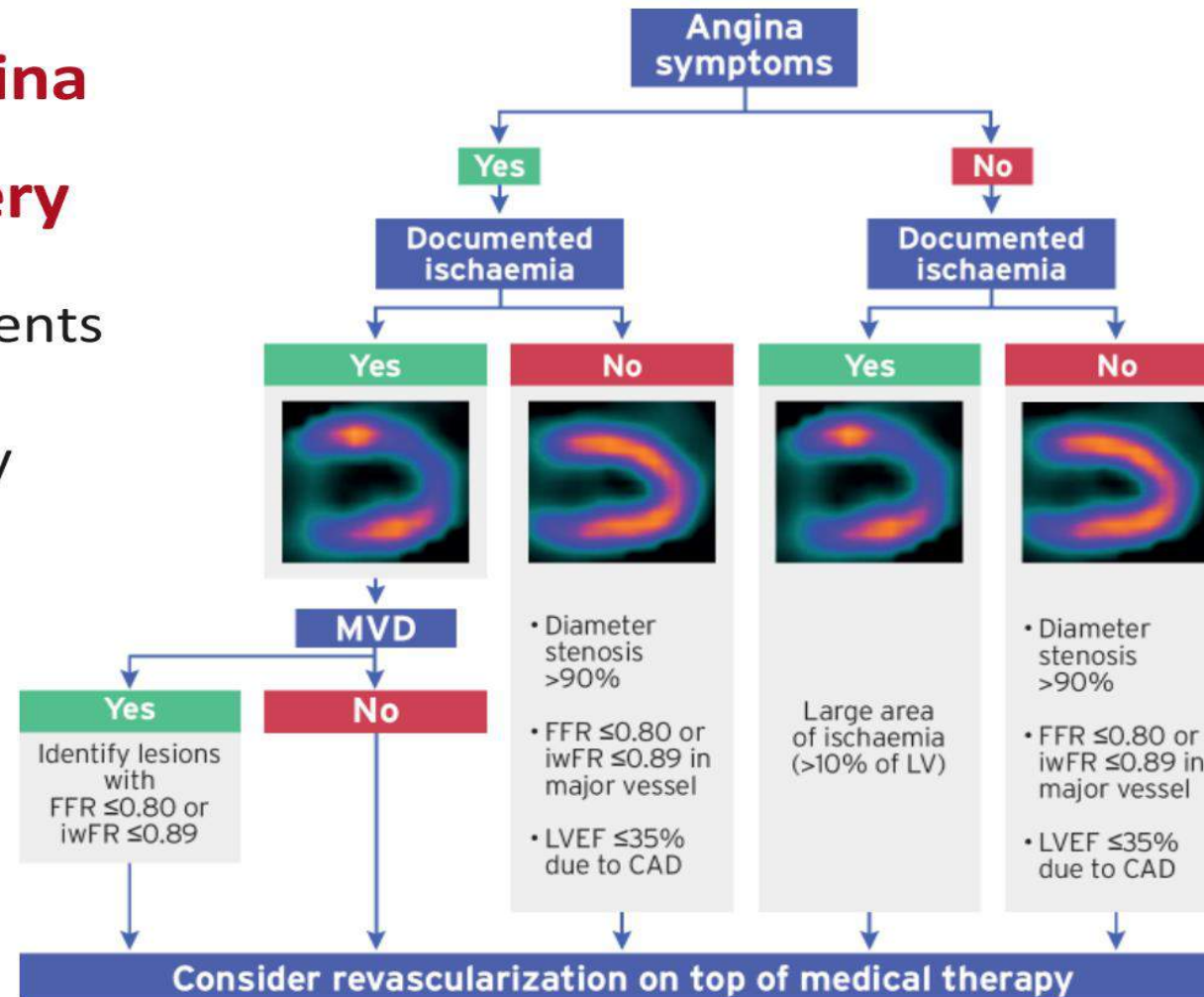
Percutaneous Coronary Intervention for Chronic Angina



Percutaneous Coronary Intervention for Chronic Angina

Patients with angina and/or dyspnoea and coronary artery disease

Decision tree for patients undergoing invasive coronary angiography

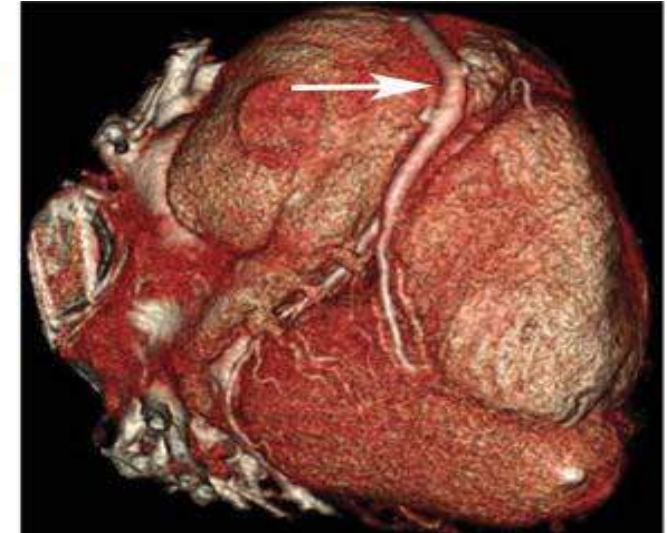
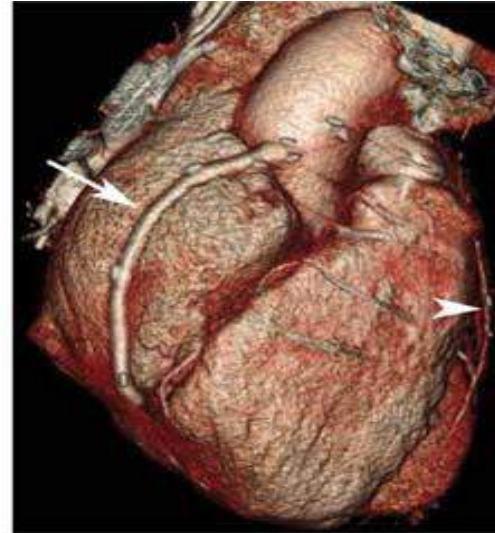
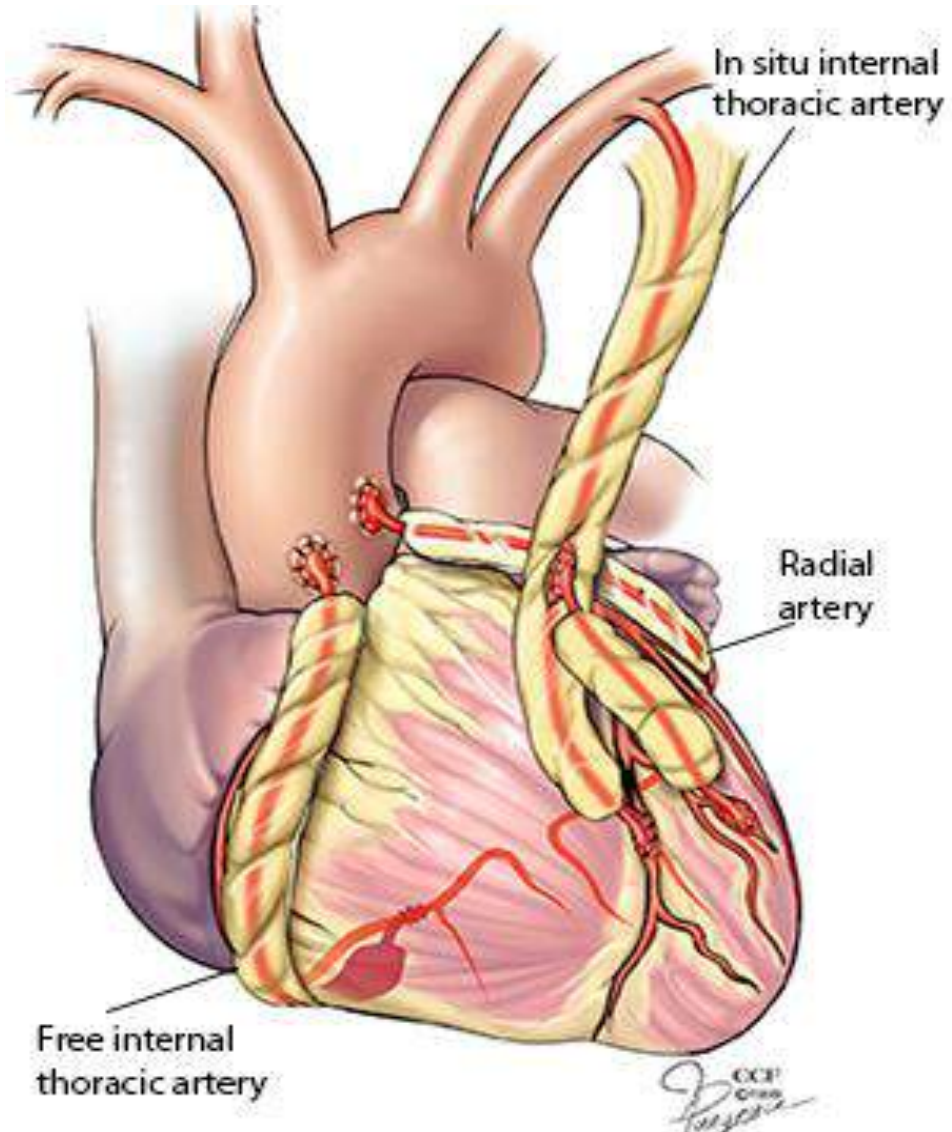


CAD = coronary artery disease;
FFR = fractional flow reserve;
iwFR = instantaneous wave-free ratio;
LV = left ventricle;
LVEF = left ventricular ejection fraction;
MVD = multivessel disease.

www.escardio.org/guidelines

2019 ESC Guidelines for the diagnosis and management of chronic coronary syndromes
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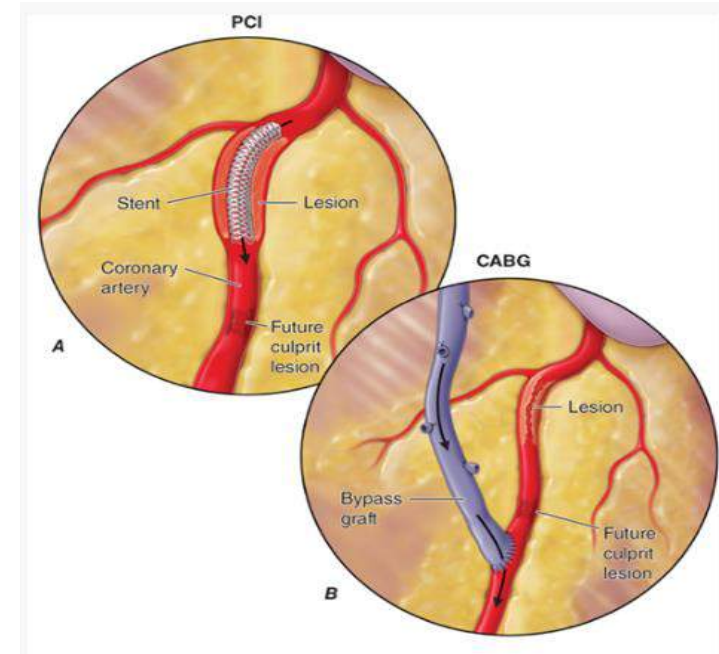
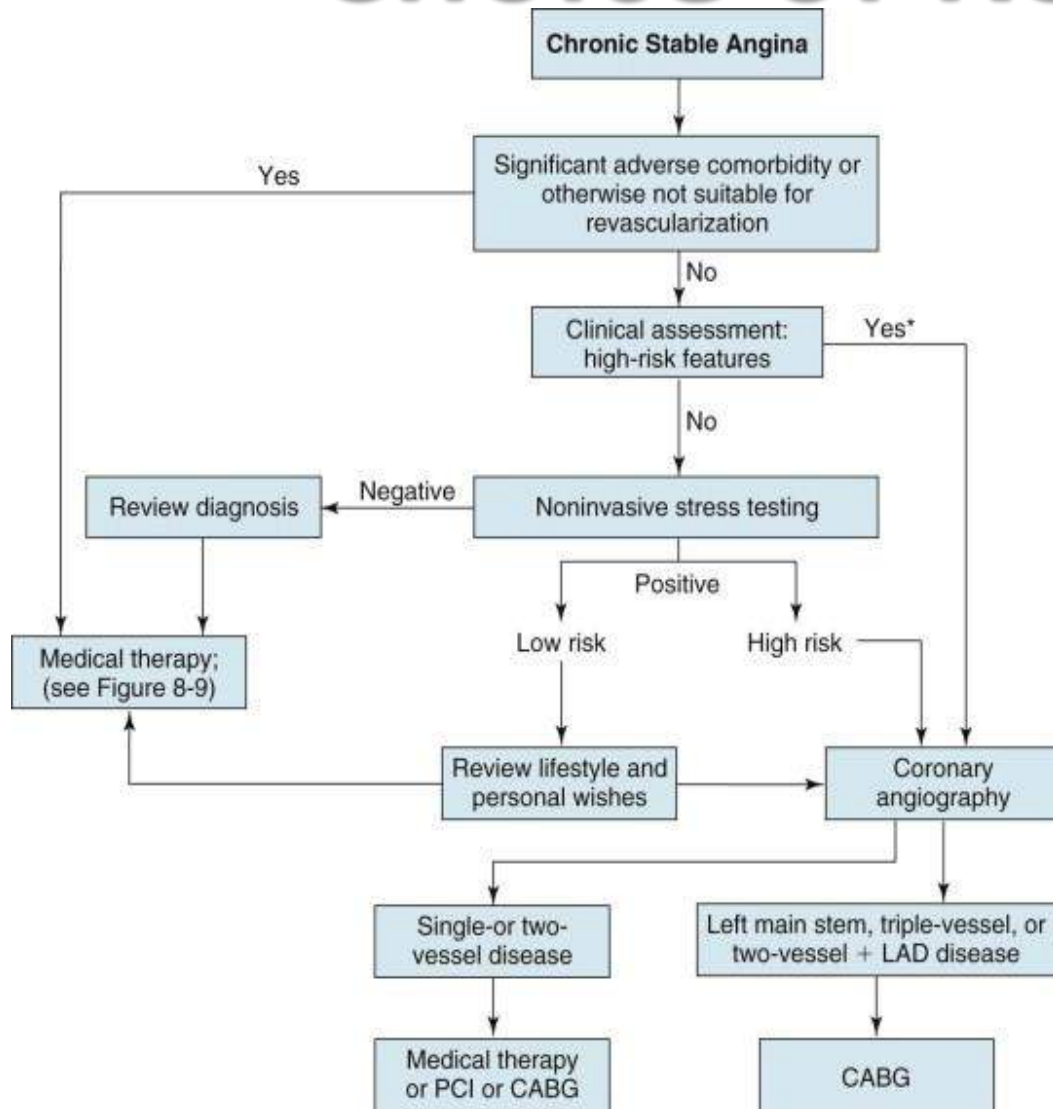
CABG for Chronic Angina



B

C

Choice of Revascularization

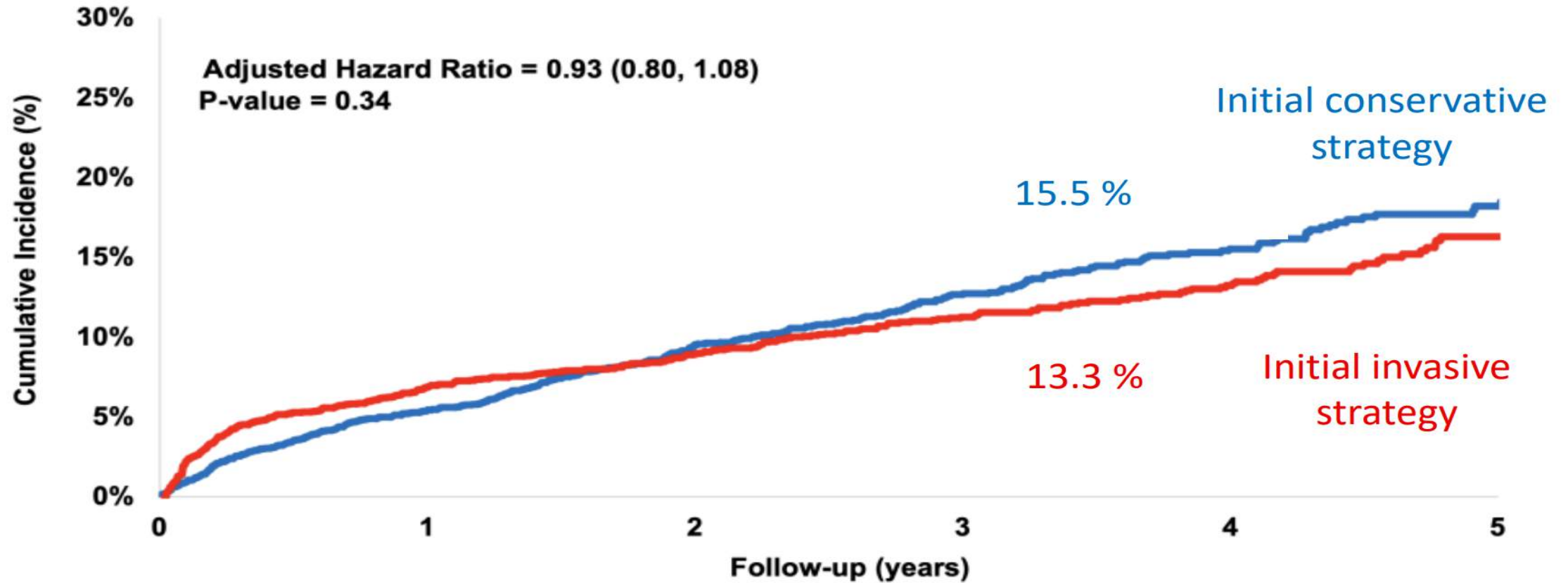


CABG is preferable than PCI for revascularization to improve survival in:

- 1. unprotected left main,**
- 2. three-vessel disease,**
- 3. those with two-vessel disease and a proximal LAD lesion**

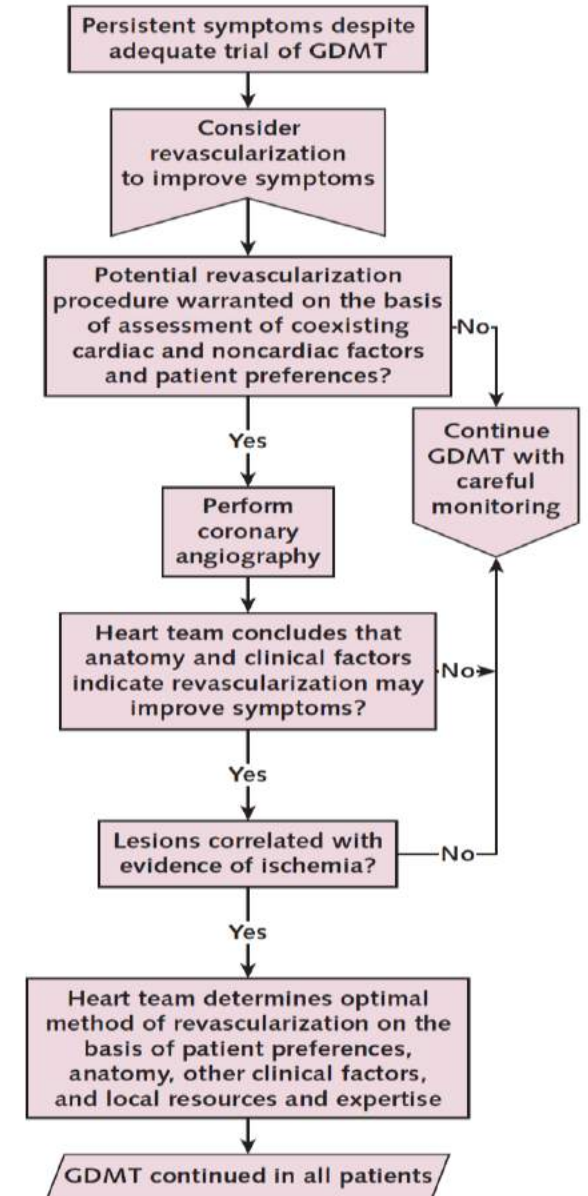
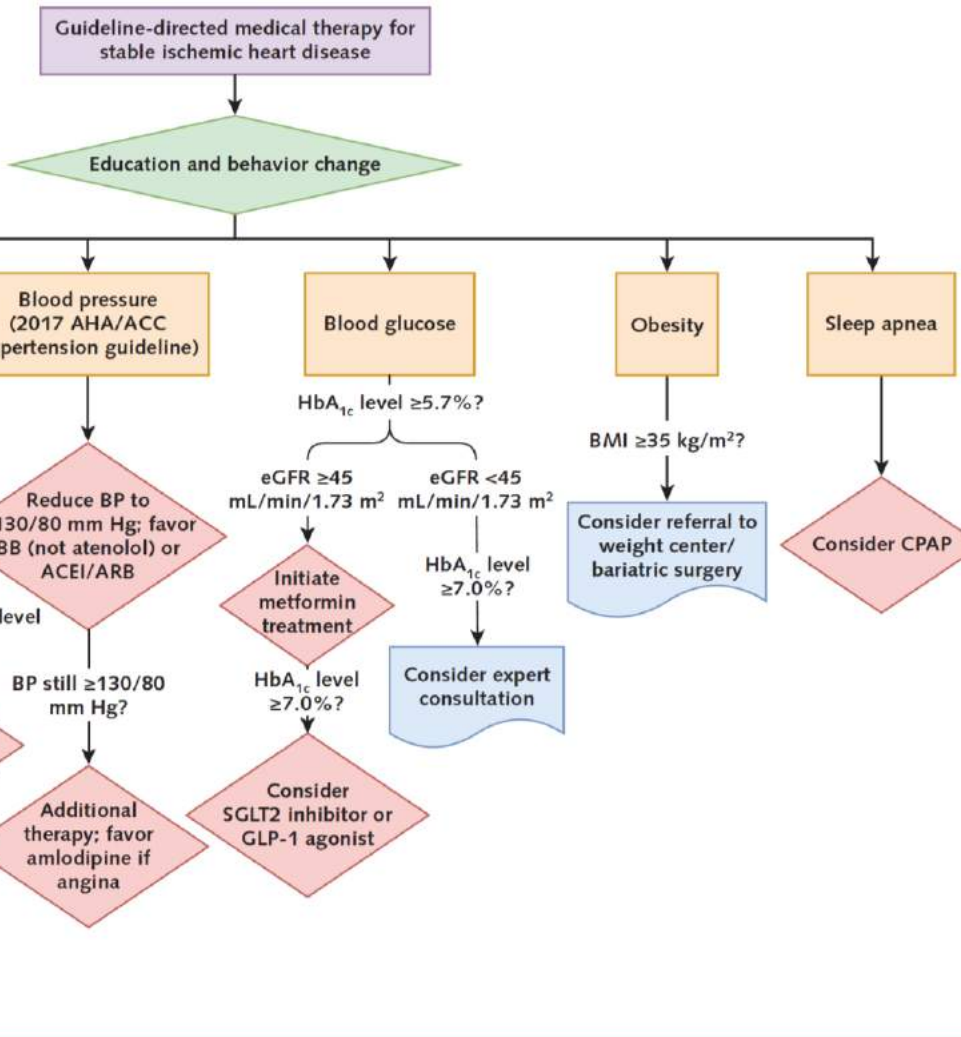
Management:

ISCHEMIA trial – Primary endpoint results

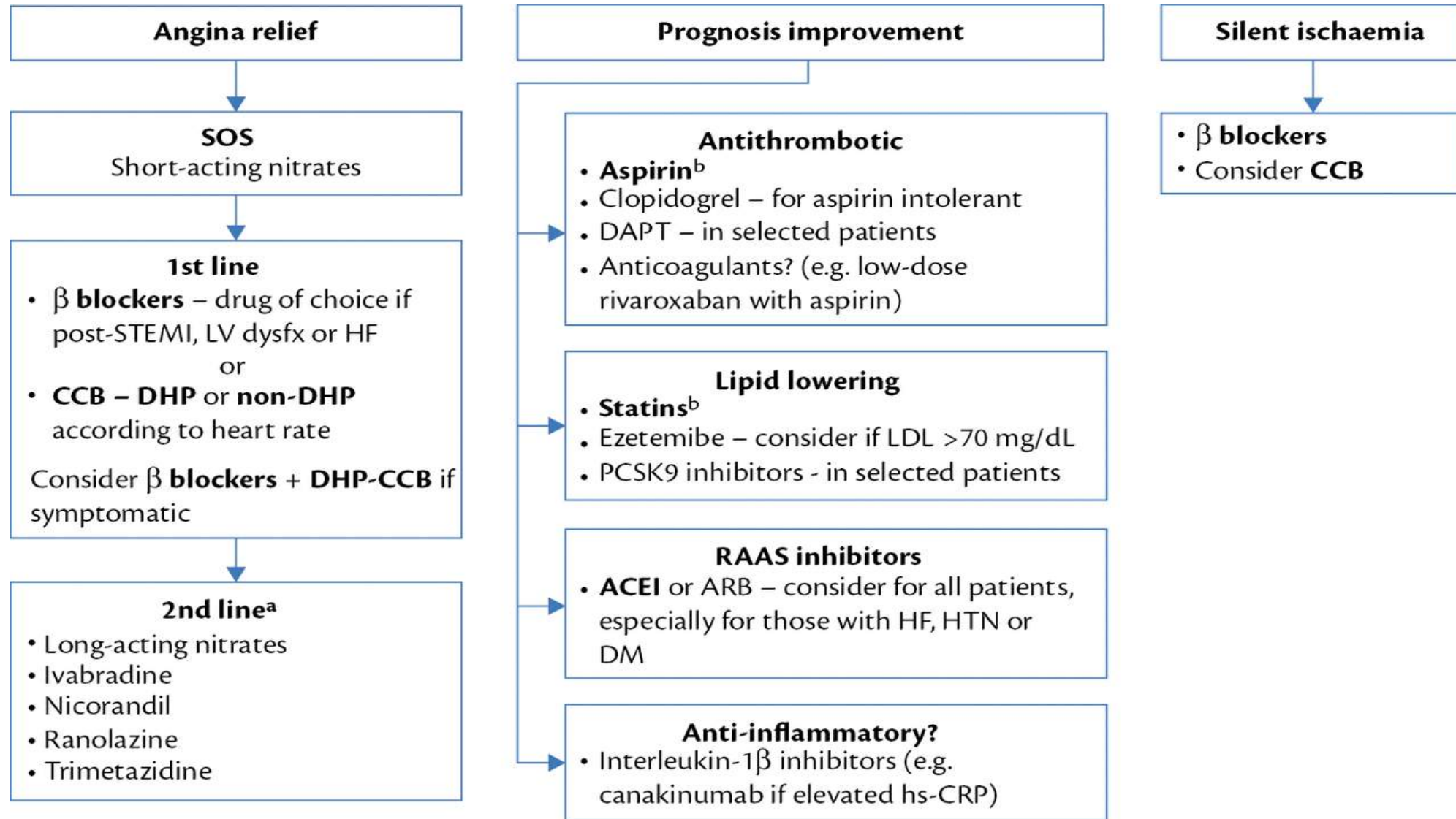


Maron DJ et al NEJM. 2020 Apr 9;382(15):1395-1407

Management



Management:



CTM, M/49 (5)

Ix:

- CTA showed triple vessel disease and suspected >70% occlusion of mid LAD
- Coronary angiogram: TVD with 30-50% LAD/LCX/RCA lesions

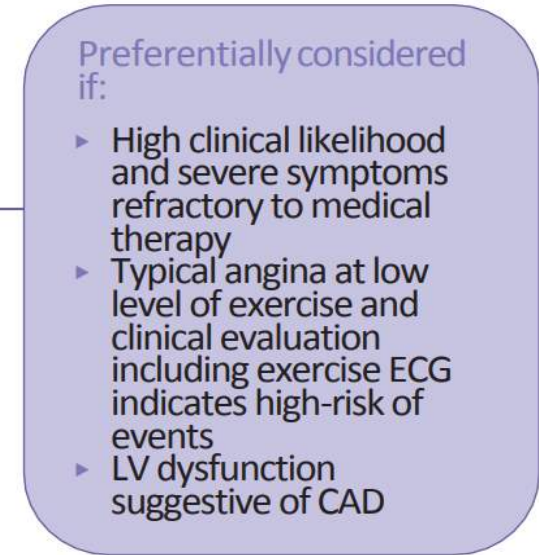
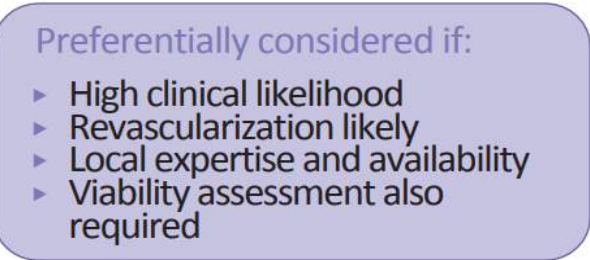
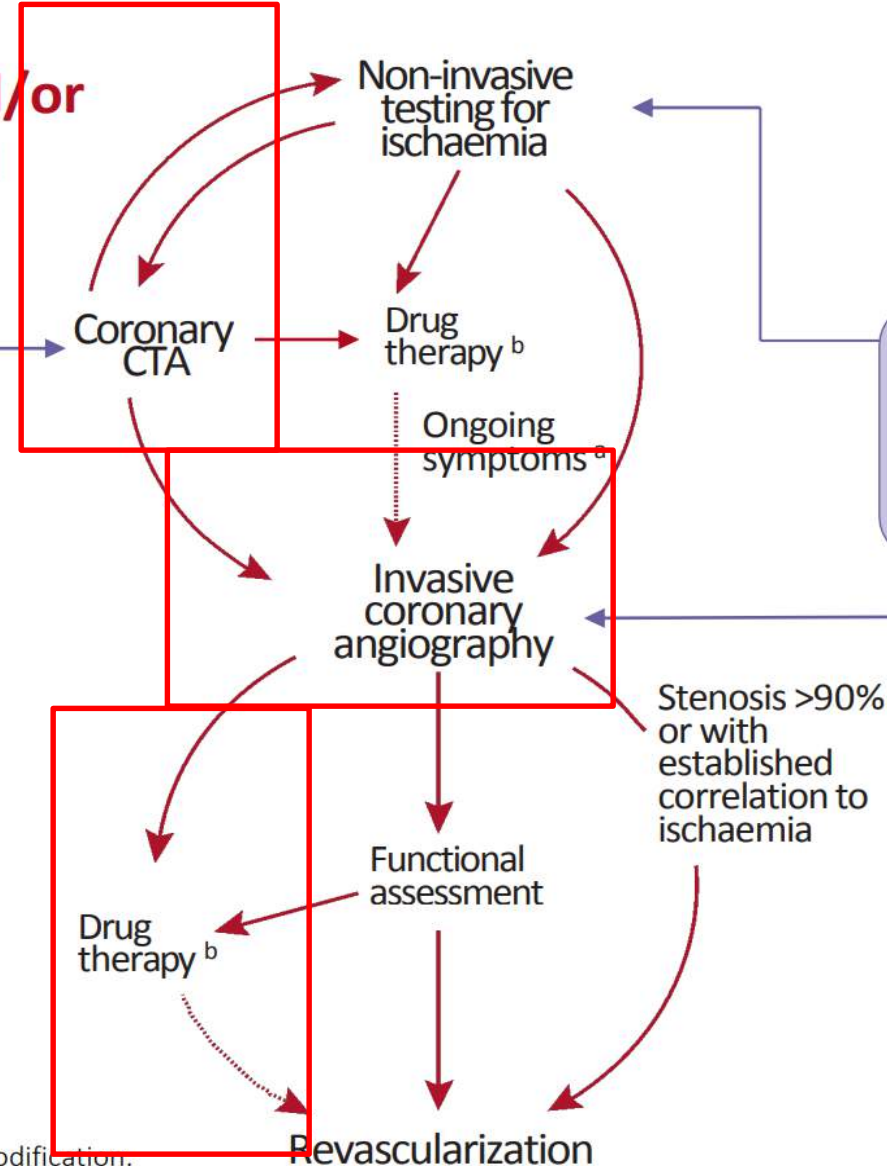
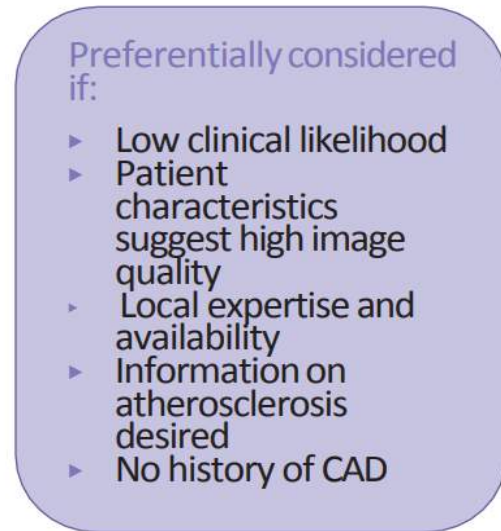
Treatment:

- Advice on diet and smoking cessation
- Aspirin 100mg daily, betaloc 25mg daily
- Started on Atorvastatin 40mg and ezetimide 10mg daily
- Repeated Blood test showed LDL-C 1.4 mmol/L



Patients with angina and/or dyspnoea and suspected coronary artery disease

Main diagnostic pathways



^a Consider microvascular angina.

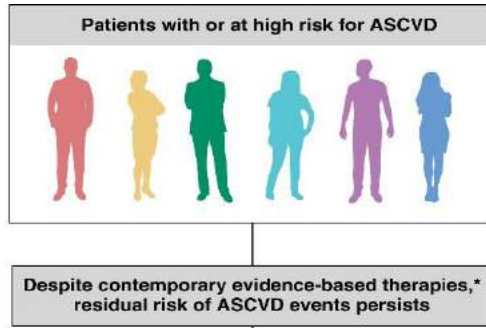
^b Antianginal medications and/or risk-factor modification.

www.escardio.org/guidelines

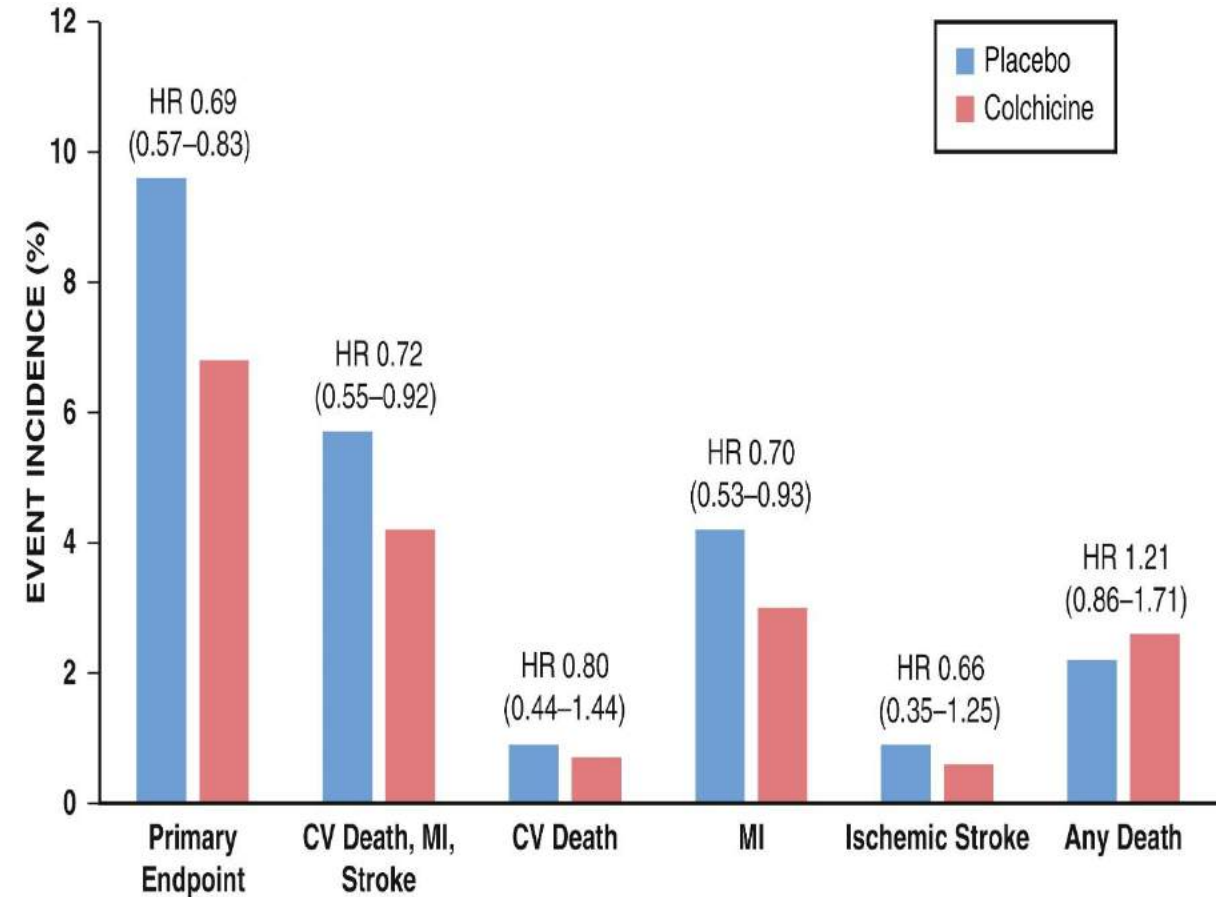
ESC Guidelines on the diagnosis and management of chronic coronary syndromes
(European Heart Journal 2019; 10.1093/eurheartj/ehz425)

Management:

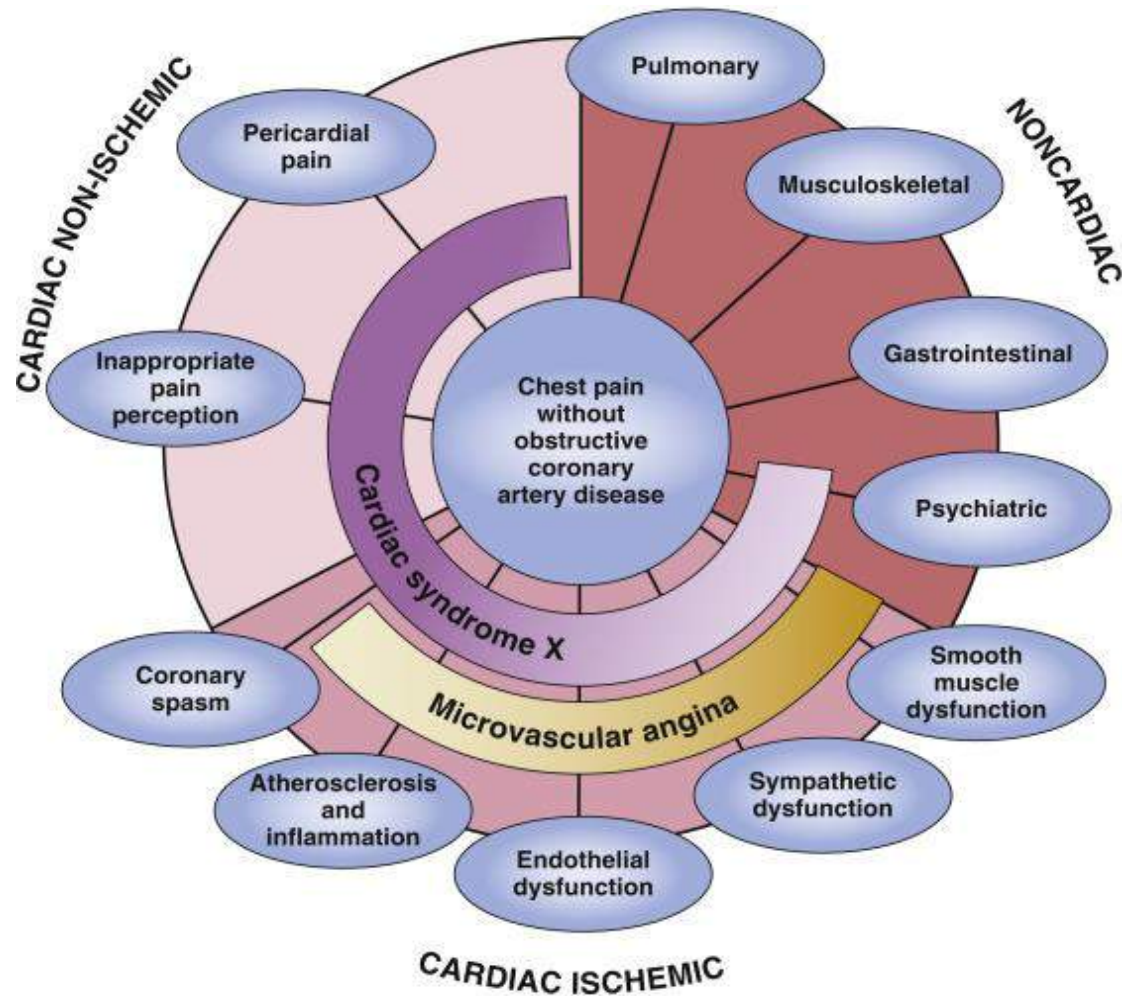
Emerging Approaches for CAD Prevention



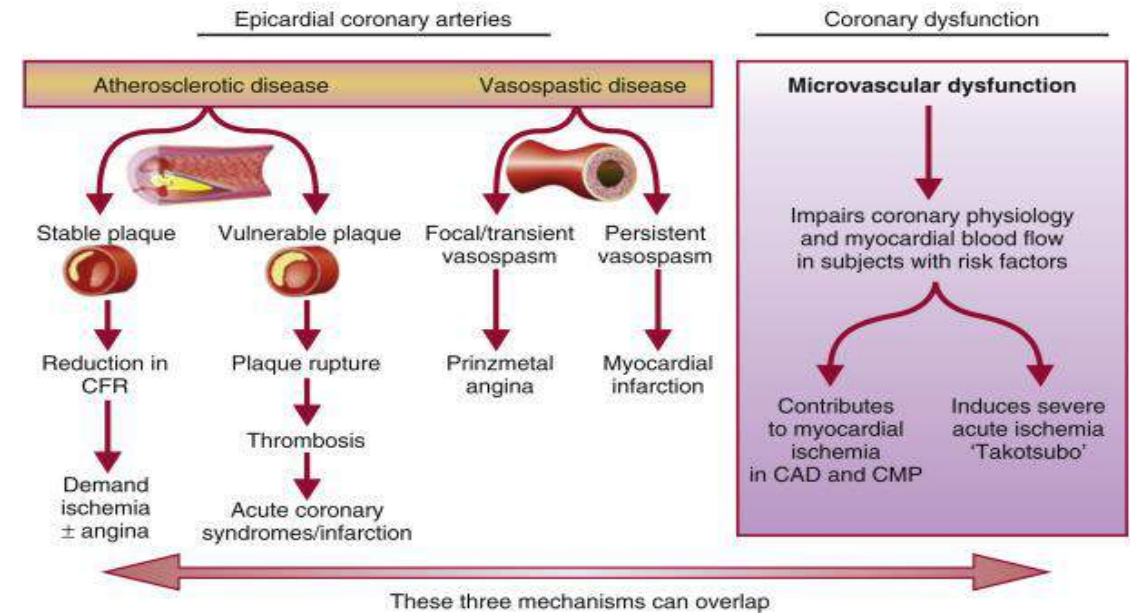
Biological Issue	Residual Cholesterol Risk	Residual Inflammatory Risk	Residual Thrombotic Risk	Residual Tryglyceride Risk	Residual Lp(a) Risk	Residual Diabetes Risk
Critical Biomarker	LDL-C $\geq$ 100 mg/dL	InsCRP $\geq$ 2mg/L	No simple biomarker	TG $\geq$ 150mg/dL	Lp(a) $\geq$ 50mg/dL	HbA1c Fasting glucose
Potential Intervention	Targeted LDL/Apo B Reduction	Targeted Inflammation Reduction	Targeted Antithrombotic Reduction	Targeted Triglyceride Reduction	Targeted Lp(a) Reduction	SGLT2 Inhibitors GLP-1 Agonists
Randomized Trial Evidence	IMPROVE-IT FOURIER SPIRE ODYSSEY	CANTOS COLCOT LoDoCo2 OASIS-9	PEGASUS COMPASS THEMIS	REDUCE-IT PROMINENT	Planned	EMPA-REG CANVAS DECLARE CREDENCE LEADER SUSTAIN-6 REWIND



Angina in Patients with Evidence of Myocardial Ischemia and No Obstructive Coronary Artery Disease



Mechanisms of myocardial ischemia



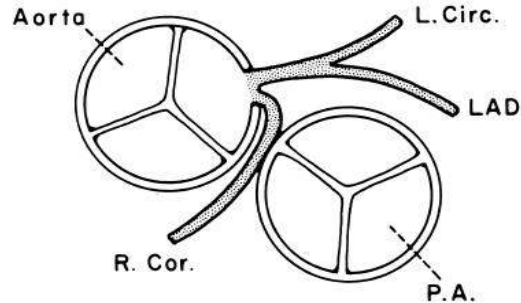
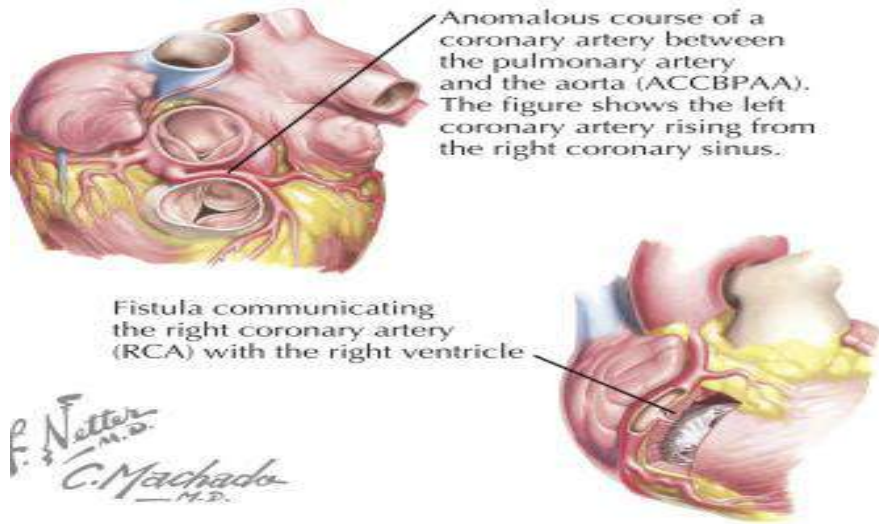
Cardiac syndrome X:

- traditional pharmacologic antianginal, anti-ischemic medications
- enhanced external counterpulsation,
- cognitive behavioural therapy, tricyclic medication, and neurostimulation

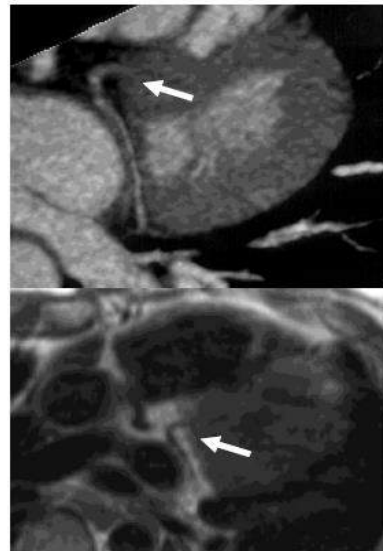
Cardiovascular Therapeutics: A Companion to Braunwald's Heart Disease

Chronic Coronary Artery Disease: A Companion to Braunwald's Heart Disease

Congenital Coronary Anomalies



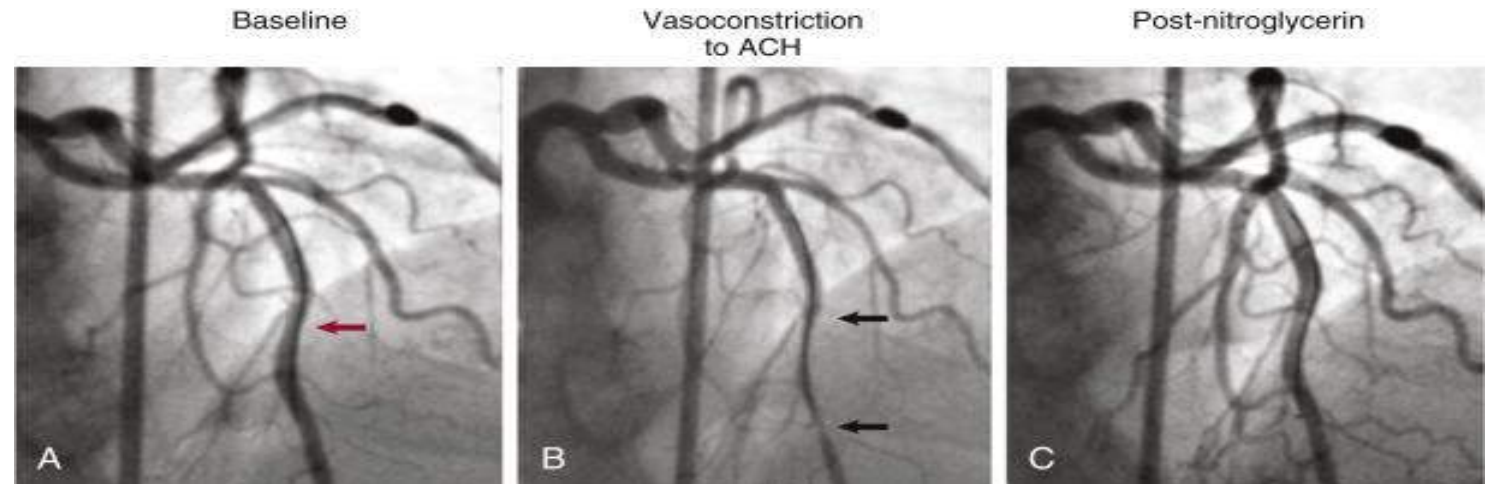
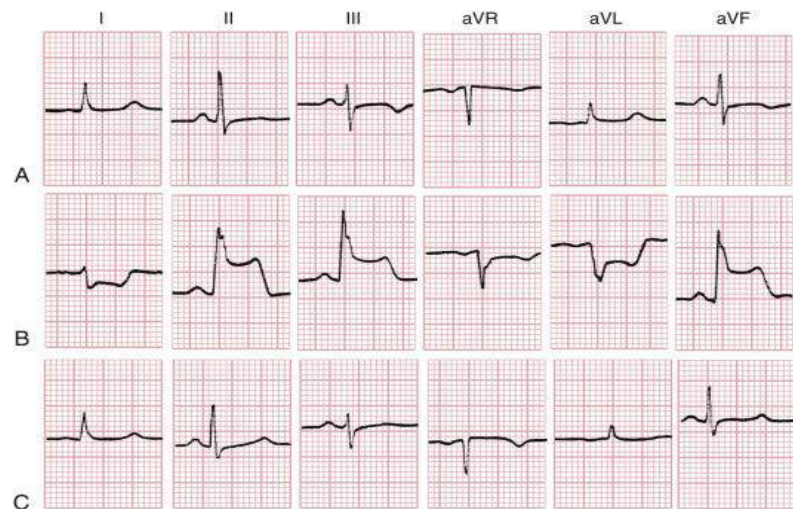
- ▶ Anomalous origin of coronary artery
- ▶ Acute angle bend causing compression between pulmonary artery and aorta



- Account for 6-17% of sudden death in athletes
- Symptoms: chest pain, exertional syncope and sudden cardiac death
- Physical examination and ECG: NAD
- Diagnosis: CT coronary angiogram, MRI or coronary angiogram
- Avoid of competitive sport, consider surgical intervention if +ve for myocardial ischemia

Prinzmetal Variant

- Occurs at rest, common after cold exposure.
- Cyclical in nature.
- ECG finding of episodic ST-segment elevations.
- Caused by coronary artery spasm with or without superimposed CAD.
- Patients are more likely to develop ventricular arrhythmias
- Rx: CCBs+/- Nitrate.



Conclusions:

1 Exclusion of acute coronary syndrome and noncardiac chest pain

▶ Perform detailed patient history and physical examination

Determine effect of symptoms on quality of life
Determine if there are signs of heart failure or valvular disease
Estimate pretest probability for coronary artery disease (CAD)

2 Assessment of objective risk markers for CAD

▶ Perform electrocardiography

Identify left bundle-branch block or major ST- or T-wave abnormality to determine what CAD assessment should be performed first
Detect prior infarction or left ventricular (LV) hypertrophy

▶ Consider echocardiography

Detect LV systolic dysfunction
Detect valve abnormalities

▶ Consider laboratory tests to determine other risks

Detect kidney impairment, diabetes, dyslipidemia, cardiac stress, or cardiac injury (abnormal or worsening hs-cTnT or NT-proBNP)

3 Consideration of anatomical or functional testing for CAD

▶ Defer testing

If symptoms are infrequent and without major effect on quality of life
If there are no high-risk findings from initial objective evaluation, such as prior infarction or ST-segment abnormalities detected on electrocardiogram, LV dysfunction seen on echocardiogram, or abnormal hs-cTnT or NT-proBNP

▶ Coronary computed tomographic angiography

Detect obstructive or nonobstructive CAD
Exclude left main or triple vessel disease

▶ Stress testing

If functional assessment is desired
If there is known CAD

▶ Invasive angiography

Further evaluate equivocal findings from noninvasive testing
Determine suitability for coronary revascularization

4 Management of stable angina

▶ Initiate optimal medical therapy

Use secondary prevention medications such as high-potency statin and low-dose aspirin to reduce risk for cardiovascular events
Antianginal medications such as β -blocker, calcium channel blockers, or nitrates to improve angina-related quality of life
Consider SGLT2i or GLP-1R agonist if patient has diabetes

▶ Perform coronary artery bypass graft surgery if indicated

If there is left main or triple vessel disease with diabetes
If there is left main or triple vessel disease with LV systolic dysfunction

▶ Titrate antianginal therapies

Based on symptoms, adverse effects, heart rate, and blood pressure

▶ Consider percutaneous coronary intervention

When there is persistent angina that impairs quality of life

Conclusions:

ABCDE Approach Rx for IHD

A→ **A**nti-platelet agents, **A**nti-anginal agents & **A**CEI/**A**RB, **A**ngioplasty

B→ **B**eta-blocker, **B**P control & Bypass

C→ **C**holesterol lowering agents, **C**igarette cessation & **C**alcium channel blockers

D→ **D**iabetes & **D**iet management

E→ **E**ducation & **E**xercise



TELL US WHAT

MBBS Whole Class Session

YOU THINK

THANK YOU!



Look for today's teacher [name of the teacher]
and the others you may have missed.
(HKU Portal login required)